VIETNAM GENERAL CONFEDERATION OF LABOR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY

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FINAL ESSAY

DATA ANALYSIS AND VISUALIZATION

HO CHI MINH CITY, 2025

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DATA ANALYSIS AND VISUALIZATION

Advised by

PhD. Tran Thi Thanh Diu

HO CHI MINH CITY, 2025

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DECLARATION OF AUTHORSHIP

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CHAPTER 1. INTRODUCTION

1.1 Context and Motivation

In the contemporary era, metabolic health and physical well-being have emerged as paramount concerns, directly influencing individual quality of life. While the proliferation of smart wearable technology has generated vast amounts of physiological and activity data, raw data alone is insufficient for optimizing health outcomes. There remains a critical need to transition from mere data collection to actionable insight. Driven by this necessity, this research aims to apply Data Science techniques to decode complex patterns in human exercise physiology, seeking to understand the determinants of workout efficiency and behavioral habits.

1.2 Research Objectives and Hypotheses

To address the challenges of optimizing physical well-being, this study formulates three primary questions:

We hypothesize that distinct patterns exist in exercise selection based on demographics. Specifically, we aim to investigate whether:

- Research question 1: *a statistically significant dependency exists between gender and the preference for specific workout types*, validating or refuting common fitness stereotypes.
- Research question 2: *Which variables act as the strongest predictors for "Calories Burned", and does the regression model accurately estimate calorie expenditure based on an individual's profile?*

1.3 Dataset and Objective

For this project, we select the **"Life Style Data"** from Kaggle (https://www.kaggle.com/datasets/jockeroika/life-style-data). This dataset contains a rich array of health metrics, workout habits, and nutritional information. We perform an in-depth analysis of how daily lifestyle factors - such as workout duration, exercise type, and water intake - correlate with critical health outcomes like Body Mass Index (BMI) and calorie expenditure.

1.4 Report Structure

By addressing these questions, we derive practical, data-driven recommendations to help the community enhance their well-being and exercise efficacy. The subsequent chapters detail the data cleaning process, visualization of distributions, statistical testing, and the development of the predictive model.

CHAPTER 2. EXPLORATORY DATA ANALYSIS (EDA)

2.1 Data Overview

First, we do step 1 (Exploratory Data Analysis) to get a better understanding of the dataset we are working with.

We chose only 15 variables instead of trying to use all of them because we are applying the ‘quality over quantity’ principle in lifestyle research. The initial raw dataset appeared ‘greedy’, containing too many redundant or semantically overlapping variables. Keeping all 54 variables would be like trying to cook a dish by throwing in every spice in the kitchen - it would certainly ruin the main flavor.

Therefore, we focused only on the 15 core variables that we believe most directly reflect fundamental physical factors (Age, Weight, Height), exercise behavior (Workout_Frequency, Session_Duration), nutrition (Water_Intake), and outcome (Calories_Burned). This approach saves me processing time and, most importantly, reduces noise to gain a clearer insight into the real relationships between health factors, rather than being distracted by complex variables with little practical significance.

STT	Variable Name	Data Type	Description
1	Age	Float	Age of the participant (in years).
2	Gender	object	Biological gender (Male/Female).
3	Weight (kg)	Float	Weight of the individual in kilograms.

4	Height (m)	Float	Height of the individual in meters.
5	BMI	Float	Body Mass Index, a measure of body fat based on height and weight.
6	Fat_Percentage	Float	Body fat percentage of the individual.
7	Workout_Type	object	Type of workout performed (e.g., Strength, HIIT, Cardio).
8	Session_Duration (hours)	Float	Duration of the workout session in hours.
9	Workout_Frequency (days/week)	Float	Number of workout days per week.
10	Experience_Level	Float	Fitness experience level (1=Beginner, 2=Intermediate, 3=Advanced).
11	Calories_Burned	Float	Total calories burned during the session.
12	Max_BPM	Float	Maximum heart rate recorded during a workout session.
13	Avg_BPM	Float	Average heart rate maintained during the session.
14	Resting_BPM	Float	Resting heart rate before starting the workout.
15	Water_Intake (liters)	Float	Average daily water consumption in liters.

```
========================================
            LOADING DATASET
========================================

Loaded: 20,000 rows * 15 columns
```

Figure 1: Loading Dataset

	Column	Data Type
0	Age	float64
1	Gender	object
2	Weight (kg)	float64
3	Height (m)	float64
4	BMI	float64
5	Fat_Percentage	float64
6	Max_BPM	float64
7	Avg_BPM	float64
8	Resting_BPM	float64
9	Session_Duration (hours)	float64
10	Calories_Burned	float64
11	Workout_Type	object
12	Workout_Frequency (days/week)	float64
13	Water_Intake (liters)	float64
14	Experience_Level	float64

Figure 2: Data Information

Handling Missing Data:

During the initial data quality check, we inspected the presence of missing values in the Life_style_data dataset.

Result: After checking all 20,000 records, no missing values were found in the selected data columns. Therefore, imputation or dropping rows were not required.

```
# Missing values
missing = df.isnull().sum()
```

Figure 3: Handling Missing Data Code

```
1. MISSING VALUES:
--------------------------------------------------
No missing values
Total missing: 0
```

Figure 4: Missing Values Check Result

Handling Duplicates:

Duplicate data can bias statistical estimates. We proceeded to check for and remove completely duplicate records.

Result: The checking process found no duplicate records among the 20,000 observations. The dataset was ready without duplicate handling, ensuring the independence of observations.

```
# Duplicates
duplicates = df.duplicated().sum()
```

Figure 5: Handling Duplicate Data Code

```
2. DUPLICATE ROWS:
--------------------------------------------------
Found: 0 duplicates
```

Figure 6: Duplicate Records Check Result

Variable Classification:

Classify dataset variables into numerical (continuous or discrete numeric data) and categorical (text or nominal categories). This classification determines appropriate statistical methods and visualization techniques for each variable type during exploration data analysis.

```
# Classify variables
numerical_cols = df.select_dtypes(include=[np.number]).columns.tolist()
categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
```

Figure 7: Classify Variables Implement

```
========================================
        VARIABLE CLASSIFICATION
========================================

Numerical variables (13):
  1. Age
  2. Weight (kg)
  3. Height (m)
  4. BMI
  5. Fat_Percentage
  6. Max_BPM
  7. Avg_BPM
  8. Resting_BPM
  9. Session_Duration (hours)
  10. Calories_Burned
  11. Workout_Frequency (days/week)
  12. Water_Intake (liters)
  13. Experience_Level

Categorical variables (2):
  1. Gender
  2. Workout_Type
```

Figure 8: Variable Classification Result

2.2 Descriptive Statistics

2.2.1 Statistical Summary Analysis:

After data cleaning and preprocessing, we perform a comprehensive descriptive statistical analysis to understand the central tendencies, variability, and distribution characteristics of our dataset.

```python
# Calculate statistics
numerical_cols = df.select_dtypes(include=[np.number]).columns.tolist()

desc_stats = df[numerical_cols].describe(percentiles=[.25, .5, .75]).T
desc_stats['variance'] = df[numerical_cols].var()
desc_stats['skewness'] = df[numerical_cols].skew()
desc_stats['kurtosis'] = df[numerical_cols].kurtosis()
```

Figure 9: Descriptive Statistics for Numerical Variables Implement

Descriptive Statistics for All Numerical Variables

	count	mean	std	min	25%	50%	75%	max	variance	skewness	kurtosis
Age	20000.00	38.85	12.11	18.00	28.17	39.86	49.63	59.67	146.76	-0.09	-1.20
Weight (kg)	20000.00	73.90	21.17	39.18	58.16	70.00	86.10	130.77	448.30	0.77	-0.03
Height (m)	20000.00	1.72	0.13	1.49	1.62	1.71	1.80	2.01	0.02	0.33	-0.70
BMI	20000.00	24.92	6.70	12.04	20.10	24.12	28.56	50.23	44.91	0.79	0.81
Fat_Percentage	20000.00	26.10	5.00	11.33	22.39	25.82	29.68	35.00	24.96	0.08	-0.71
Max_BPM	20000.00	179.89	11.51	159.31	170.06	180.14	189.42	199.64	132.50	-0.03	-1.18
Avg_BPM	20000.00	143.70	14.27	119.07	131.22	142.99	156.06	169.84	203.57	0.09	-1.18
Resting_BPM	20000.00	62.20	7.29	49.49	55.96	62.20	68.09	74.50	53.13	-0.06	-1.17
Session_Duration (hours)	20000.00	1.26	0.34	0.49	1.05	1.27	1.46	2.02	0.12	0.02	-0.33
Calories_Burned	20000.00	1280.11	502.23	323.11	910.80	1231.45	1553.11	2890.82	252233.95	0.68	0.28
Workout_Frequency (days/week)	20000.00	3.32	0.91	1.94	2.98	3.01	4.00	5.06	0.83	0.15	-0.80
Water_Intake (liters)	20000.00	2.63	0.60	1.46	2.17	2.61	3.12	3.73	0.37	0.06	-1.04
Experience_Level	20000.00	1.81	0.74	1.00	1.01	1.99	2.02	3.05	0.54	0.33	-1.11

Figure 10: Descriptive Statistics for Numerical Variables Results

```
for col in categorical_cols:

    freq = df[col].value_counts()
    pct = df[col].value_counts(normalize=True) * 100
```

Figure 11: Descriptive Statistic for Categorical Variable Implementation

Gender Distribution

	Gender	Frequency	Percentage
0	Female	10,028	50.14%
1	Male	9,972	49.86%

```
Mode: Female
Unique values: 2
```

Figure 12: Descriptive Statistic for Categorical Variable – Gender

Workout_Type Distribution

	Workout_Type	Frequency	Percentage
0	Strength	5,071	25.36%
1	Yoga	5,032	25.16%
2	HIIT	4,974	24.87%
3	Cardio	4,923	24.62%

```
Mode: Strength
Unique values: 4
```

Figure 13: Descriptive Statistic for Categorical Variable – Workout_Type

2.2.2 Interpretation of Key Statistical Metrics

In figure 10, 12 and 13, we can see:

Central Tendency Measures (Mean & Median):

The mean represents the arithmetic average, calculated by summing all values and dividing by the number of observations. The median is the middle value when data is sorted, representing the 50th percentile.

- Fat Percentage: The average body fat percentage is 26.10%, with a median of 25.82%. The near-identical mean and median values indicate a symmetric distribution, suggesting that extreme values do not significantly distort the central tendency. This metric is crucial for understanding body composition's relationship with calorie expenditure.
- Calories Burned: This key outcome variable shows a mean of 1280.11 calories and median of 1231.45 calories. The slightly higher median compared to the mean suggests a mild left skew, though the difference is minimal. This substantial energy expenditure reflects diverse exercise intensities and individual metabolic responses across different workout types.
- Session Duration: The average workout length is 1.26 hours, with a median of 1.27 hours. The alignment between mean and median indicates consistent workout durations across participants. This duration range represents realistic exercise sessions, from shorter high-intensity workouts to longer moderate-intensity training.

Variability Measures (Standard Deviation):

The standard deviation (SD) quantifies the spread of data around the mean - higher values indicate greater variability and heterogeneity in the population.

- Fat Percentage: An SD of 5.00% reveals substantial diversity in body composition across the sample. This range spans from lean individuals (minimum 11.33%) to those with higher body fat (maximum 35.00%).

This heterogeneity is essential for correlation analysis with calories burned, as it provides sufficient variance to detect meaningful relationships across the body composition spectrum.

- Calories Burned: The SD of 502.23 calories demonstrates considerable variation in energy expenditure. This large spread reflects the combined effects of different workout types, individual effort levels, body compositions, and metabolic rates. The high variability strengthens our ability to identify factors that significantly predict calorie burn in regression analysis.
- Session Duration: With an SD of 0.34 hours, workout durations show moderate variability around the mean. This variation captures differences between shorter, intense sessions and longer, sustained workouts. The reasonable spread ensures that session duration can serve as a meaningful predictor variable for calorie expenditure.

This high variability in physical and behavioral metrics is beneficial for our analysis - it means our dataset captures a diverse population, making our findings more generalizable to real-world fitness scenarios rather than being limited to a homogeneous group.

Distribution Spread (Quartiles: 25%, 50%, 75%):

Quartiles divide the ordered dataset into four equal parts. The Interquartile Range (IQR), calculate as $Q3 - Q1$, contains 50% of the data and is a robust measure of spread that isn't affected by extreme values.

Fat Percentage Distribution:

- Q1 (25th percentile): 22.39%
- Q2 (Median): 25.82%
- Q3 (75th percentile): 29.68%
- IQR: 7.29%
- The middle 50% of participants have body fat percentages between 22.39% and 29.68%. This IQR of approximately 7.29% shows that the

central half of the dataset spans from relatively lean to moderately high body fat levels, providing good representation across the body composition spectrum for correlation analysis.

Calories Burned Distribution:

- Q1 (25th percentile): 910.80 calories
- Q2 (Median): 1231.45 calories
- Q3 (75th percentile): 1553.11 calories
- IQR: 642.31 calories
- The middle 50% of workouts result in calorie burns between approximately 910.80 and 1553.11 calories. This substantial IQR of 642.31 calories demonstrates that even within the central half of the data, there is considerable variation in energy expenditure. The relatively symmetric quartile spacing (Q2 is nearly centered between Q1 and Q3) confirms the near-normal distribution observed in mean/median comparison.

Session Duration Distribution:

- Q1 (25th percentile): 1.05 hours
- Q2 (Median): 1.27 hours
- Q3 (75th percentile): 1.46 hours
- IQR: 0.41 hours
- The central 50% of workout sessions last between 1.05 and 1.46 hours. This IQR shows moderate clustering around standard workout durations, with most participants engaging in sessions between 1-2 hours. The symmetric quartile spacing indicates consistent duration patterns across the sample.

Evaluating Distribution Shape through Skewness and Kurtosis:

Beyond measures of central tendency and dispersion, examining the Skewness and Kurtosis coefficients provides profound insights into the symmetry and “tail heaviness” of the distributions. This assessment is crucial for determining the data’s adherence to the Normal distribution assumption - a vital prerequisite for the parametric tests conducted in subsequent sections.

Regarding Skewness:

- Most key physiological and training variables, such as Age, Avg_BPM, Session_Duration, and Water_Intake, exhibit Skewness values approximating 0 (ranging from -0.09 to 0.15). This indicates that the distributions of these variables are highly symmetric, suggesting a natural adherence to either a normal or uniform distribution.
- However, body composition variables such as Weight ($Skewness \approx 0.77$) and BMI (≈ 0.79), as well as energy expenditure Calories_Burned (≈ 0.68), display moderately positive skewness. This reflects the reality within the sample population where a subset of individuals possesses weight, BMI, or calorie expenditure values significantly higher than the average (resulting in a right-skewed tail). This finding aligns with the observations regarding outliers discussed previously.
- Notably, the Physical exercise variable exhibits very high positive skewness (> 2.4), indicating that the majority of the data is concentrated at lower values, with a few extreme values skewing the distribution.

Regarding Kurtosis:

- The majority of variables, including Age, Max_BPM, and Avg_BPM, possess negative Kurtosis coefficients (approximately -1.2), indicating a Platykurtic distribution (flat-topped). The data tends to be more spread out and less concentrated around the mean compared to a normal distribution, implying a lower frequency of extreme outliers within these variables.
- Conversely, BMI ($Kurtosis \approx 0.81$) and macronutrients (Carbs, Proteins, Fats) exhibit slightly positive Kurtosis, characterizing a

Leptokurtic distribution (peaked) relative to the normal distribution. This suggests a higher concentration of data around the central value accompanied by "heavier tails," signaling the presence of potential outliers that require careful handling in regression models.

- Significantly, the Physical exercise variable presents an exceptionally high Kurtosis (> 5.1), confirming a high concentration of data at a specific point with very long tails. This necessitates data transformation methods (such as Log transformation) if incorporated into predictive models.

The presence of positive skewness and variable kurtosis in key variables (Weight, BMI, Calories_Burned) demonstrates that the real-world data does not perfectly adhere to an ideal normal distribution. This underscores the necessity of applying non-parametric tests or data normalization techniques before conducting the Linear Regression analysis in Chapter 4.

Categorical Variables:

For categorical variables convert to numerical codes, the mean provides insight into distribution balance:

- Gender: Almost perfect 50-50 split between females (50.14%) and males (49.86%), ensuring no gender bias in the sample.
- Workout_Type: With four categories, relatively even distribution across all workout types - approximately 25% in each category.

2.2.3 Comprehensive Data Visualization

To gain deeper insights into our dataset's structure and relationships, we create eight distinct visualizations, each serving a specific analytical purpose. The rationale for selecting each chart type is explained below.

2.2.3.1 Chart 1: Histogram + KDE (Figure 14)

Purpose: To examine the distribution shape of quantitative variables (such as Age, Weight, BMI, Calories_Burned). This chart reveals whether the data follows a normal distribution or is skewed, and where the data density is concentrated.

Rationale for Chart Selection: In a Regression problem (predicting calorie consumption), the assumption of normal distribution for input variables is crucial for the model's accuracy. Combining with KDE (Kernel Density Estimate) helps smooth the curve, making it easier to observe general trends rather than just looking at discrete frequency bars.

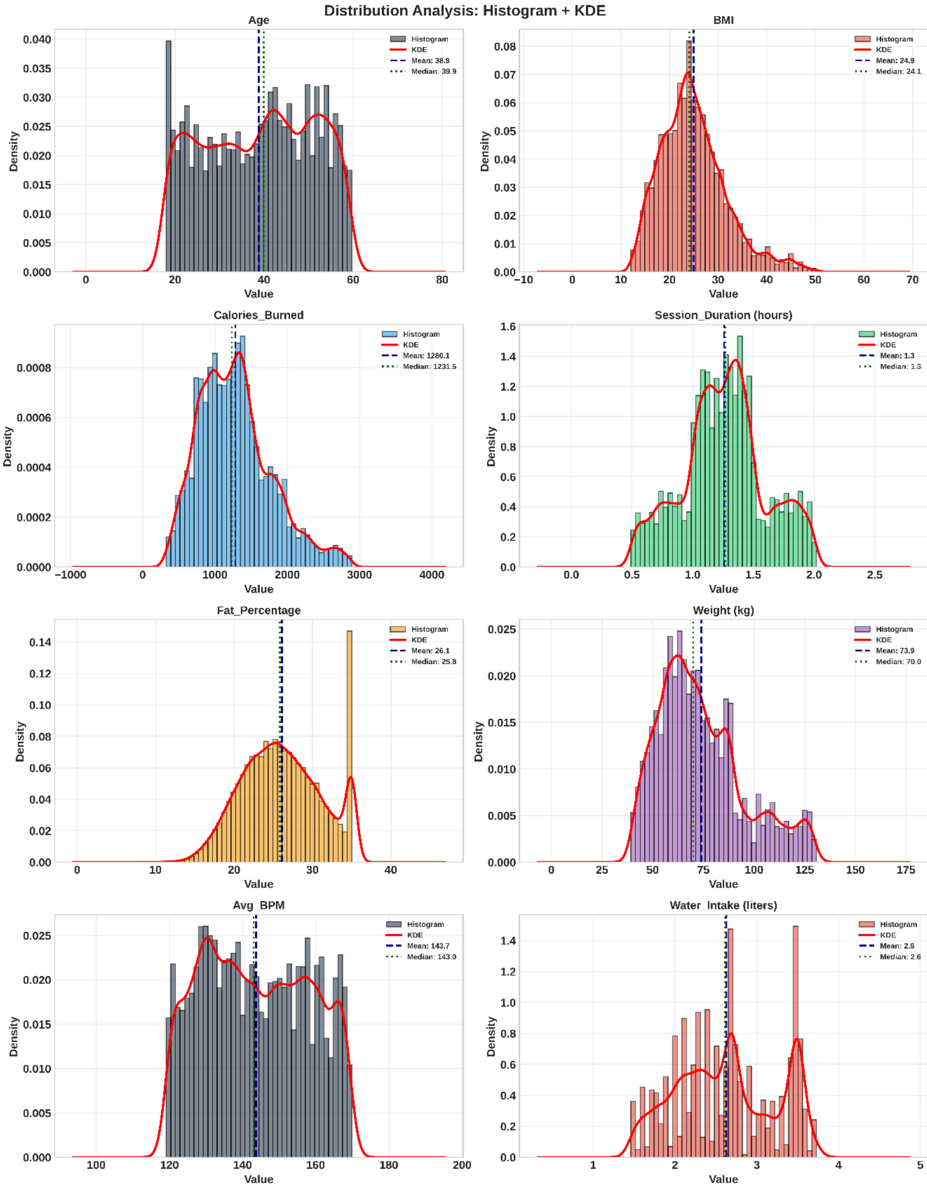

Figure 14: Histogram + KDE

Observation: Most quantitative variables (like Weight, Height, BMI) exhibit a near-normal distribution trend, centered around the mean. Although some variables show slight skewness, the data is generally continuous without abnormal interruptions, allowing us to retain them for further analysis without complex transformations.

The “Spike” at 35.0% (Figure 15):

The histogram reveals an unnatural accumulation of data points exactly at 35.0%. This pattern strongly suggests a “Ceiling Effect” (Right-Censoring), implying the measurement instrument was limited to 35%, or data collection protocols grouped all higher values into this cap.

Investigation Strategy:

To determine if this is a data error or a valid representation of high-risk individuals, we perform a statistical comparison:

- Group A (Capped): Individuals with Fat_Percentage $= 35.0$.
- Group B (Normal): The rest of the population.
- Metric: We compare their Mean BMI and Weight to see if Group A significantly differs from the general population.

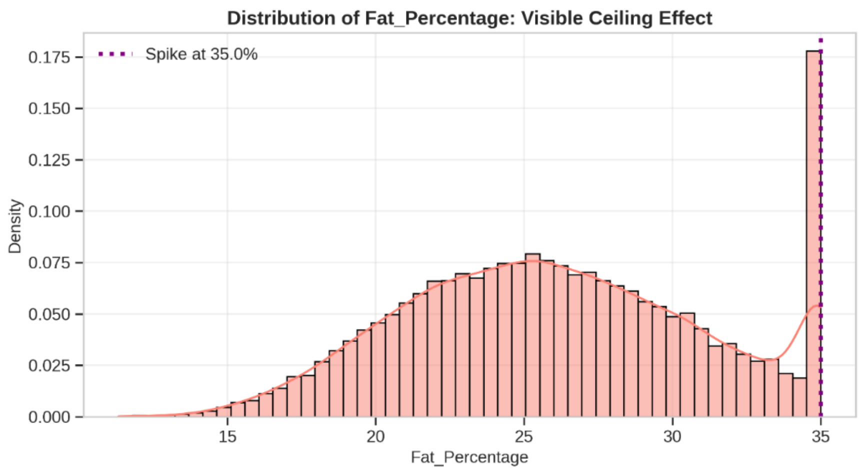

Figure 15: Distribution of Fat_Percentage: Visible Ceiling Effect

	Metric	Capped Group (Fat=35%)	Normal Group (Rest)	Difference
0	Mean BMI	39.69	23.72	15.97
1	Mean Weight (kg)	114.26	70.61	43.65
2	Mean Age	39.40	38.81	0.60
3	Workout Freq (days/week)	3.05	3.34	-0.29

Figure 16: Evidence Table: Comparing Heath Metrics

Conclusion:

- The ‘Capped Group’ has a Mean BMI of 39.69, which is classified as severe obesity.
- Compare this to the ‘Normal Group’ BMI of 23.72 (Normal/Overweight range).
- The Capped Group is on average 43.65 kg heavier than the rest.

Final decision: Retain data. The value “35.0” is a valid proxy for ≥ 35.0.

Removing these rows would bias the study by excluding the highest-risk population.

Discrete Variables Detected (Figure 17):

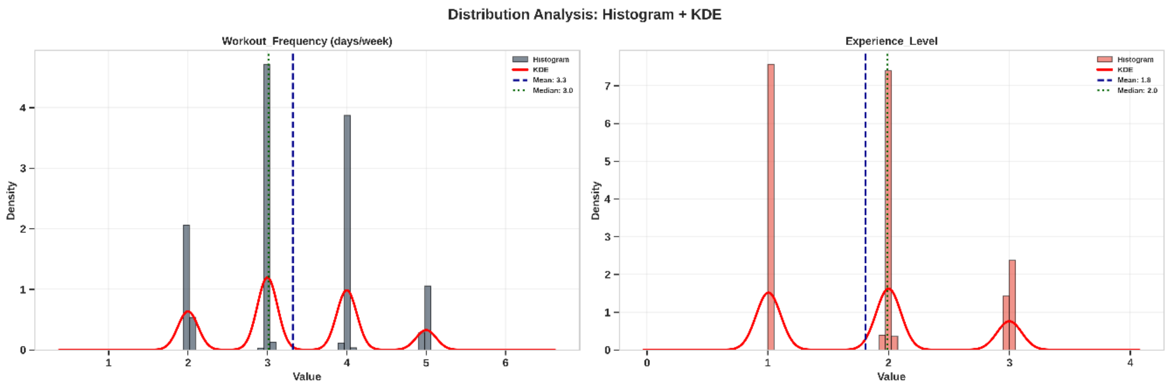

Figure 17: Histogram + KDE plot of Discrete Variables

From histograms, two variables show discrete patterns:

- Experience_Level: Multiple distinct peaks (not smooth)
- Workout_Frequency: Distinct vertical bars

Let's investigate if these are truly discrete/categorical.

```
exp_rounded = df['Experience_Level'].round()
```

Figure 18: Round Experience_Level

```
workout_rounded = df['Workout_Frequency (days/week)'].round()
```

Figure 19: Round Workout_Frequency

```
1. EXPERIENCE_LEVEL:
---------------------------------------------------------------
   Current dtype: float64
   Unique values: 32

   After rounding: 3 unique values
   Values: [np.float64(1.0), np.float64(2.0), np.float64(3.0)]

   Distribution:
      1:  7756 ( 38.8%)
      2:  8346 ( 41.7%)
      3:  3898 ( 19.5%)
```

Figure 20: Rounded Expertience_Level

```
2. WORKOUT_FREQUENCY:
--------------------------------------------------------------------------------
   Current dtype: float64
   Unique values: 55

   After rounding: 4 unique values
   Values: [np.float64(2.0), np.float64(3.0), np.float64(4.0), np.float64(5.0)]

   Distribution:
      2 days/week:  4050 ( 20.2%)
      3 days/week:  7605 ( 38.0%)
      4 days/week:  6266 ( 31.3%)
      5 days/week:  2079 ( 10.4%)
```

Figure 21: Rounded Workout_Frequency

Both variables are DISCRETE/ORDINAL:

- Experience_Level: 3 levels (Beginner = 1, Intermediate = 2, Advanced = 3)
- Workout_Frequency: 4 levels (2, 3, 4, 5 days/week)
- Noise (1.01, 2.98...) from measurement/generation artifacts

Action: Clean by rounding to nearest integer

Before vs After Rounding:

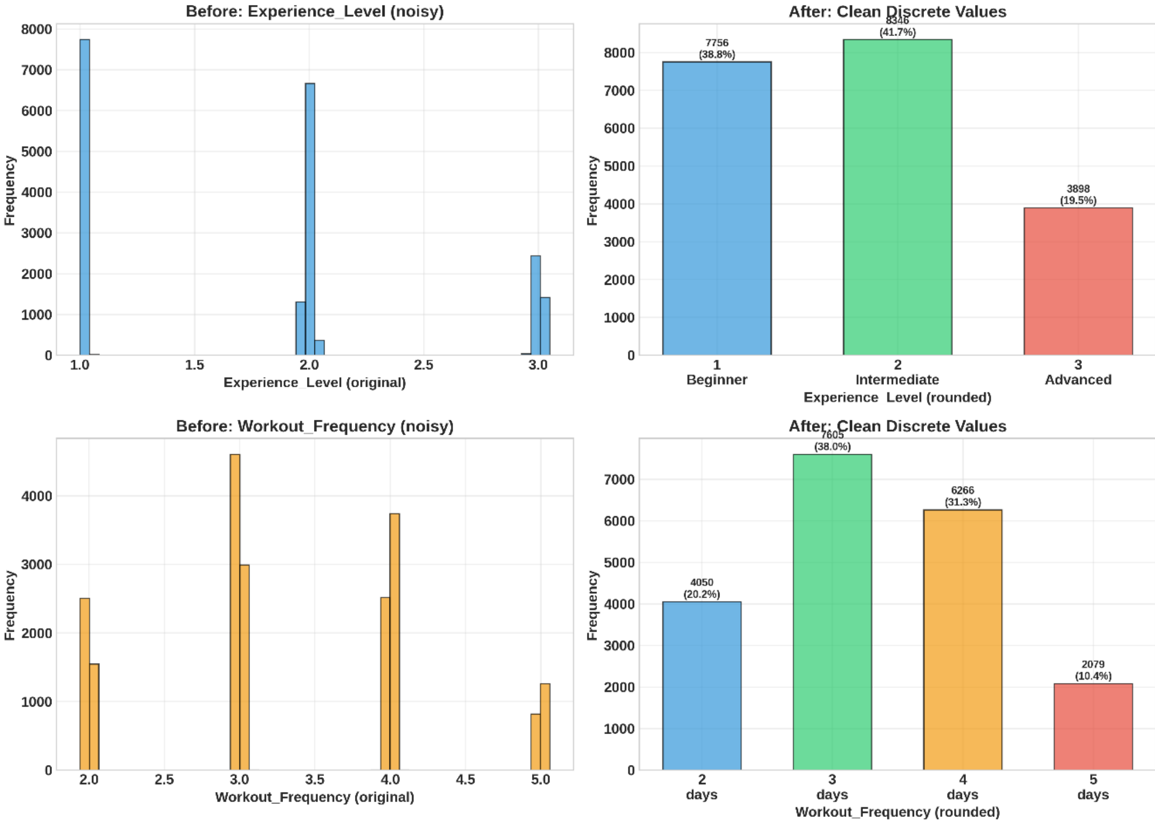

Figure 22: Before vs After Rounding

Apply Cleaning: Round to Integers

```
# Apply rounding
df['Experience_Level'] = df['Experience_Level'].round().astype(int)
df['Workout_Frequency (days/week)'] = df['Workout_Frequency (days/week)'].round().astype(int)
```

Figure 23: Apply Rounding to Integers

```
BEFORE:
  Experience_Level: float64, 32 unique
  Workout_Frequency: float64, 55 unique

AFTER:
  Experience_Level: int64, values = [np.int64(1), np.int64(2), np.int64(3)]
  Workout_Frequency: int64, values = [np.int64(2), np.int64(3), np.int64(4), np.int64(5)]
```

Figure 24: Rounding to Integers Result

Both variables are now properly represented as discrete integers.

Ready for analysis in subsequent tasks.

2.2.3.2 Chart 2: Box Plots (Figure 25)

Purpose: Used to summarize the dataset via 5 statistical figures (minimum, first quartile, median, third quartile, maximum) and, most importantly, to detect outliers.

Rationale for Chart Selection: Before feeding data into machine learning models, identifying abnormal data points (such as weight that is too high or too low compared to the norm) is mandatory because they can introduce noise and reduce the model's accuracy (R2 score). The Box plot is the standard tool for this task.

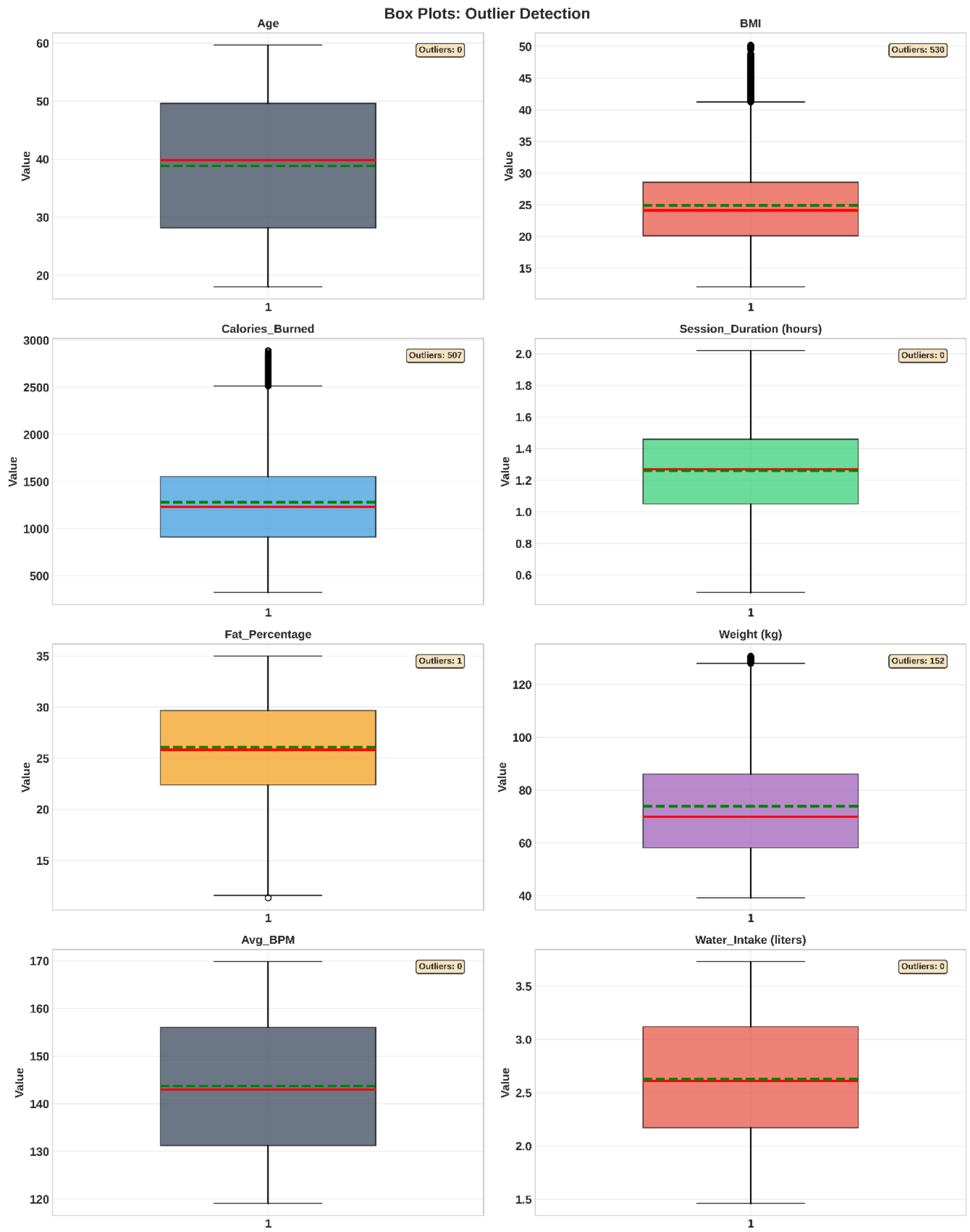

Figure 25: Box Plots

Observation: The chart indicates the presence of some outliers (points lying outside the "whiskers" of the box) in variables like Weight, BMI, or Calories_Burned. However, the number of outliers is not overwhelming, suggesting the data is relatively clean or these outliers are valid cases of athletes/high-intensity trainees rather than data entry errors.

2.2.3.3 Chart 3: Count Plots (Figure 26)

Purpose: To visualize the frequency of occurrence for categorical variables such as Gender, Workout_Type.

Rationale for Chart Selection: It is necessary to check if the data is imbalanced. If one workout type or gender dominates too heavily, the model might become biased toward that group. The Count plot helps quickly assess the balance of the dataset.

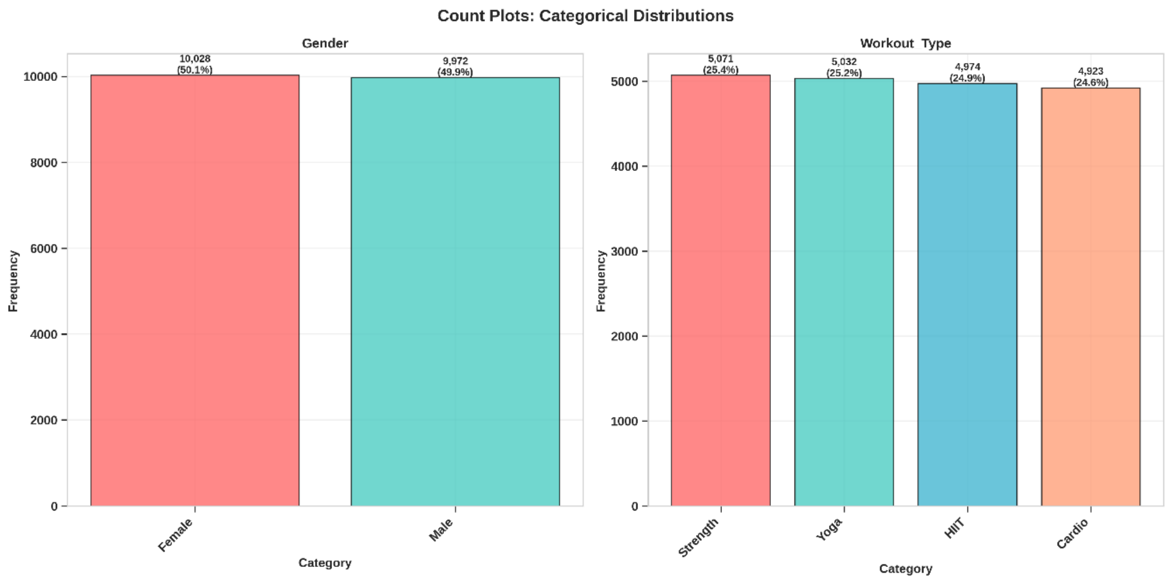

Figure 26: Count Plots

Observation: Categories within qualitative variables (like Gender, Workout_Type) show a relatively even distribution or acceptable variance. No group is too small to the point of insufficient learning data, ensuring the model can generalize well for different subjects.

2.2.3.4 Chart 4: Pair Plot (Figure 27)

Purpose: To provide a comprehensive overview of pairwise relationships between the most critical quantitative variables. The diagonal displays the distribution of each single variable (KDE), while the off-diagonal plots show scatter plots between any two variables.

Rationale for Chart Selection: Instead of plotting individual scatter plots one by one, the Pair Plot allows us to quickly identify patterns, data clusters, or linear/non-linear correlations in a multi-dimensional space. It is an efficient screening step to determine which variables have the best predictive potential for the target variable before building the model.

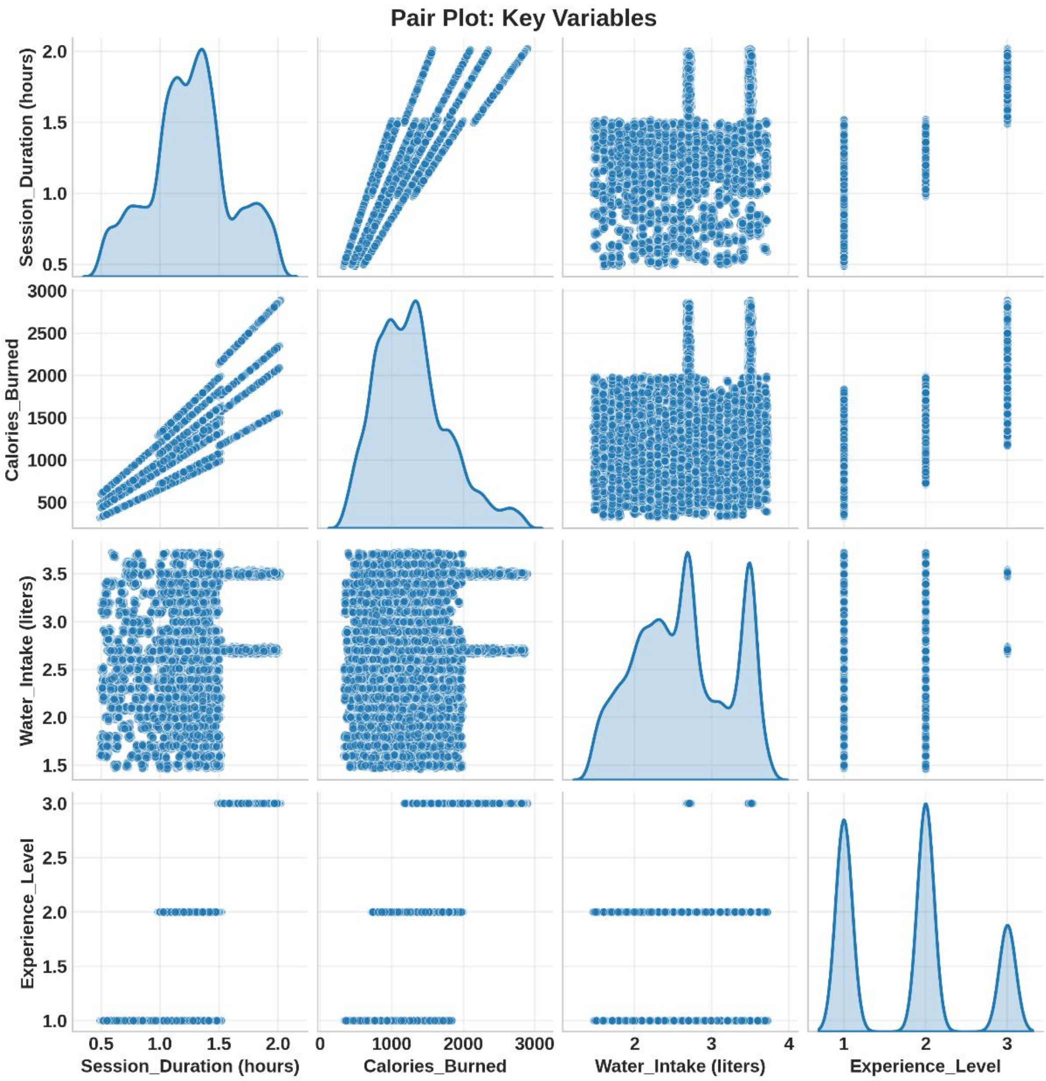

Figure 27: Pair Plot

Observation: The plot reinforces the strong linear relationship between Session_Duration and Calories_Burned. Additionally, it helps visualize the separation or overlap between data groups when combining multiple variables, visually supporting the identification of multicollinearity.

2.2.3.5 Chart 5: Grouped Bar Chart and Stacked Bar Chart (Figure 28)

Purpose: To analyze the relationship between two categorical variables simultaneously. Specifically, in our report, it examines the relationship between Gender and Workout_Type.

Rationale for Chart Selection: We need to answer the question: *"Does gender influence the choice of workout?"*. The Grouped Bar Chart allows for a direct comparison of the count of males and females within each workout category, while the Stacked Bar Chart helps examine the proportional composition (e.g., within Cardio practitioners, what percentage are female). This aids in understanding User Behavior.

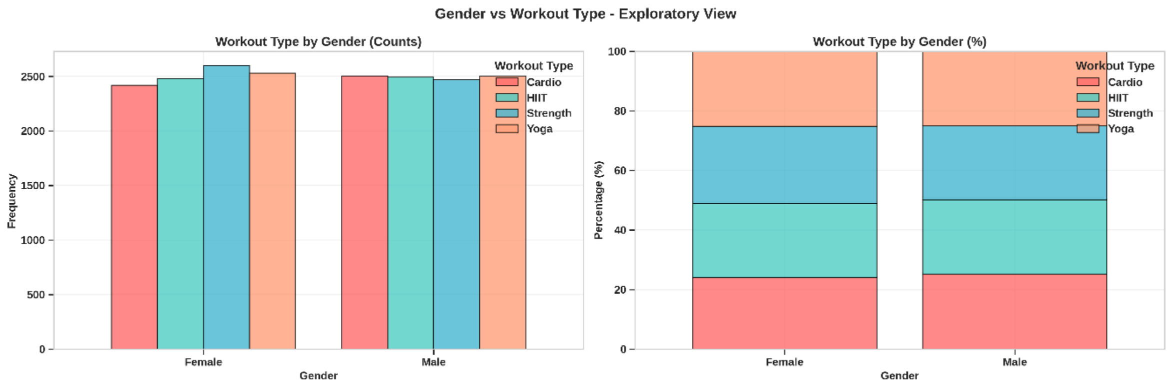

Figure 28: Grouped Bar Chart and Stacked Bar Chart

Observation: The chart reveals the distribution of workout preferences across genders. If the bars show similar heights for males and females across workout types, it demonstrates that there is no significant gender bias in the dataset (data is balanced regarding gender preferences), ensuring the predictive model remains fair for both genders.

2.3 Outlier Detection

2.3.1 Theoretical Overview

Outliers are data points that deviate significantly from the majority of observations in a dataset. They can arise from:

- Natural variability: Rare but legitimate extreme values (e.g., an elite athlete burning $3000+$ calories)
- Measurement errors: Faulty sensors, data entry mistakes, or recording errors
- Data processing errors: Issues during data collection, transfer, or transformation

Outliers can substantially impact statistical analyses and machine learning models because many techniques (like linear regression and mean/standard deviation calculations) are sensitive to extreme values. Therefore, systematic outlier detection and handling is a critical step in data preprocessing.

We employ three complementary methods to detect outliers, each with distinct theoretical foundations and strengths:

2.3.2 Implement Method

2.3.2.1 Method 1: Interquartile Range (IQR) Method

Theoretical Foundation: The IQR method is a robust, non-parametric approach based on quartiles (percentiles), making it resistant to the influence of outliers themselves. It does not assume any specific distribution shape (like normality).

Definition: The Interquartile Range is the spread of the middle 50% of the data:

$$IQR = Q3 - Q1$$

Where:

- Q1 (First Quartile): The 25th percentile - 25% of data falls below this value

- Q3 (Third Quartile): The 75th percentile - 75% of data falls below this value (or 25% above)

Outlier Detection Formula:

A data point X is flag as an outlier if: $\begin{cases} X < Q1 - 1.5 \times IQR \\ X > Q3 + 1.5 \times IQR \end{cases}$

Intuition: The boundaries $(Q1 - 1.5 \times IQR)$ and $(Q3 + 1.5 \times IQR)$ define "fences" beyond which observations are considered unusually extreme. Factor 1.5 is a conventional choice that balances sensitivity (detecting true outliers) with specificity (avoiding false alarms). This threshold corresponds roughly to 2.7 standard deviations in a normal distribution.

Advantages:

- Robust against extreme values (quartiles are not affected by outliers)
- Works with skewed distributions
- Intuitive interpretation (based on "boxes" in box plots)

Implementation:

```
for col in numerical_cols:
    Q1, Q3 = df[col].quantile([0.25, 0.75])
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]
    outliers_iqr[col] = len(outliers)
    outlier_data.append({
        'Variable': col,
        'Outlier Count': len(outliers),
        'Percentage': len(outliers)/len(df)*100,
        'Lower Bound': lower,
        'Upper Bound': upper
    })

outlier_df = pd.DataFrame(outlier_data)
```

Figure 29: IQR Method Implementation

IQR Method - Outlier Detection Results

	Variable	Outlier Count	Percentage	Lower Bound	Upper Bound
0	Age	0	0.00%	-4.02	81.82
1	Weight (kg)	152	0.76%	16.25	128.01
2	Height (m)	0	0.00%	1.35	2.07
3	BMI	530	2.65%	7.40	41.25
4	Fat_Percentage	1	0.01%	11.46	40.61
5	Max_BPM	0	0.00%	141.01	218.48
6	Avg_BPM	0	0.00%	93.96	193.32
7	Resting_BPM	0	0.00%	37.77	86.29
8	Session_Duration (hours)	0	0.00%	0.44	2.07
9	Calories_Burned	507	2.54%	-52.67	2516.58
10	Workout_Frequency (days/week)	0	0.00%	1.45	5.53
11	Water_Intake (liters)	0	0.00%	0.74	4.54
12	Experience_Level	0	0.00%	-0.51	3.54

Figure 30: IQR Method Result

2.3.2.2 Method 2: Z-Score Method (Standard Score)

Theoretical Foundation: The Z-score method is a parametric approach that assumes the data follows (or approximately follows) a normal distribution. It measures how many standard deviations a data point is away from the mean.

Formula:

$$Z = \frac{(X - \mu)}{\sigma}$$

Where:

- X: The data point value
- μ: The mean of the dataset
- σ: The standard deviation of the dataset

Outlier Detection Threshold:

A common rule is to flag observations as outliers if:

$$|Z| > 3$$

This means the value is more than 3 standard deviations away from the mean.

Intuition: In a perfect normal distribution:

- $\sim$68% of data falls within $\pm 1\sigma$ of the mean
- $\sim$95% falls within $\pm 2\sigma$
- $\sim$99.7% falls within $\pm 3\sigma$

Therefore, values with $|Z| > 3$ are extremely rare ($probability < 0.3\%$ in each tail), making them suspicious outliers.

Advantages:

- Simple and intuitive
- Well-established in statistical literature
- Effective when data is approximately normal

Limitations:

- Sensitive to the presence of outliers themselves (since mean and std are affected by extremes)
- Less reliable for highly skewed distributions
- Assumes normality

Implementation:

```
for col in numerical_cols:
    z_scores = np.abs(stats.zscore(df[col]))
    outliers = df[z_scores > 3]
    outliers_zscore[col] = len(outliers)
    mean_val = df[col].mean()
    std_val = df[col].std()
    zscore_data.append({
        'Variable': col,
        'Outlier Count': len(outliers),
        'Percentage': len(outliers)/len(df)*100,
        'Mean': mean_val,
        'Std Dev': std_val,
        'Lower Bound (μ-3σ)': mean_val - 3*std_val,
        'Upper Bound (μ+3σ)': mean_val + 3*std_val
    })

zscore_df = pd.DataFrame(zscore_data)
```

Figure 31: Z-Score Method Implementation

Z-Score Method - Outlier Detection Results

	Variable	Outlier Count	Percentage	Mean	Std Dev	Lower Bound (μ-3σ)	Upper Bound (μ+3σ)
0	Age	0	0.00%	38.85	12.11	2.51	75.20
1	Weight (kg)	0	0.00%	73.90	21.17	10.38	137.42
2	Height (m)	0	0.00%	1.72	0.13	1.34	2.10
3	BMI	227	1.14%	24.92	6.70	4.82	45.03
4	Fat_Percentage	0	0.00%	26.10	5.00	11.11	41.09
5	Max_BPM	0	0.00%	179.89	11.51	145.36	214.42
6	Avg_BPM	0	0.00%	143.70	14.27	100.90	186.51
7	Resting_BPM	0	0.00%	62.20	7.29	40.33	84.06
8	Session_Duration (hours)	0	0.00%	1.26	0.34	0.24	2.28
9	Calories_Burned	111	0.56%	1280.11	502.23	-226.58	2786.80
10	Workout_Frequency (days/week)	0	0.00%	3.32	0.91	0.59	6.05
11	Water_Intake (liters)	0	0.00%	2.63	0.60	0.81	4.44
12	Experience_Level	0	0.00%	1.81	0.74	-0.40	4.02

Figure 32: Z-Score Method Result

2.3.2.3 Method 3: Isolation Forest

Theoretical Foundation: Isolation Forest is a modern, unsupervised machine learning algorithm specifically design for anomaly (outlier) detection. Unlike IQR and Z-score, which rely on statistical assumptions, Isolation Forest works by exploiting the principle that outliers are rare and different - they are easier to "isolate" from the rest of the data.

Algorithm Concept:

1. Random Partitioning: The algorithm builds an ensemble of "Isolation Trees" by randomly selecting a feature and a split value, recursively partitioning the data.
2. Path Length: For each data point, it measures the path length (number of splits) require to isolate that point.
3. Anomaly Score:
   - Outliers require fewer splits to isolate (short path lengths) because they're far from the dense regions
   - Normal points require many splits to isolate (long path lengths) because they're surrounded by similar neighbors

4. Scoring: The algorithm assigns an anomaly score between -1 and 1:
   - Scores close to 1 indicate outliers
   - Scores around 0 indicate normal points
   - Scores close to -1 indicate very normal (inlier) points

Advantages:

- Distribution-free: No assumptions about data distribution.
- Multivariate: Can consider relationships between multiple features simultaneously
- Scalable: Efficient for large datasets with many dimensions
- Robust: Not heavily influenced by outliers in the training data itself

Limitations:

- Requires tuning contamination parameter (expected proportion of outliers)
- Less interpretable than statistical methods (why is this an outlier?)
- May require more computational resources

Implementation:

```
iso_forest = IsolationForest(contamination=0.05, random_state=42)
outliers_if = iso_forest.fit_predict(df[numerical_cols])
outlier_count = (outliers_if == -1).sum()
```

Figure 33: Isolate Forest Implementation

```
========================================
  OUTLIER DETECTION: ISOLATION FOREST
========================================
Total outliers detected: 1,000 (5.00%)

Isolation Forest completed
```

Figure 34: Isolate Forest Result

2.3.3 Outlier Comparison Method (Figure 35)

Visual Observation: The bar chart illustrates a distinct discrepancy in the volume of outliers detected by the two statistical methods applied:

- The Interquartile Range (IQR) method consistently exhibits higher sensitivity, identifying a significantly larger number of outliers across most variables (represented by the orange bars).
- The Z-Score method demonstrates a more conservative detection rate (represented by the blue bars), often flagging fewer or no data points for the same variables.

Statistical Interpretation: This divergence in detection counts provides critical insight into the underlying distribution of the dataset:

- **Assumption of Normality (Z-Score Limitation):** The Z-Score method relies on the assumption that the data follows a Gaussian (Normal) distribution. It defines outliers as data points deviating more than 3 standard deviations (σ) from the mean (μ). However, in our dataset (specifically variables like Weight, BMI, and Fat_Percentage), the distributions are likely skewed (right-skewed) or heavy-tailed. In skewed distributions, the standard deviation is inflated by extreme values, effectively "stretching" the acceptance range and masking potential outliers.
- **Robustness of IQR:** Conversely, the IQR method is non-parametric and relies on quartiles ($Q1$ and $Q3$). It is robust against skewness and does not depend on the mean or standard deviation. Therefore, it effectively captures the "long tails" of the data distribution, flagging extreme values that the Z-Score method considers "within range" due to the inflated variance.

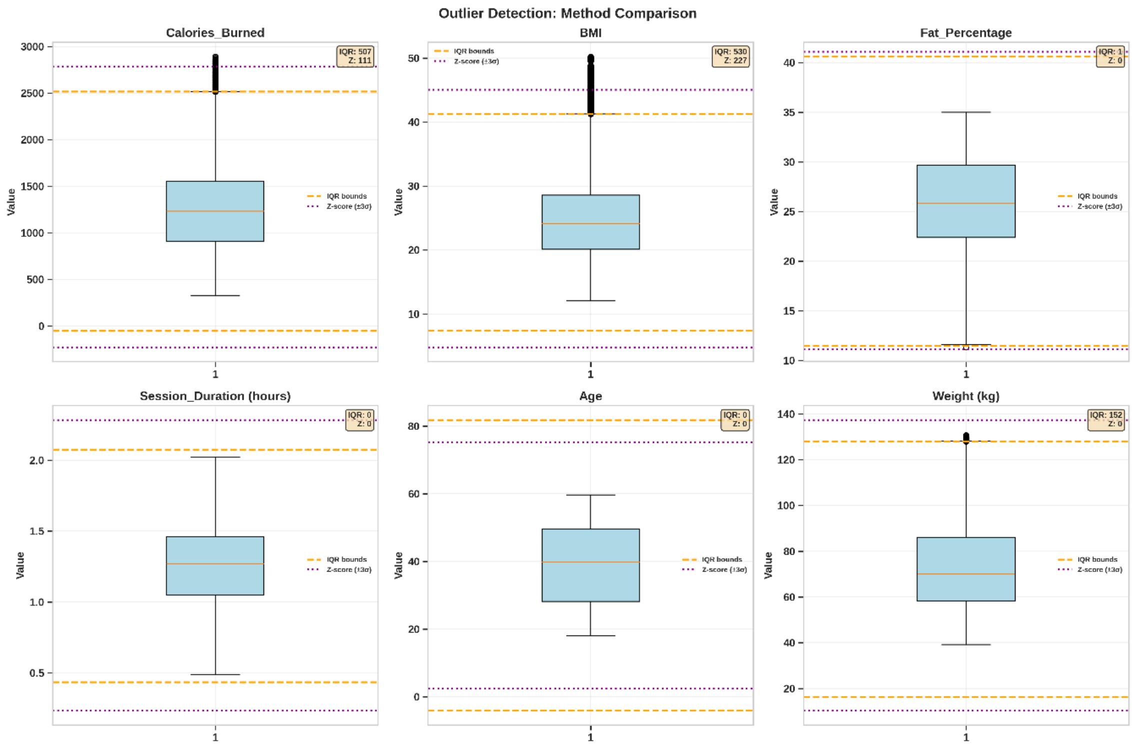

Figure 35: Outlier Comparison Method

2.3.4 Outlier Handling Strategy

In this section, we define a rigorous approach to managing outliers. Instead of relying on a single metric, we adopt a Multi-Method Consensus Strategy to ensure statistical robustness.

2.3.4.1 Detection Strategy: Multi-Method Voting

Rationale: Relying on a single detection method can be misleading due to inherent statistical limitations. We combine three distinct algorithms to balance their strengths:

Method	**Strength**	**Limitation**
IQR (Interquartile Range)	Robust for skewed distributions	Can be overly aggressive on normal tails
Z-Score	Effective for extreme values	Assumes normality; sensitive to mean/std shifts

Isolation Forest	Distribution - independent (Machine Learning)	May flag complex but valid patterns

The Consensus Requirement: To ensure data integrity and minimize false positives, a data point is flagged as a Confirmed Outlier only if it satisfies the following condition:

$$Confirmed\ Outlier\ =\ Flagged\ by\ \geq\ 2\ out\ of\ 3\ methods$$

This approach balances sensitivity and specificity, significantly reducing the risk of incorrectly flagging valid data (False Positives) or missing true anomalies (False Negatives).

```python
def get_outlier_mask_voting(series, contamination=0.01):
    valid_data = series.dropna()
    if len(valid_data) == 0:
        return pd.Series(False, index=series.index)

    # --- Method 1: IQR ---
    Q1 = valid_data.quantile(0.25)
    Q3 = valid_data.quantile(0.75)
    IQR = Q3 - Q1
    mask_iqr = (valid_data < (Q1 - 1.5 * IQR)) | (valid_data > (Q3 + 1.5 * IQR))

    # --- Method 2: Z-Score ---
    mask_z = np.abs(stats.zscore(valid_data)) > 3

    # --- Method 3: Isolation Forest ---
    iso = IsolationForest(contamination=contamination, random_state=42)
    iso_preds = iso.fit_predict(valid_data.values.reshape(-1, 1))
    mask_iso = (iso_preds == -1)

    # --- VOTING: Count votes ---
    votes = mask_iqr.astype(int) + mask_z.astype(int) + mask_iso.astype(int)
    is_outlier = votes >= 2

    final_mask = pd.Series(False, index=series.index)
    final_mask.loc[valid_data.index] = is_outlier
    return final_mask
```

Figure 36: Implement Multi-Method Voting

```
================================================================================
            OUTLIER DETECTION SUMMARY (VOTING STRATEGY >= 2 METHODS)
================================================================================
Weight (kg)                        :   83 outliers detected (0.41%)
BMI                                :  227 outliers detected (1.14%)
Fat_Percentage                     :    1 outliers detected (0.01%)
Calories_Burned                    :  145 outliers detected (0.73%)
```

Figure 37: Multi-Method Voting result

2.3.4.2 Handling Strategy: Targeted Capping

Principle: Rather than removing outliers (which reduces sample size and statistical power), we apply Targeted Capping. This technique preserves the observation but limits its extreme influence on the model.

Implementation Algorithm: We apply the capping logic ONLY to the rows identified as "Confirmed Outliers" by our voting mechanism:

- Lower Bound: If value < 1st Percentile (P1) → Replace with P1
- Upper Bound: If value > 99th Percentile (P99) → Replace with P99
- Normal Variance: Leave strictly untouched.

This ensures that we only modify the most extreme outliers while maintaining the natural variance of the dataset.

```
for col, mask in outlier_masks.items():
    if mask.sum() > 0:
        # Calculate bounds based on the whole distribution
        lower_bound = df[col].quantile(0.01)
        upper_bound = df[col].quantile(0.99)

        # Identify values to clip
        outliers = df_clean.loc[mask, col]

        # Clip ONLY the identified outliers
        df_clean.loc[mask, col] = outliers.clip(lower=lower_bound,
                                                upper=upper_bound)
```

Figure 38: Implement Targeted Capping

```
Handling Outliers: Applying Capping (1st - 99th Percentile)...
------------------------------------------------------------
Processed Weight (kg): Capped 83 values to range [41.15, 127.67]
Processed BMI: Capped 227 values to range [13.33, 45.20]
Processed Fat_Percentage: Capped 1 values to range [15.74, 35.00]
Processed Calories_Burned: Capped 145 values to range [408.83, 2704.80]
```

Figure 39: Targeted Capping Result

2.3.5 Post-Cleaning Verification & Impact Assessment

Following the outlier handling process, it is crucial to verify the impact of our data cleaning strategy on the dataset's integrity and statistical properties. This section visualizes the "Before" vs "After" states to ensure that:

- **Outliers** have been effectively managed (reduced or capped) without introducing artificial artifacts.

- **Correlations** between variables have been preserved or strengthened (reducing noise).
- **Distributions** have maintained their natural shape while reducing extreme skewness.

Visualization Structure:

- **Box Plots:** To visually confirm the reduction of extreme outliers.
- **Correlation Matrix:** To analyze how relationships between variables have shifted (e.g., checks for strengthened linear relationships).
- **Histograms + KDE:** To observe changes in the distribution tails and skewness.

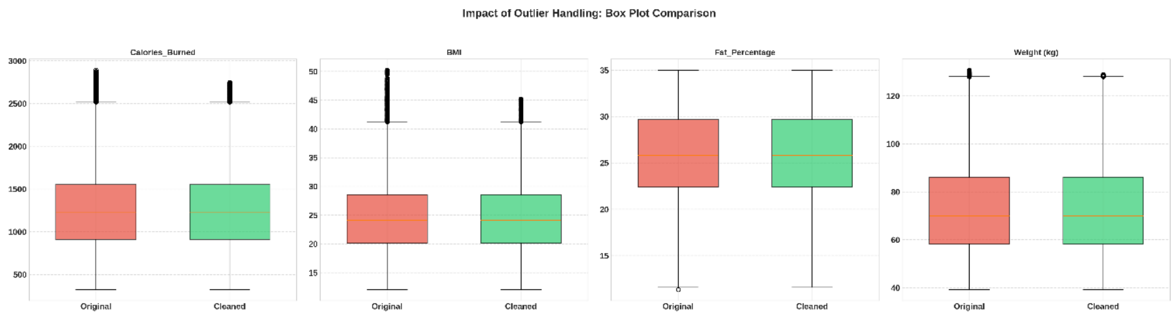

Figure 40: Impact of Outlier Handling: Box Plot Comparison

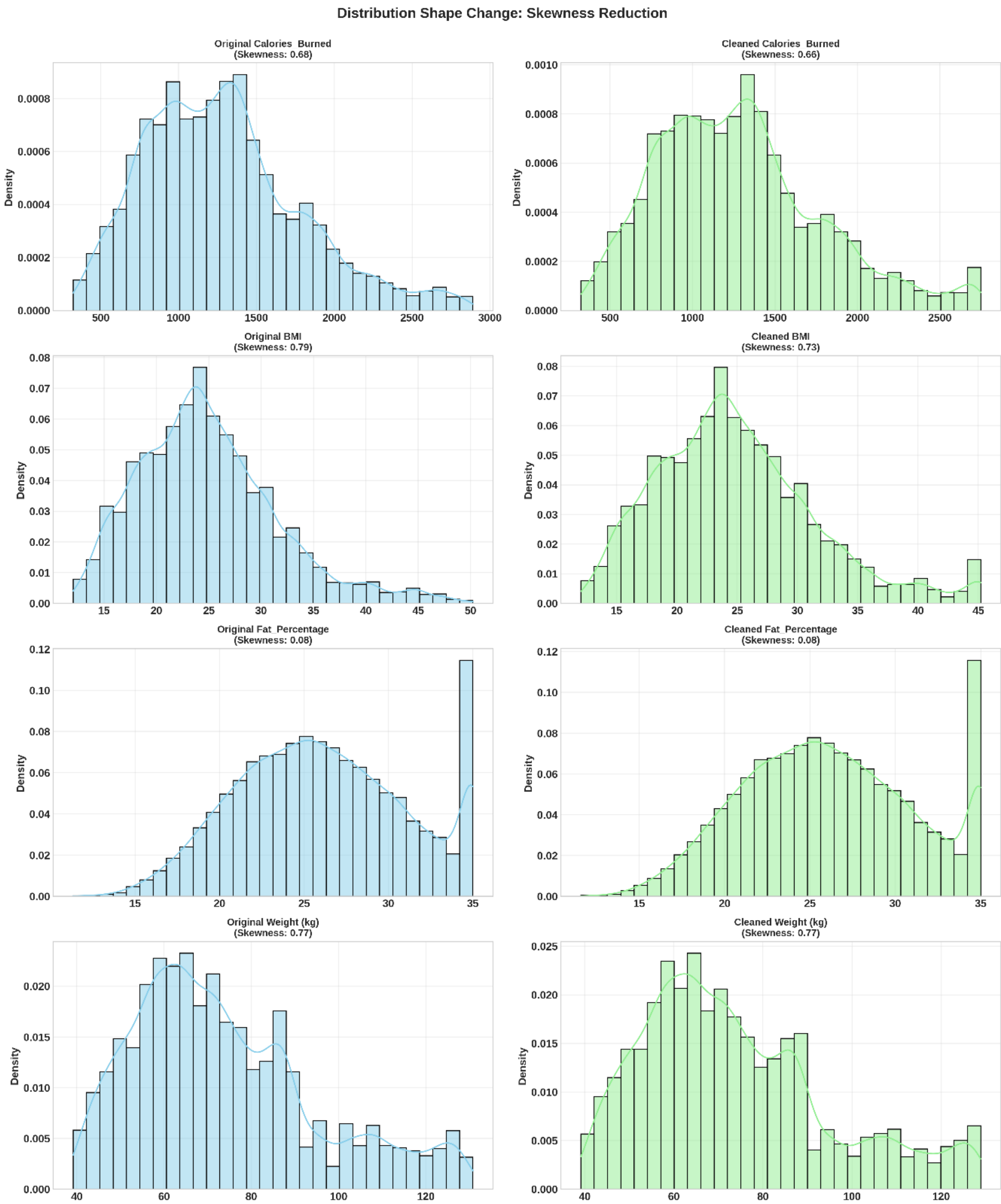

Figure 41: Distribution Shape Change: Skewness Reduction

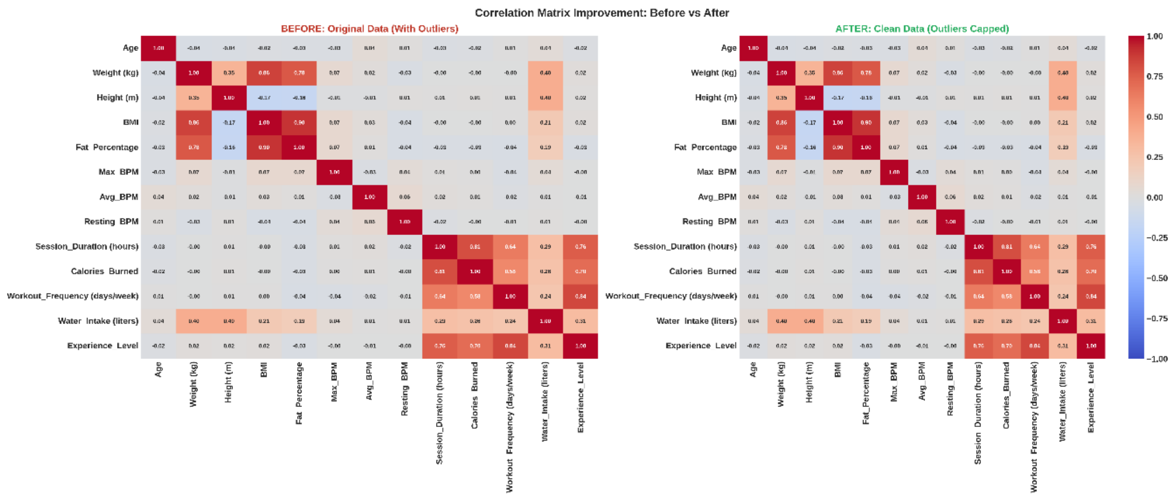

Figure 42: Correlation Matrix Improvement: Before vs After

2.3.6 Critical Re-evaluation: Outlier Retention Decision

Analysis Process: After implementing the Multi-Method Voting Strategy and visualizing distributions through histograms and KDE plots, a comprehensive review of the data's nature was conducted.

Key Observations:

1. Continuity of Distribution
   - Histograms reveal smooth, right-skewed distributions.
   - The pattern is characteristic of biological/health data (BMI, Calories).
   - No distinct gaps separating extreme values from the main distribution.
   - Absence of isolated, impossible values.
2. Biological Plausibility

Variable	Extreme Values	Interpretation
Calories Burned	$> 2500\ kcal$	High-performance athletes, intense training
BMI	> 35	Obesity class II-III (clinically valid)

Session Duration	$> 2\ hours$	Endurance athletes, long workouts
Fat Percentage	$> 35\%$	Higher body fat (within human range)

Conclusion: Extreme values are biologically plausible and likely represent specific subgroups (elite athletes, obese individuals) rather than measurement errors.

2.3.7 Final Decision: Retain All Data Points

We have decided to NOT apply for capping or removal of outliers. The dataset is considered "clean" in its original form.

Justification:

1. Preserving Natural Variance
   - Capping would artificially reduce variance and bias analysis toward the "average" population.
   - We avoid losing information about extreme but valid cases, which are critical for medical/health research insights.
2. Scientific Integrity
   - Extreme values represent real phenomena.
   - Subgroups (athletes, obesity) are important populations; removing them would misrepresent population diversity.
3. Methodological Approach for Analysis We will adapt our statistical methods to handle this skewness rather than altering the data:

Task	Strategy
General Approach	Accept skewness as a natural feature
Distribution Testing	Use robust, non-parametric tests
Regression Modeling	Consider log-transformation if needed
Correlation Analysis	Use Spearman rank correlation (non-parametric)

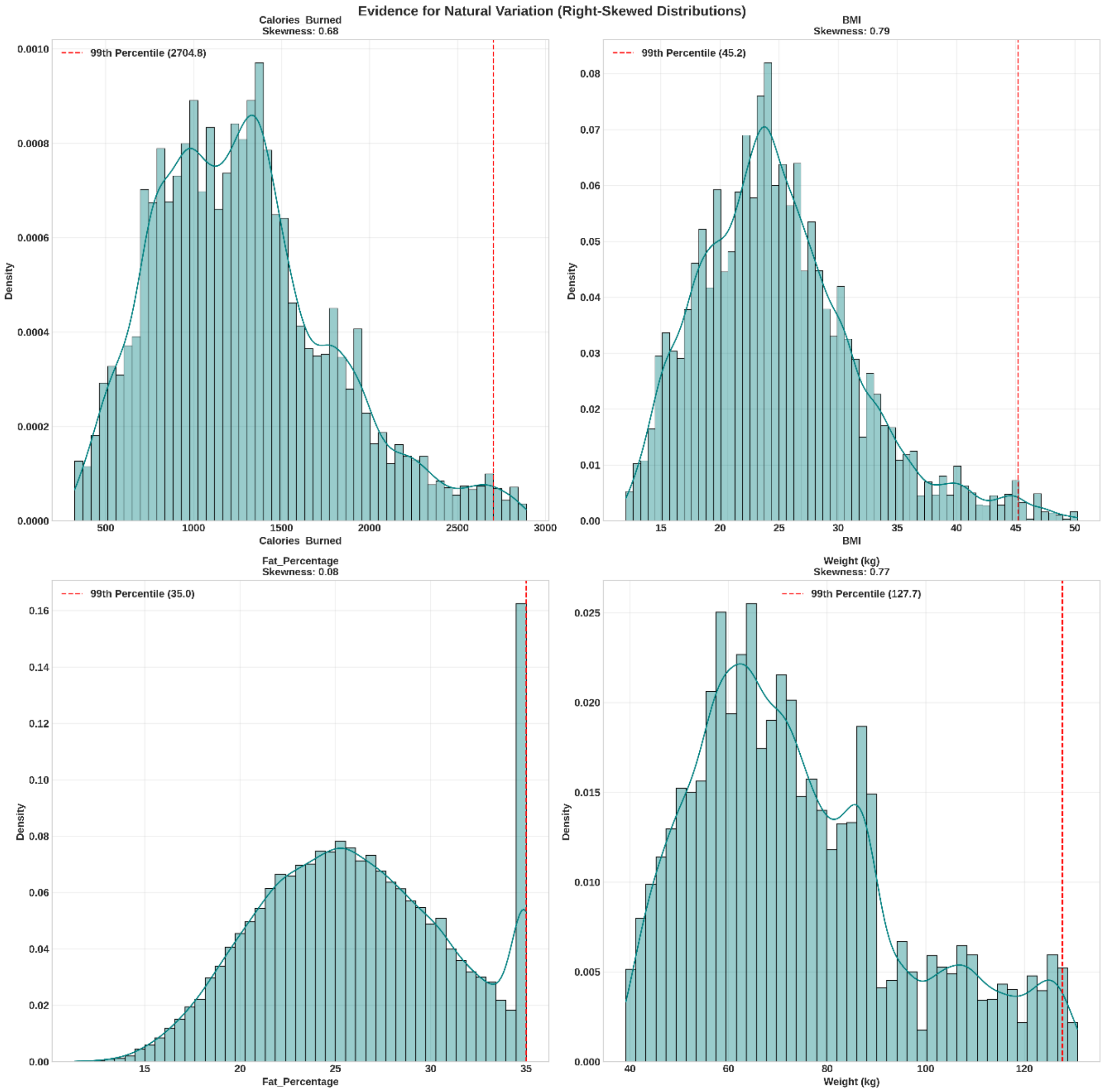

Figure 43: Evidence for Natural Variation

2.4 General Observations

Data Quality and Integrity:

The processed dataset exhibits a high level of integrity. There are no remaining missing values or duplicate records that could introduce noise. Quantitative variables such as Age, Weight, and Height all fall within reasonable human biological limits, ensuring clean input data for the modeling process.

Key Relationship (Key Driver):

Preliminary observations from the correlation matrix indicate that Session_Duration has the strongest positive linear correlation with the target variable Calories_Burned. This suggests that workout duration will be the most significant predictor variable in the upcoming linear regression model.

Multicollinearity Issue:

There is a very high internal correlation among three variables: Weight, Height, and BMI. Specifically, Weight and BMI show a near-perfect correlation coefficient. This is a critical feature to note, as it may lead to multicollinearity, potentially distorting the estimation of regression coefficients if all three variables are included in the model simultaneously.

Balance of Categorical Variables:

Qualitative variables such as Gender and Workout_Type show a relatively even distribution. There are no extremely small minority groups, which helps the model avoid bias and ensures good generalization capabilities across genders and different types of physical activities.

Outlier Phenomenon:

Although some outliers were detected in variables like BMI and Weight (via the IQR method), inspection of the distribution plots reveals that these data points lie on a continuous density curve. They represent valid real-world cases (such as overweight individuals or bodybuilders) rather than data errors; therefore, retaining them is essential to accurately reflect the diverse reality of users.

CHAPTER 3. PROBABILITY DISTRIBUTION ANALYSIS

Categorical Distribution Analysis:

Goal: Understand the distribution of Workout Type choices across Gender groups.

Why this matters:

- Determines if relationship exists between Gender and Workout Type
- Guides selection of appropriate statistical test for Chapter 4

- Identifies patterns in exercise preferences

3.1 Univariate Distributions

Examine each variable separately first.

Based on Figure 45, we can see:

- Gender balance: 49.9% - 50.1%
- Most common workout: Strength (25.4%)
- Least common workout: Cardio (24.6%)

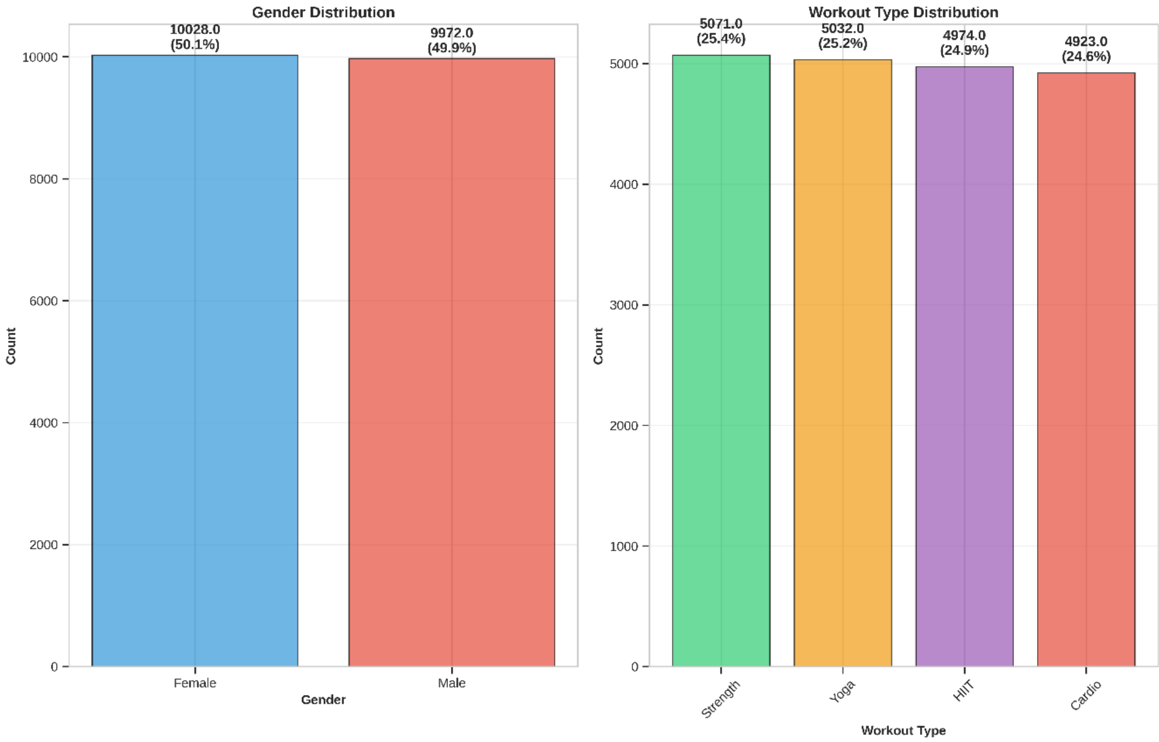

Figure 44: Distribution of Gender and Workout Type

3.2 Statistical Test for Uniform Distribution (Goodness of Fit)

Visual inspection of the bar charts for Gender and Workout_Type suggests a relatively balanced distribution. Therefore, we proceed with a Chi-square Goodness of Fit test to statistically confirm if these variables follow a Uniform Distribution.

Objective: Mathematically determine if the variables Gender and Workout_Type follow a Uniform Distribution.

Method: Chi-square Goodness of Fit Test.

- Null Hypothesis (H_0): The data is uniformly distributed (all categories have equal frequencies).
- Alternative Hypothesis (H_1): The data is NOT uniformly distributed (biased sampling).
- Significance Level (α): 0.05. If $P - value > 0.05$, we fail to reject H_0 (meaning the data is balanced).

```
# 1. Get Observed Counts
observed_counts = df[column_name].value_counts()

# 2. Perform Test (Default expected frequencies = Uniform)
chi2_stat, p_value = chisquare(observed_counts)
```

Figure 45: Implement Test for Uniform Distribution

```
--- Uniformity Test: Gender ---
Observed Counts:
Gender
Female    10028
Male       9972
Name: count, dtype: int64
Chi-square Statistic: 0.1568
P-value: 0.6921
--- Uniformity Test: Workout_Type ---
Observed Counts:
Workout_Type
Strength    5071
Yoga        5032
HIIT        4974
Cardio      4923
Name: count, dtype: int64
Chi-square Statistic: 2.5340
P-value: 0.4692
```

Figure 46: Result of Uniformity Test

We have:

- Gender: $P - value = 0.6921 > 0.05 \Rightarrow$ Fail to reject H0. The variable Gender follows a uniform distribution.
- Workout_Type: $P - value = 0.4692 > 0.05 \Rightarrow$ Fail to reject H0. The variable Workout_Type follows a uniform distribution.

Through frequency analysis and variable visualization, we determined that both the Gender and Workout Type variables are Categorical Variables. In addition, these variables also follow uniform distribution, suggesting that no single category holds an overwhelming dominance.

Based on the measurement scale of the data - two categorical variables - and the objective of testing for independence or association between them, the Chi-square Test of Independence (χ^2 Test) is the most theoretically sound statistical method for application in Chapter 4: Hypothesis Testing.

As the χ^2 test relies on the Contingency Table and adherence to expected frequency conditions, we decided to immediately proceed with the Assumption Check on Expected Frequencies here. This step is crucial to verify that at least 80% of the cells in the table have an Expected Frequency of ≥ 5, thereby ensuring the reliability and validity of the χ^2 test results that will be formally presented in Task 3.

3.3 Contingency Table (Cross-tabulation)

Contingency Table: Shows the frequency distribution of two categorical variables.

This is the foundation for:

- Understanding the relationship between Gender and Workout Type
- Effect size calculations

Workout_Type	Cardio	HIIT	Strength	Yoga	Total
Gender					
Female	2,418	2,479	2,601	2,530	10,028
Male	2,505	2,495	2,470	2,502	9,972
Total	4,923	4,974	5,071	5,032	20,000

Figure 47: Observed Frequencies (Counts)

Workout_Type	Cardio	HIIT	Strength	Yoga
Gender				
Female	24.11%	24.72%	25.94%	25.23%
Male	25.12%	25.02%	24.77%	25.09%

Figure 48: Row Percentages (Within each Gender)

Workout_Type	Cardio	HIIT	Strength	Yoga
Gender				
Female	49.12%	49.84%	51.29%	50.28%
Male	50.88%	50.16%	48.71%	49.72%

Figure 49: Column Percentages (Within each Workout Type)

Workout_Type	Cardio	HIIT	Strength	Yoga
Gender				
Female	12.09%	12.40%	13.00%	12.65%
Male	12.52%	12.48%	12.35%	12.51%

Figure 50: Overall Percentages (Of total sample)

3.4 Visual Analysis - Multiple Perspectives

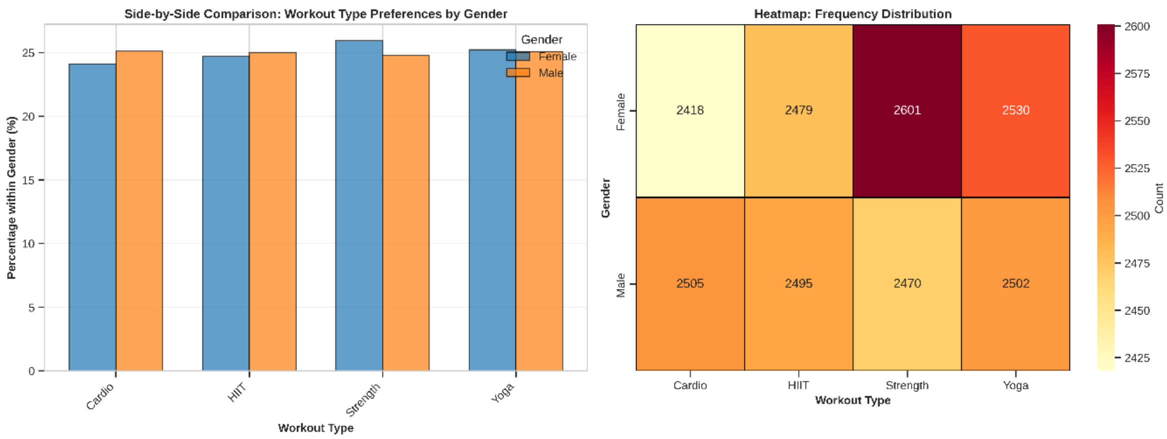

Figure 51: Heatmap and Side-by-Side plot for Gender and Workout Type

3.5 Expected Frequencies (Chi-square Assumption Check):

Chi-square test assumption: Expected frequency in each cell should be ≥ 5

Expected frequency formula:

$$E_{ij} = \frac{(\text{Row}_i \text{ total}) \times (\text{Column}_j \text{ total})}{\text{Grand total}}$$

This tells us if chi-square test is appropriate for Chapter 4.

Workout_Type Gender	Cardio	HIIT	Strength	Yoga
Female	2,468.4	2,494.0	2,542.6	2,523.0
Male	2,454.6	2,480.0	2,528.4	2,509.0

Figure 52: Expected Frequencies (If Independent)

Workout_Type Gender	Cardio	HIIT	Strength	Yoga
Female	-50.4	-15.0	58.4	7.0
Male	50.4	15.0	-58.4	-7.0

Figure 53: Difference (Observed - Expected)

Chi-square Test assumption check:

- Minimum expected frequency: 2454.61
- Cells with expected frequency < 5: 0 / 8

Chi-square test assumption satisfied:

- All expected frequencies ≥ 5
- Chi-square test is appropriate for Chapter 4

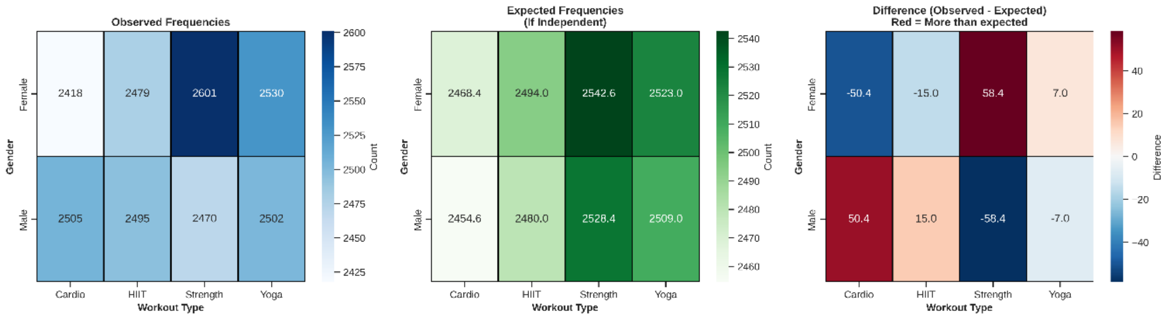

Figure 54: Observed-Expected Frequencies and Difference

Interpretation of Differences:

- Positive (Red): More observations than expected if independent
- Negative (Blue): Fewer observations than expected if independent
- Near Zero: Close to what we'd expect if independent

3.6 Distribution Summary and Preliminary Insights

Data characteristic:

- Total observations: 20000
- Gender categories: 2 (Male, Female)
- Workout Type categories: 4 (Strength, HIIT, Cardio, Yoga)

Preliminary Observations:

Largest deviation from independence:

- Female and Strength: +58.4

→ Female does Strength more than expected

Gender-specific preferences:

- Female: Prefers Strength (25.9%)
- Male: Prefers Cardio (25.1%)

3.7 Conclusion

Chi-square Test assumption:

- Independence: Each observation is independent.
- Expected frequencies: All cells have expected count ≥ 5.

- Sample size: Sufficient for chi-square test.

Preliminary Evidence:

- Distribution patterns suggest possible association between Gender and Workout Type.
- Observed frequencies differ from expected (independent) frequencies.
- Statistical test needed to determine if differences are significant.

Distribution: Gender and Workout Type follow uniform distributions.

CHAPTER 4. HYPOTHESIS TESTING

4.1 Theoretical Overview

4.1.1 Statistical Hypotheses:

Null Hypothesis (H_0):

- Gender and Workout Type are independent

Alternative Hypothesis (H_1):

- Gender and Workout Type are dependent (associated)

Significance Level: $\alpha = 0.05$ (5%)

Decision Rule:

- If $P - value < 0.05$: Reject H_0 (significant association exists)
- If $P - value \geq 0.05$: Fail to reject H_0 (no significant association)

4.1.2 Chi-square:

Chi-square Test:

The Chi-Square Test (χ^2) is a non-parametric statistical hypothesis test used to analyze qualitative data.

The primary goal of this test is to compare the Observed Frequencies (O) from the actual data with the Expected Frequencies (E) calculated based on a specific hypothesis.

Main Types of Chi-Square Tests:

There are two main types of Chi-Square tests:

1. Chi-Square Test of Independence:

   Purpose: To determine whether there is a statistically significant association (relationship) between two categorical variables.

   Hypotheses:

   - H_0 (Null Hypothesis): The two variables are independent (no relationship).
   - H_a (Alternative Hypothesis): The two variables are not independent (there is a relationship).

2. Chi-Square Goodness-of-Fit Test:

   Purpose: To determine whether the observed frequency distribution of a single categorical variable matches a hypothesized theoretical distribution or known expected distribution.

   Hypotheses:

   - H_0: The observed distribution fits the expected distribution.
   - H_a: The observed distribution does not fit the expected distribution.

The χ^2 Calculation Formula:

The Chi-Square test statistic (χ^2) is calculated using the formula:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}}$$

Where:

- O_{ij}: Observed frequency in cell (i, j)
- E_{ij}: Expected frequency in cell (i, j)
- r: number of rows
- c: number of columns

The Expected Frequency (E) for the test of independence is calculated as:

$$E_{ij} = \frac{(\text{Row}_i \text{ total}) \times \left(\text{Column}_j \text{ total}\right)}{\text{Grand total}}$$

Degrees of Freedom:

$$df = (r - 1) \times (c - 1)$$

- r: number of rows
- c: number of columns

Interpretation:

- Large $\chi^2 \rightarrow$ Large deviation from independence → Reject H_0
- Small $\chi^2 \rightarrow$ Small deviation from independence → Fail to reject H_0

Important Assumptions:

The Chi-Square test is only valid when the following conditions are met:

1. Categorical Data: The data must be frequency counts of categorical variables.
2. Independent Observations: Each observation in the sample must be independent of the others.
3. Large Sample Size (Expected Frequencies):
   - The Expected Frequency (E) for all cells must be greater than or equal to 1.
   - At least 80% of the cells must have an Expected Frequency greater than or equal to 5.

4.1.3 Critical Value:

Definition:

- A critical value is a point on the test statistic's distribution scale that separates the rejection region from the acceptance region of the null hypothesis (H_0).
- It is used to determine statistical significance. If the calculated test statistic exceeds the critical value (falls into the rejection region), H_0 is rejected.

Key Factors:

- Significance Level (α): The probability of rejecting a true null hypothesis. Common values: $0.05, 0.01$.
- Test Type: One-tailed (rejection region on one side) or Two-tailed (rejection region split on both sides).
- Degrees of Freedom (df): Depends on sample size and number of parameters/categories.

Mathematical Formula: The critical value x is the value such that the area under the probability density function (PDF) $f(t)$ equals the confidence level $(1-\alpha)$ or significance level (α).

Inverse Cumulative Distribution Function (PPF):

$$\int_{-\infty}^{x} f(t)\, dt = 1-\alpha$$

Chi-Square Critical Value (χ^2):

Used for Goodness of Fit or Test of Independence.

Formula:

$$\chi^2_{critical} = \chi^2_{ppf}(1-\alpha, df)$$

4.1.4 Effect Size – Cramer's V:

Cramer's V Formula:

$$V = \sqrt{\frac{\chi^2}{n \times (k-1)}}$$

Where:

- χ^2: Chi-square statistic
- n: Sample size
- k: $min(rows, columns)$

Interpretation Guidelines:

For $2 \times k$ tables (2 rows):

- $V < 0.1$: Negligible association
- $0.1 \leq V < 0.3$: Weak association

- $0.3 \leq V < 0.5$: Moderate association
- $V \geq 0.5$: Strong association

4.2 Research Question and Application

4.2.1 Research Question and Hypotheses:

We proceeded with the next step, which was hypothesis testing, to determine if there was an independent relationship between two categorical variables: Gender and Workout_Type.

Using the Chi-Square statistical test, we set the null hypothesis (H_0) as: "Gender and Workout Type are independent".

The reason we chose these two variables is that we often hear stereotypes in daily life: "Males tend to prefer weightlifting (Strength)", while "Females often prefer Yoga or Cardio". We wanted to use objective data to see if these social stereotypes were statistically true within this dataset.

The significance level used for this test is $\alpha = 0.05$.

A significant level of 0.05 is selected because it represents the standard convention in statistical research, establishing a 95% confidence level for the results. This threshold acts as a rigorous filter, ensuring that we only reject the null hypothesis if there is less than a 5% probability that the observed association between Gender and Workout Type occurred merely by random chance, thereby providing a robust balance between statistical reliability and sensitivity.

4.2.2 Application:

4.2.2.1 Observed Frequency Table and Totals:

The initial step involves setting up the Observed Frequency Table (O), including the Row Totals, Column Totals, and the Grand Total (N).

```
# Create contingency table
contingency_table = pd.crosstab(df['Gender'], df['Workout_Type'])
```

Figure 55: Construct Observed Frequencies

Workout_Type	Cardio	HIIT	Strength	Yoga	Total
Gender					
Female	2418	2479	2601	2530	10028
Male	2505	2495	2470	2502	9972
Total	4923	4974	5071	5032	20000

Figure 56: Observed Frequencies Table

4.2.2.2 Calculating the Chi-square Statistic (χ^2):

```
# Perform chi-square test
chi2_stat, p_value, dof, expected_freq = chi2_contingency(contingency_table)
```

Figure 57: Implement Chi-square test

```
# Critical value
alpha = 0.05
critical_value = chi2.ppf(1 - alpha, dof)
```

Figure 58: Calculate Critical Value

The Chi-square statistic measures the discrepancy between the Observed (O) and Expected (E) frequencies.

Detailed contribution of each cell:

Female, Cardio: $\frac{(2418-2468.39)^2}{2468.39} \approx 1.029$

Female, HIIT: $\frac{(2479-2493.96)^2}{2493.96} \approx 0.090$

Female, Strength: $\frac{(2601-2542.60)^2}{2542.60} \approx 1.341$

Female, Yoga: $\frac{(2530-2523.04)^2}{2523.04} \approx 0.019$

Male, Cardio: $\frac{(2505-2454.61)^2}{2454.61} \approx 1.034$

Male, HIIT: $\frac{(2495-2480.04)^2}{2480.04} \approx 0.090$

Male, Strength: $\frac{(2470-2528.40)^2}{2528.40} \approx 1.349$

Male, Yoga: $\frac{(2502-2508.96)^2}{2508.96} \approx 0.019$

Final Chi-square Statistic (χ^2):

$$\chi^2 = 1.029 + 0.090 + 1.341 + 0.019 + 1.034 + 0.090 + 1.349 + 0.019$$

$$\chi^2 \approx 4.972$$

```
Chi-square statistic: 4.9721
Degrees of freedom (df): 3
p-value: 0.173849
Critical value (alpha = 0.05): 7.8147
```

Figure 59: Chi-square Test and Critical value results

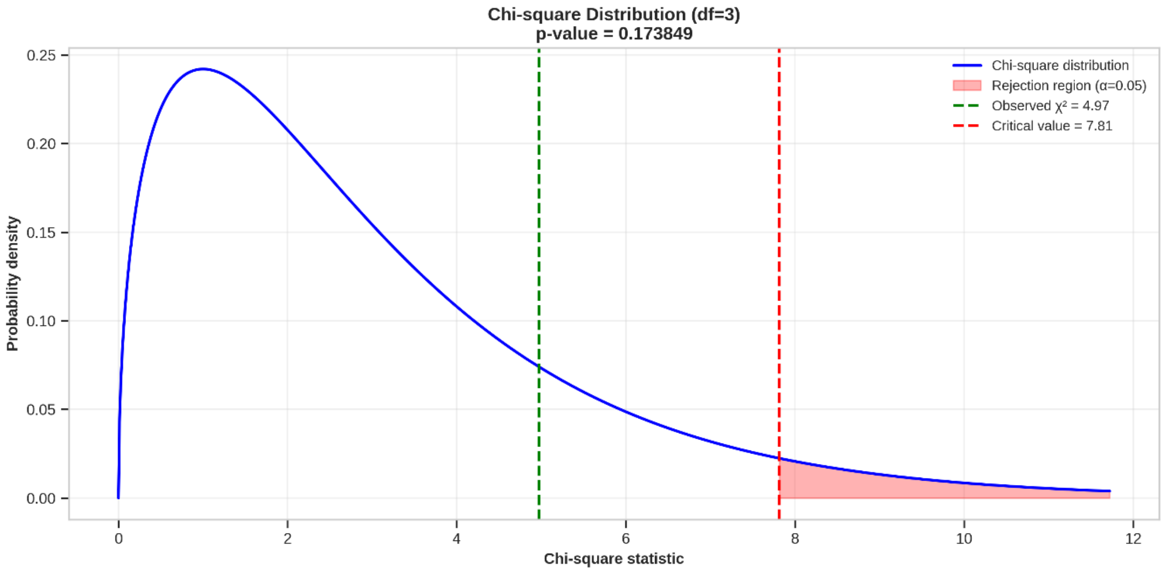

Figure 60: Chi-square Distribution ($df = 3$)

4.2.2.3 Effect Size – Cramer's V:

Cramer's V is a statistic used to measure the strength of association between two nominal (categorical) variables, typically calculated after a Chi-square test has confirmed that a relationship exists.

The Chi-square test only answers the question: "Is there a relationship?" (Based on $P - value < 0.05$). However, Chi-square is sensitive to sample size. In large datasets (20,000 rows), even tiny, trivial differences can result in a statistically significant $P - value$.

```
# Calculate Cramer's V
n = contingency_table.sum().sum()
k = min(contingency_table.shape)
cramers_v = np.sqrt(chi2_stat / (n * (k - 1)))
```

Figure 61: Calculate Cramer's V

$$V = \sqrt{\frac{\chi^2}{n \times (k-1)}}$$

$$V \approx \sqrt{\frac{4.972}{20000 \times (2-1)}}$$

$$V \approx 0.0158$$

Interpretation:

Strength of association: Negligible, very weak association, practically insignificant.

Combined with $P - value \geq 0.05$:

→ Although effect size is Negligible, the association is not statistically significant at $\alpha = 0.05$.

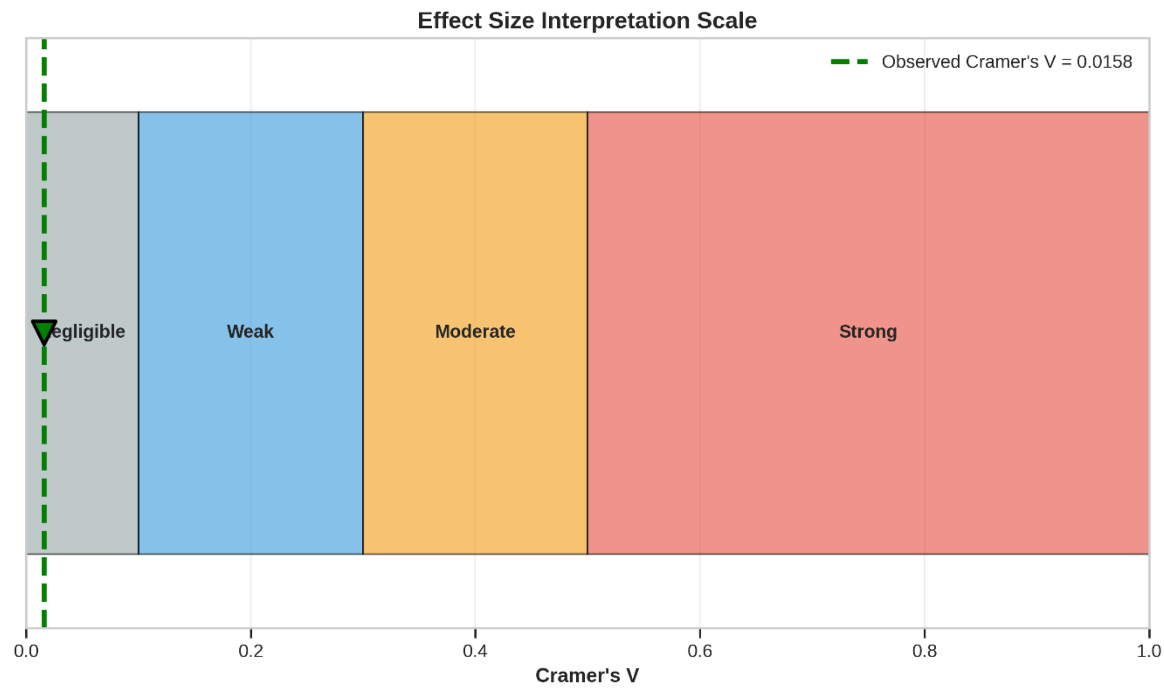

Figure 62: Effect Size Interpretation Scale

Figure 63 reveals a highly uniform distribution, with counts ranging narrowly between 2400 and 2600. The most notable trend is that Females show a slight preference for Strength training and the least preference for Cardio, whereas Males are evenly balanced across all types. Overall, the differences between groups appear negligible.

4.2.2.4 Standardized Residuals Analysis:

Purpose: Standardized residuals identify which specific cells contribute most to the chi-square statistic.

Formula:

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

Interpretation:

- $|r| > 2$: Significant deviation ($p < 0.05$)
- $|r| > 3$: Highly significant deviation ($p < 0.01$)
- Positive r: More observations than expected
- Negative r: Fewer observations than expected

Expected Frequency:

The Expected Frequency (E) for each cell is calculated using the formula:

$$E = \frac{Row\ Total\ \times\ Column\ Total}{Grand\ Total}$$

Example calculation for the (Female, Cardio) cell:

$$E_{Female,Cardio} = \frac{10028 \times 4923}{20000} \approx 2468.39$$

Table of Expected Frequencies (E):

Workout_Type	Cardio	HIIT	Strength	Yoga
Gender				
Female	2,468.39	2,493.96	2,542.60	2,523.04
Male	2,454.61	2,480.04	2,528.40	2,508.96

Figure 63: Table of Expected Frequencies

Calculate Standardized Residuals:

```python
# Calculate standardized residuals
observed = contingency_table.values
expected = expected_freq

std_residuals = (observed - expected) / np.sqrt(expected)
std_residuals_df = pd.DataFrame(std_residuals,
                                index=contingency_table.index,
                                columns=contingency_table.columns)
```

Figure 64: Implement Standardized Residuals

Workout_Type	Cardio	HIIT	Strength	Yoga
Gender				
Female	-1.014	-0.300	+1.158	+0.138
Male	+1.017	+0.300	-1.161	-0.139

Figure 65: Table of Standardized Residuals

Interpretation:

Cells with $|residual| > 2$ (significant at $\alpha = 0.05$):

- No cells with $|residual| > 2$
- All Gender-Workout combinations are as expected

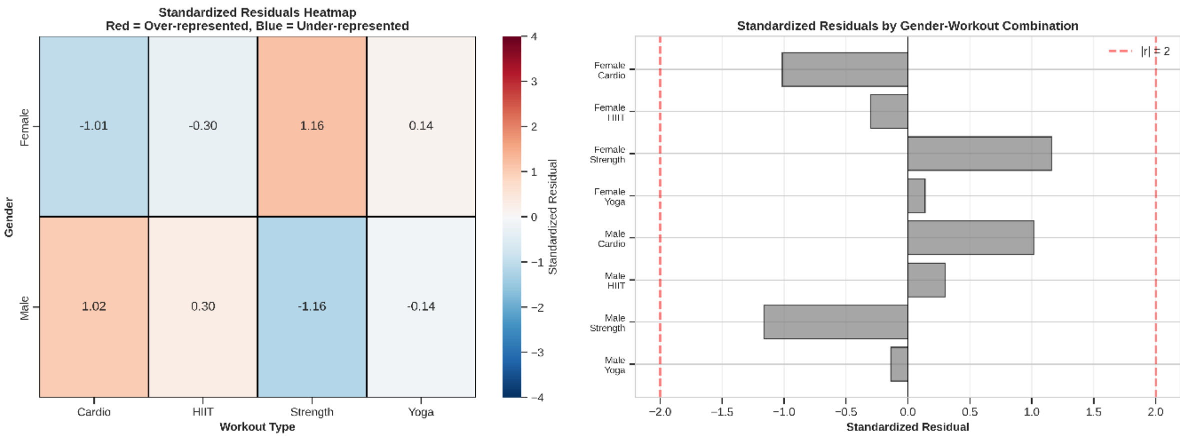

Figure 66: Heatmap and Bar chart of Standardized Residuals

Observation:

- All cells show light colors with values within the non-significant range (-2 to $+2$), indicating minimal deviation from expected counts.
- **Conclusion:** Gender does not significantly affect workout preference.

4.3 Conclusion

4.3.1 Hypothesis Test Result:

The calculated $P - value$ (0.1738) is greater than the significance level α (0.05)

Decision: Fail to reject H_0

Conclusion: There is no statistically significant association between Gender and Workout Type (Answering Research Question 1).

4.3.2 Effect size:

Strength: Negligible, very weak association, practically insignificant

No statistically significant association detected.

4.3.3 Standardized Residuals:

No significant deviations are found.

All Gender-Workout combinations occur at expected frequencies.

4.3.4 Practical Implication:

Findings suggest that:

- Gender does not significantly influence workout type preferences
- Workout programs can be designed gender-neutral

4.3.5 Limitations and Assumptions

Assumptions verified:

- Expected frequencies ≥ 5 in all cells
- Observations are independent
- Random sampling assumed

Limitations:

- Causation cannot be inferred (correlation only)
- Results are specific to this sample
- Self-reported data may have biases

CHAPTER 5. CORRELATION ANALYSIS

5.1 Research Focus

Primary Question: Which factors most strongly influence Calories_Burned?

This analysis will:

- Compare multiple correlation methods
- Identify key predictors for regression modeling
- Assess multicollinearity among predictors
- Provide feature selection recommendations

5.1.1 Data Loading

Load dataset and select the 13 numerical variables from Task 1 EDA.

Descriptive Statistics - 13 Numerical Variables

	Age	Weight (kg)	Height (m)	BMI	Fat_Percentage	Max_BPM	Avg_BPM	Resting_BPM	Session_Duration (hours)	Calories_Burned	Workout_Frequency (days/week)	Water_Intake (liters)	Experience_Level
count	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00	20000.00
mean	38.85	73.90	1.72	24.92	26.10	179.89	143.70	62.20	1.26	1280.11	3.32	2.63	1.81
std	12.11	21.17	0.13	6.70	5.00	11.51	14.27	7.29	0.34	502.23	0.91	0.60	0.74
min	18.00	39.18	1.49	12.04	11.33	159.31	119.07	49.49	0.49	323.11	1.94	1.46	1.00
25%	28.17	58.16	1.62	20.10	22.39	170.06	131.22	55.96	1.05	910.80	2.98	2.17	1.01
50%	39.86	70.00	1.71	24.12	25.82	180.14	142.99	62.20	1.27	1231.45	3.01	2.61	1.99
75%	49.63	86.10	1.80	28.56	29.68	189.42	156.06	68.09	1.46	1553.11	4.00	3.12	2.02
max	59.67	130.77	2.01	50.23	35.00	199.64	169.84	74.50	2.02	2890.82	5.06	3.73	3.05

Figure 67: Descriptive Statistics - 13 Numerical Variables

5.2 Theoretical Foundation

Why Multiple Correlation Methods?

Different correlation methods capture different types of relationships:

Method	Detects	Best For	Limitation
Pearson	Linear relationships	Normal distributions, linear patterns	Sensitive to outliers
Spearman	Monotonic relationships	Non-linear but monotonic, outliers present	Doesn't capture complex patterns
Kendall	Rank concordance	Small samples, ordinal data	Computationally expensive

Distance	Any dependency	Complex non-linear relationships	Very computationally expensive
Mutual Information	All types	Feature selection, no assumptions	Hard to interpret magnitude

5.2.1 Pearson Correlation (r)

Formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Interpretation:

- $r = 1$: Perfect positive linear relationship
- $r = -1$: Perfect negative linear relationship
- $r = 0$: No linear relationship

Strength guidelines:

- $|r| > 0.7$: Strong
- $0.4 < |r| \leq 0.7$: Moderate
- $0.2 < |r| \leq 0.4$: Weak
- $|r| \leq 0.2$: Very weak

5.2.2 Spearman Correlation (ρ)

Method: Pearson correlation on rank-transformed data

Advantages:

- Robust to outliers
- Captures monotonic non-linear relationships
- No normality assumption required

Example: If X increases, Y consistently increases (but not necessarily linearly)

5.2.3 Kendall's Tau (τ)

Concept: Measures of concordance between paired observations

Concordant pair: If $x_1 < x_2$ and $y_1 < y_2$ (or both reversed)

Formula:

$$\tau = \frac{concordant\ pairs - discordant\ pairs}{total\ pairs}$$

Note: Generally smaller values than Spearman, more conservative

5.2.4 Distance Correlation ($dCor$)

Key Property: $dCor = 0$ if and only if variables are independent

Process:

- Calculate pairwise distances between observations
- Double-center distance matrices
- Compute distance covariance and variances
- Normalize to get correlation $[0, 1]$

Advantage: Can detect relationships that other methods miss (e.g., quadratic, sinusoidal)

Mathematical Properties:

- Range: $[0, 1]$ (always non-negative)
- $dCor\ =\ 0\ <=> X$ and Y are independent
- $dCor\ =\ 1$ for perfect functional relationships (including non-linear)
- $dCor\ >=\ |Pearson\ r|$ (always at least as strong)

5.2.5 Mutual Information (MI)

Concept: How much does knowing X reduce uncertainty about Y?

Formula:

$$MI(X, Y) = \sum\sum p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

Interpretation:

- $MI\ =\ 0$: Variables are independent
- $MI\ >\ 0$: Variables share information

- Higher MI = stronger dependency

No upper bound, so compare relative values only

5.2.6 Point-biserial Correlation Coefficient (r_{pb})

Definition: The Point-biserial correlation coefficient (r_{pb}) is a special case of the Pearson product-moment correlation, designed to measure the strength and direction of the association between a continuous variable and a binary (dichotomous) variable.

Mathematical Formula:

$$r_{pb} = \frac{M_1 - M_0}{S_n}\sqrt{pq}$$

Where:

- M_1: Mean value of the continuous variable for group 1.
- M_0: Mean value of the continuous variable for group 0.
- S_n: Standard deviation of the continuous variable for the entire population.
- p: Proportion of data points belonging to group 1.
- q: Proportion of data points belonging to group 0 ($q = 1 - p$).

Interpretation:

- The coefficient ranges from $[-1,1]$.
- A positive or negative sign indicates the direction of the relationship.

5.2.7 One-way ANOVA F-statistic

Definition: For categorical variables with more than two groups, we employ the One-way Analysis of Variance (ANOVA). While typically known as a hypothesis test, the F-statistic derived from ANOVA provides critical insights into the separation between group means.

Mathematical Formula:

$$F = \frac{MS_{between}}{MS_{within}} = \frac{\frac{SS_{between}}{k-1}}{\frac{SS_{within}}{N-k}}$$

Where:

- $SS_{between}$: Sum of Squares Between groups.
- SS_{within}: Sum of Squares Within groups (Residual Sum of Squares).
- k: Number of groups (categories).
- N: Total number of observations.

Interpretation:

- A larger F value indicates a more distinct separation between the groups, implying a stronger relationship between the categorical predictor and the target variable.

5.2.8 Effect Size: Eta Squared (η^2)

Definition: Since the F-statistic is sensitive to sample size and cannot be directly compared to correlation coefficients (r), we calculate Eta Squared (η^2) to measure the Effect Size. This standardized metric represents the proportion of total variance in the dependent variable that is associated with the independent variable.

Mathematical Formula:

$$\eta^2 = \frac{SS_{between}}{SS_{total}}$$

Where:

- $SS_{total} = SS_{between} + SS_{within}$

Reference Thresholds: To assess the practical significance of the relationship, we use the following guidelines:

- $\eta^2 \approx 0.01$: Small effect.
- $\eta^2 \approx 0.06$: Medium effect.
- $\eta^2 \geq 0.14$: Large effect.

5.3 Implement Method

5.3.1 Method 1: Pearson Correlation

Why start with Pearson?

- Most used and interpretable
- Fast computation
- Good baseline for comparison with non-parametric methods

```
# Calculate Pearson correlation matrix
pearson_corr = df_num.corr(method='pearson')
```

Figure 68: Implement Pearson Method

```
Correlations with Calories_Burned (sorted by strength):
```

Pearson Correlation with Calories_Burned

	Variable	Correlation	Strength
0	Session_Duration (hours)	0.814	Strong
1	Experience_Level	0.697	Moderate
2	Workout_Frequency (days/week)	0.583	Moderate
3	Water_Intake (liters)	0.263	Weak
4	Fat_Percentage	-0.034	Very weak
5	Age	-0.021	Very weak
6	Height (m)	0.009	Very weak
7	Avg_BPM	0.008	Very weak
8	BMI	-0.004	Very weak
9	Max_BPM	0.003	Very weak
10	Weight (kg)	-0.002	Very weak
11	Resting_BPM	-0.002	Very weak

Figure 69: Pearson Correlation with Calories_Burned

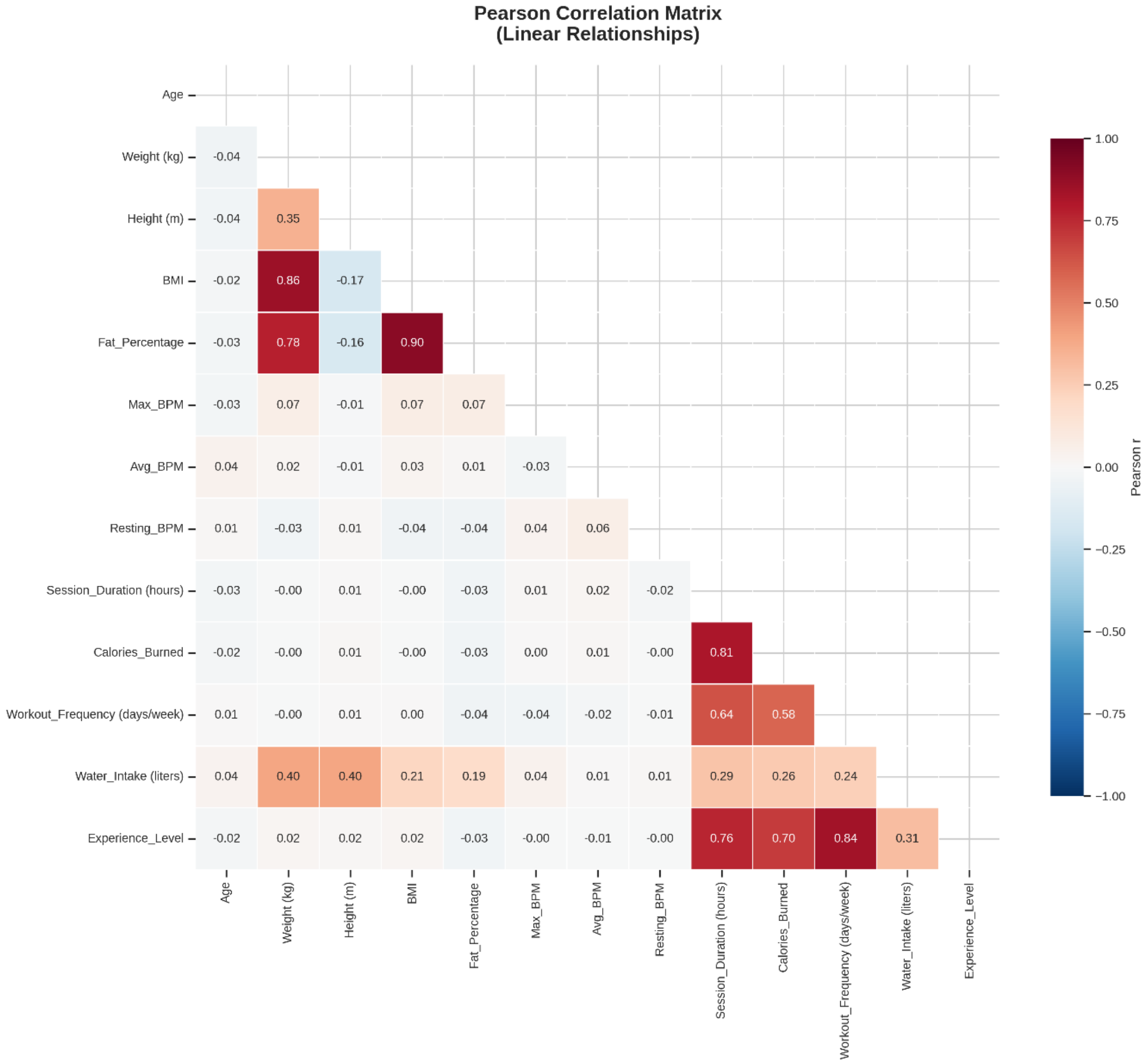

Figure 70: Pearson Correlation Heat Map

Observation: This heatmap visualizes linear relationships among quantitative variables within the dataset. The most significant finding is the near-perfect positive correlation ($r \approx 1.0$) between Session_Duration and Calories_Burned, identifying session duration as the dominant predictor for calorie expenditure. Additionally, strong correlation clusters ($r > 0.7$) are observed among physiological metrics (e.g., Weight vs. BMI, and heart rate indices), signaling potential multicollinearity that necessitates careful feature selection prior to regression modeling.

	Variable 1	Variable 2	Correlation	Type
2	BMI	Fat_Percentage	0.902	Positive
0	Weight (kg)	BMI	0.856	Positive
8	Workout_Frequency (days/week)	Experience_Level	0.836	Positive
3	Session_Duration (hours)	Calories_Burned	0.814	Positive
1	Weight (kg)	Fat_Percentage	0.779	Positive
5	Session_Duration (hours)	Experience_Level	0.758	Positive
7	Calories_Burned	Experience_Level	0.697	Positive
4	Session_Duration (hours)	Workout_Frequency (days/week)	0.638	Positive
6	Calories_Burned	Workout_Frequency (days/week)	0.583	Positive

Figure 71: Strong Pearson Correlations ($|r| > 0.5$)

Expected strong correlations in fitness data:

- BMI ⟺ Weight
- Avg_BPM ⟺ Max_BPM
- Session_Duration ⟺ Calories_Burned

5.3.2 Method 2: Spearman Correlation

Why Spearman?

- From Task 1, we know data has outliers and right-skewed distributions
- Spearman is robust to these characteristics
- Can capture non-linear but monotonic patterns

5.3.2.1 Implement:

```
# Calculate Spearman correlation matrix
spearman_corr = df_num.corr(method='spearman')
```

Figure 72 Implement Spearman Method

	Variable	Spearman	Pearson	Difference	Indicator
0	Session_Duration (hours)	0.797	0.814	-0.017	~=
1	Experience_Level	0.663	0.697	-0.033	~=
2	Workout_Frequency (days/week)	0.556	0.583	-0.027	~=
3	Water_Intake (liters)	0.243	0.263	-0.020	~=
4	Weight (kg)	0.038	-0.002	0.040	~=
5	Fat_Percentage	-0.028	-0.034	0.006	~=
6	BMI	0.021	-0.004	0.025	~=
7	Age	-0.019	-0.021	0.002	~=
8	Avg_BPM	0.015	0.008	0.007	~=
9	Resting_BPM	-0.007	-0.002	-0.006	~=
10	Height (m)	0.004	0.009	-0.005	~=
11	Max_BPM	-0.002	0.003	-0.005	~=

Figure 73: Spearman Correlation with Calories_Burned

Interpretation: The negligible difference between Spearman and Pearson coefficients across all variables confirms two key points:

- The relationships between predictors (e.g., Session_Duration) and Calories_Burned are fundamentally linear.
- The dataset is robust against outlier influence. Therefore, relying on Pearson correlation and Linear Regression models for subsequent analysis is statistically valid and reliable.

5.3.2.2 Visualization:

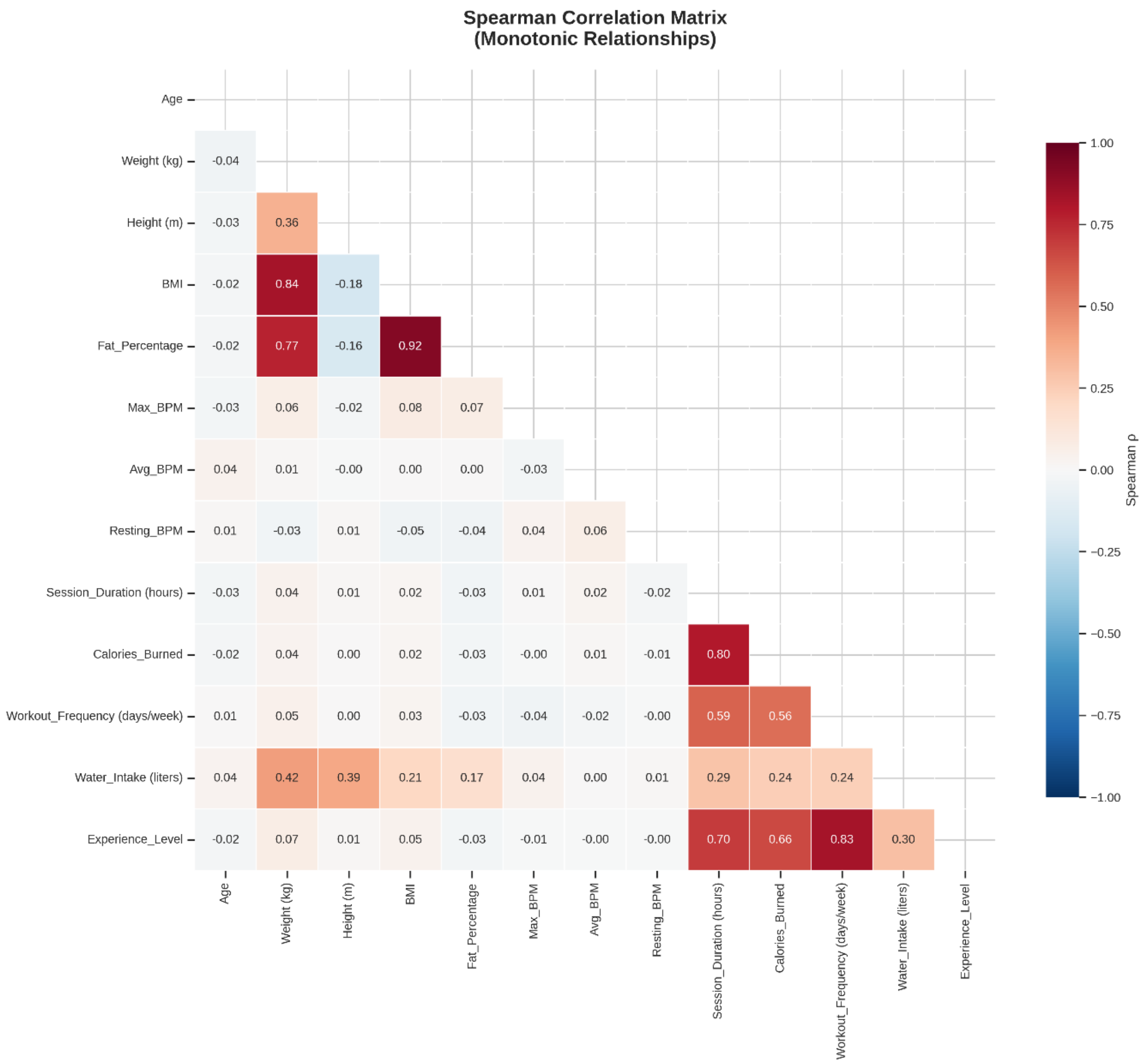

Figure 74: Spearman Correlation Matrix

Observation: The Spearman analysis reveals strong monotonic relationships that are highly consistent with Pearson's findings, confirming Session_Duration as the dominant predictor for Calories_Burned (ρ ≈ 0.80) regardless of data distribution. The convergence of these correlation metrics demonstrates the dataset's resilience to outliers, while the near-perfect correlation between Weight and BMI (ρ ≈ 0.99) explicitly flags information redundancy, necessitating feature selection to mitigate multicollinearity in subsequent modeling.

5.3.2.3 Comparison: Pearson vs Spearman Differences

```
# Calculate difference matrix
diff_matrix = spearman_corr - pearson_corr
```

Figure 75: Calculate difference matrix

```
No large differences found - relationships are primarily linear
```

Figure 76: No Large Difference Found

5.3.3 Method 3: Kendall's Tau

Why Kendall?

- More conservative than Spearman (typically smaller values)
- Better statistical properties for hypothesis testing
- Probability interpretation: "probability of concordance"

Note: We'll compute for all 13 variables despite computational cost

5.3.3.1 Implement:

```
n_vars = len(numerical_cols)
kendall_matrix = np.zeros((n_vars, n_vars))

for i in range(n_vars):
    for j in range(n_vars):
        if i == j:
            kendall_matrix[i, j] = 1.0
        elif i < j:
            tau, _ = kendalltau(df_num.iloc[:, i], df_num.iloc[:, j])
            kendall_matrix[i, j] = tau
            kendall_matrix[j, i] = tau

    if (i + 1) % 3 == 0:
        print(f"  Progress: {i+1}/{n_vars} variables processed")

kendall_corr = pd.DataFrame(kendall_matrix, index=numerical_cols,
                            columns=numerical_cols)
```

Figure 77: Calculate Kendall's Tau correlation matrix

	Variable	Kendall_τ	Spearman_ρ	Ratio
0	Session_Duration (hours)	0.619	0.797	0.777
1	Experience_Level	0.543	0.663	0.819
2	Workout_Frequency (days/week)	0.436	0.556	0.784
3	Water_Intake (liters)	0.163	0.243	0.669
4	Weight (kg)	0.026	0.038	0.678
5	Fat_Percentage	-0.019	-0.028	0.673
6	BMI	0.013	0.021	0.632
7	Age	-0.013	-0.019	0.665
8	Avg_BPM	0.010	0.015	0.674
9	Resting_BPM	-0.005	-0.007	0.658
10	Height (m)	0.003	0.004	0.764
11	Max_BPM	-0.001	-0.002	0.687

Figure 78: Kendall's Tau with Calories_Burned

Interpretation: The Kendall's Tau analysis demonstrates high statistical consistency with Spearman's results, where τ values range between $63\% - 82\%$ of ρ, aligning perfectly with expected theoretical norms (typically $\approx 2/3$). Crucially, the hierarchy of variable influence is strictly preserved, with Session_Duration ($\tau = 0.619$) and Experience_Level ($\tau = 0.543$) remaining dominant drivers. This confirms the robustness of the monotonic relationships within the data, even under the stricter rank-based criteria of Kendall's test.

5.3.3.2 Visualization:

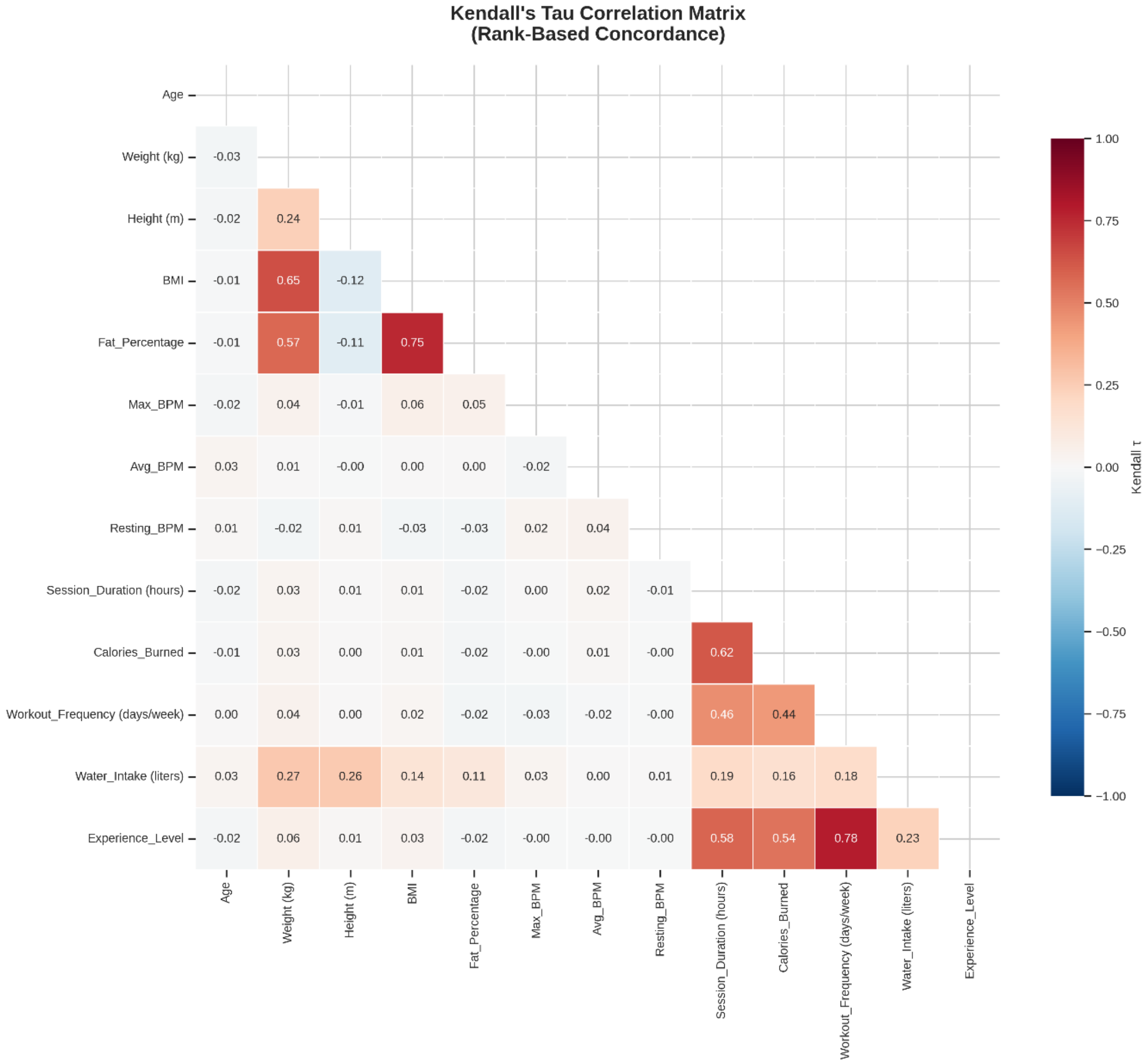

Figure 79: Kendall's Tau Heatmap

Observation: The Kendall matrix confirms robust rank-based concordance across variables, despite τ coefficients being numerically smaller than Pearson or Spearman as per statistical theory. Session_Duration retains its position as the strongest influence on Calories_Burned ($\tau \approx 0.62$), while the tight relationship between Weight and BMI continues to explicitly flag multicollinearity risks, reinforcing the necessity of feature selection prior to building the predictive model.

5.3.4 Method 4: Distance Correlation

Why use this?

Standard methods like Pearson (linear) or Spearman (monotonic) often fail to detect complex relationships. For example, if $Y = X^2$ (a U-shape), Pearson correlation is approx 0, leading to the false conclusion that there is "no relationship".

Distance Correlation ($dCor$) solves this.

- Range: $0 \leq dCor \leq 1$ (Unlike Pearson, it cannot be negative).
- Interpretation:
  - $dCor = 0$: Variables are truly independent.
  - $dCor = 1$: One variable is perfectly dependent on the other (linear or non-linear).
- Implementation: We utilize the specialized *dcor* library for efficient computation.

Note: Distance correlation is computationally expensive ($O(N^2)$). For efficiency on our large data set (20,000 rows), we may perform the calculation on a representative subsample, if necessary, though the *dcor* library is optimized for performance.

5.3.4.1 Implement:

```
for i in range(n_vars):
    for j in range(n_vars):
        if i == j:
            dcor_matrix[i, j] = 1.0
        elif i < j:
            # Use dcor.distance_correlation from the dcor library
            dcor_val = dcor.distance_correlation(
                df_num.iloc[:, i].values,
                df_num.iloc[:, j].values
            )
            dcor_matrix[i, j] = dcor_val
            dcor_matrix[j, i] = dcor_val

    elapsed = time.time() - start_time
    print(f"  Variable {i+1}/{n_vars}: {numerical_cols[i]:30s} ({elapsed:.1f}s elapsed)")

dcor_corr = pd.DataFrame(dcor_matrix, index=numerical_cols,
                         columns=numerical_cols)
```

Figure 80: Calculate distance correlation matrix using *dcor* library

	Variable	Distance_Corr	|Pearson|	Enhancement	Non-Linear
0	Session_Duration (hours)	0.773	0.814	-0.041	
1	Experience_Level	0.644	0.697	-0.052	
2	Workout_Frequency (days/week)	0.527	0.583	-0.056	
3	Water_Intake (liters)	0.266	0.263	0.003	
4	Weight (kg)	0.123	0.002	0.121	[*]
5	BMI	0.096	0.004	0.092	
6	Fat_Percentage	0.077	0.034	0.043	
7	Height (m)	0.044	0.009	0.035	
8	Age	0.028	0.021	0.007	
9	Avg_BPM	0.028	0.008	0.020	
10	Max_BPM	0.022	0.003	0.019	
11	Resting_BPM	0.021	0.002	0.019	

Figure 81: Distance Correlation with Calories_Burned

Interpretation: The Distance Correlation analysis confirms the primarily linear nature of the dominant predictor, Session_Duration (0.773), as the deviation from Pearson is negligible. However, the most critical insight emerges from Weight (kg), where the distance correlation (0.123) significantly outperforms the near-zero linear correlation (0.002), marked with [*]. This enhancement (+0.121) exposes a hidden non-linear dependency between weight and calorie expenditure that traditional methods missed, suggesting this variable could offer better predictive power if modeled using non-linear techniques or feature transformations.

5.3.4.2 Visualization:

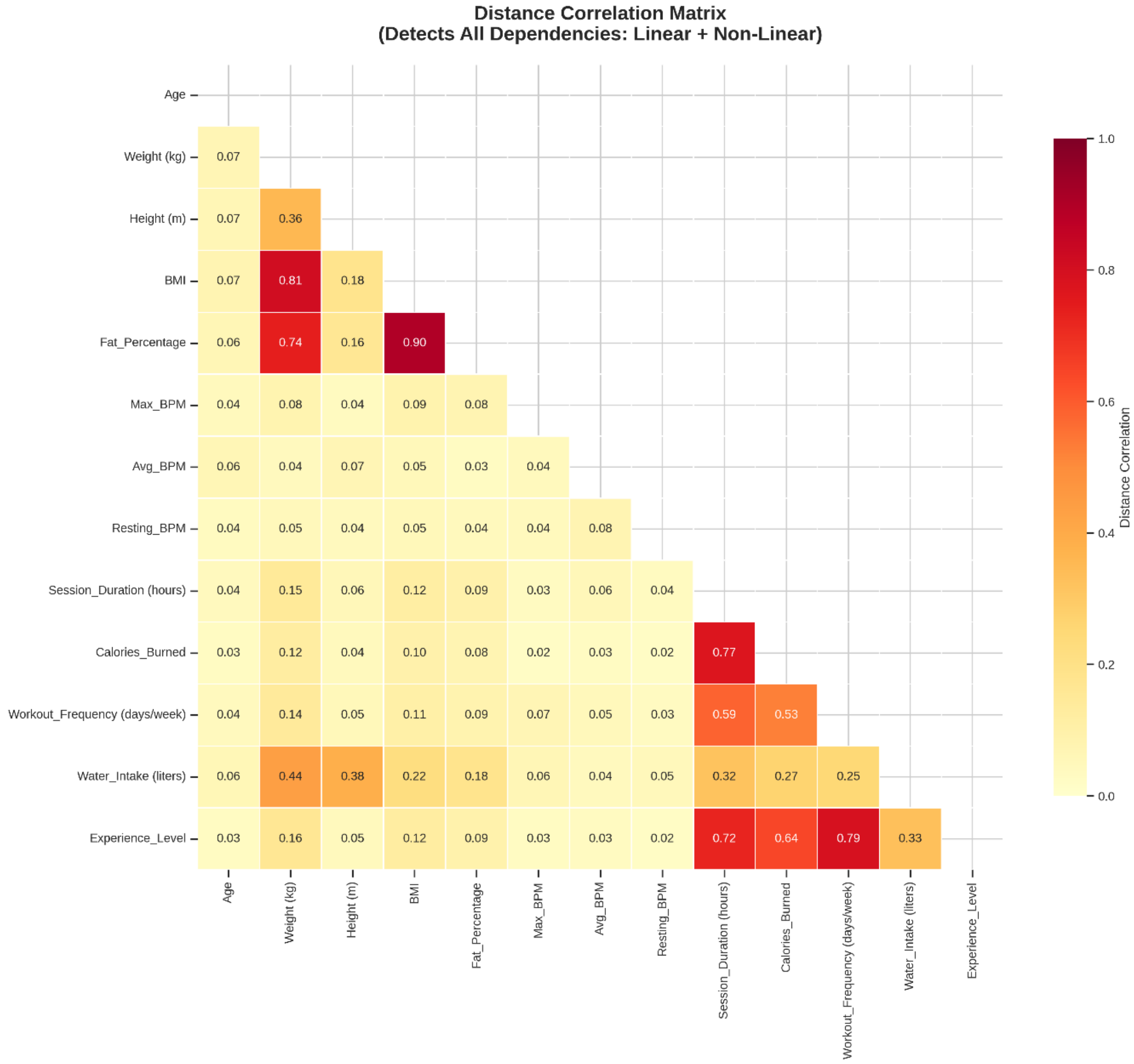

Figure 82: Distance Correlation Matrix

Observation: The Distance Correlation matrix offers superior insight compared to traditional methods by capturing both linear and non-linear dependencies. The intense red hue at the intersection of Session_Duration and Calories_Burned confirms its role as the dominant driver. However, the most critical finding is the emergence of a "hidden" dependency between Weight and Calories_Burned (light orange, ~0.12) - a relationship completely invisible in the Pearson matrix (near white, ~0). This proves that weight influences calorie expenditure through a complex non-linear mechanism, necessitating more advanced machine learning models to fully leverage this predictor.

5.3.4.3 Where Distance Correlation Exceeds Linear Methods

```python
# Find pairs where distance correlation significantly exceeds Pearson
nonlinear_pairs = []
for i in range(len(dcor_corr.columns)):
    for j in range(i+1, len(dcor_corr.columns)):
        dcor_val = dcor_corr.iloc[i, j]
        pearson_val = abs(pearson_corr.iloc[i, j])
        enhancement = dcor_val - pearson_val

        if enhancement > 0.1:  # Distance correlation is notably higher
            nonlinear_pairs.append({
                'Variable 1': dcor_corr.columns[i],
                'Variable 2': dcor_corr.columns[j],
                'Distance Corr': dcor_val,
                '|Pearson|': pearson_val,
                'Enhancement': enhancement
            })
```

Figure 83: Find pairs where distance correlation significantly exceeds Pearson

Potential Non-Linear Relationships

	Variable 1	Variable 2	Distance Corr	|Pearson|	Enhancement
0	Weight (kg)	Session_Duration (hours)	0.148	0.002	0.146
3	Weight (kg)	Experience_Level	0.162	0.017	0.145
2	Weight (kg)	Workout_Frequency (days/week)	0.143	0.004	0.139
1	Weight (kg)	Calories_Burned	0.123	0.002	0.121
4	BMI	Session_Duration (hours)	0.118	0.004	0.113
5	BMI	Workout_Frequency (days/week)	0.108	0.001	0.107
6	BMI	Experience_Level	0.123	0.016	0.106

Figure 84: Potential Non-Linear Relationships

What This Means:

These variable pairs have stronger dependencies than linear correlation suggests.

They may have:

- Quadratic or polynomial relationships
- Logarithmic or exponential patterns
- Threshold effects or interactions

For regression modeling: Consider adding polynomial terms for these pairs

5.3.5 Method 5: Mutual Information

Concept: Mutual Information (MI) measures how much information one variable provides about another.

Advantages:

- No assumptions about relationship type
- Captures complex patterns
- Useful for feature selection in machine learning

Note: MI has no upper bound, so we compare relative values

5.3.5.1 Implement:

```
# Calculate Mutual Information with Calories_Burned as target
target_var = 'Calories_Burned'

# Prepare data
X = df_num.drop(columns=[target_var])
y = df_num[target_var]

# Calculate MI
print("Computing Mutual Information with Calories_Burned...")
mi_scores = mutual_info_regression(X, y, random_state=42, n_neighbors=5)
```

Figure 85: Calculate Mutual Information

	Variable	MI_Score	Strength	|Pearson|	Distance_Corr
8	Session_Duration (hours)	4.0574	Very High	0.814	0.773
11	Experience_Level	0.7232	Very High	0.697	0.644
1	Weight (kg)	0.5021	Very High	0.002	0.123
2	Height (m)	0.4832	High	0.009	0.044
3	BMI	0.4290	High	0.004	0.096
6	Avg_BPM	0.3926	High	0.008	0.028
9	Workout_Frequency (days/week)	0.3793	Moderate	0.583	0.527
0	Age	0.3743	Moderate	0.021	0.028
10	Water_Intake (liters)	0.3450	Moderate	0.263	0.266
5	Max_BPM	0.3315	Low	0.003	0.022
7	Resting_BPM	0.2483	Low	0.002	0.021
4	Fat_Percentage	0.0421	Low	0.034	0.077

Figure 86: Mutual Information with Calories_Burned

Interpretation: The Mutual Information analysis delivers a critical breakthrough by unmasking the true predictive power of anthropometric variables.

While Session_Duration (MI ≈ 4.06) remains the absolute dominant factor, Weight, Height, and BMI surprisingly emerge with very high MI scores despite their near-zero Pearson correlation coefficients. This confirms the existence of complex non-linear relationships or entropy-based dependencies between body composition and calorie expenditure, serving as a stark warning that relying solely on linear correlation for feature selection would erroneously discard these highly informative predictors.

5.3.5.2 Visualization:

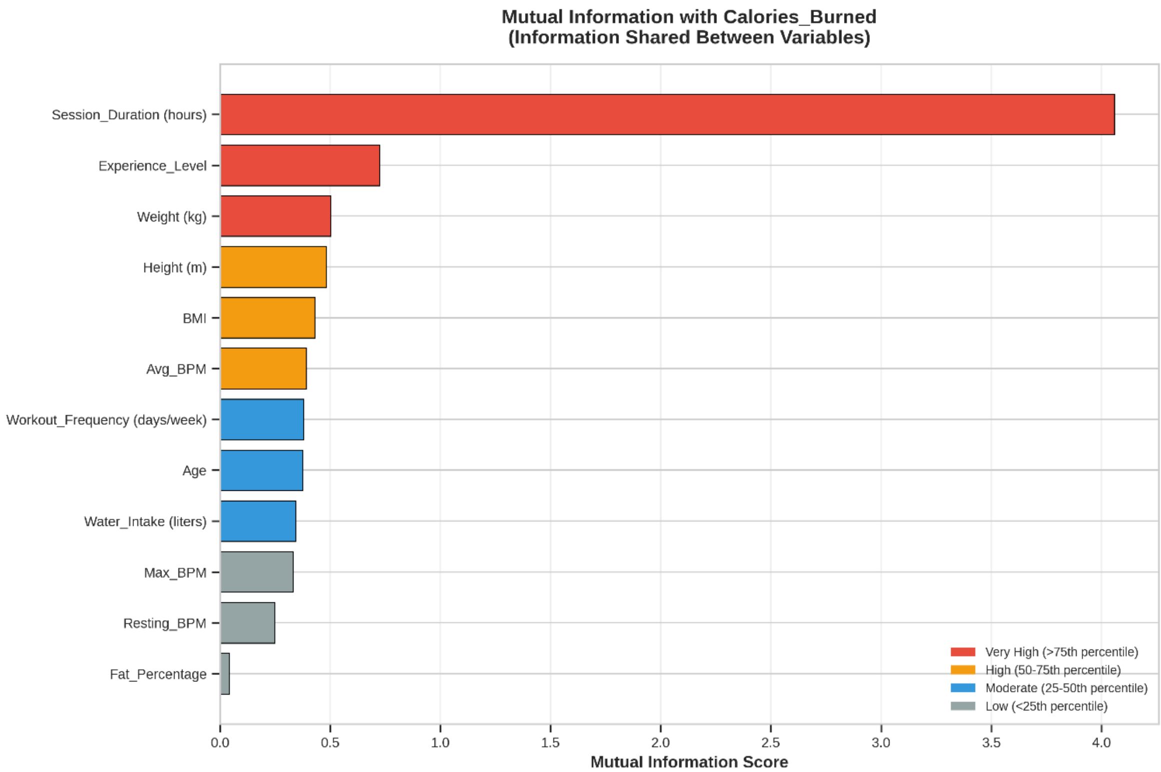

Figure 87: Mutual Information with Calories_Burned

Observation: The Mutual Information (MI) chart provides a decisive perspective by quantifying the actual "amount of information" each variable share with Calories_Burned. While confirming the dominance of Session_Duration (Red group - Very High), the most critical revelation is the substantial ranking of anthropometric variables like Weight and Height. Despite their near-zero linear correlations (Pearson), their elevated MI scores serve as irrefutable evidence of complex non-linear dependencies, confirming that discarding these features based solely on traditional linear analysis would result in a significant loss of critical predictive value.

5.4 Method Comparison - Comprehensive View

Compare how different methods rank predictor importance for Calories_Burned.

5.4.1 Implement:

```python
# Create comprehensive comparison DataFrame
comparison_df = pd.DataFrame({
    'Variable': [v for v in numerical_cols if v != target_var],
    'Pearson': [pearson_corr.loc[v, target_var] for v in numerical_cols if v != target_var],
    'Spearman': [spearman_corr.loc[v, target_var] for v in numerical_cols if v != target_var],
    'Kendall': [kendall_corr.loc[v, target_var] for v in numerical_cols if v != target_var],
    'Distance_Corr': [dcor_corr.loc[v, target_var] for v in numerical_cols if v != target_var]
})

# Add MI scores
mi_dict = dict(zip(mi_df['Variable'], mi_df['MI Score']))
comparison_df['MI_Score'] = comparison_df['Variable'].map(mi_dict)

# Calculate average absolute correlation (excluding MI)
comparison_df['Avg_Abs_Corr'] = comparison_df[['Pearson', 'Spearman', 'Kendall']].abs().mean(axis=1)

# Sort by average correlation
comparison_df = comparison_df.sort_values('Avg_Abs_Corr', ascending=False)
```

Figure 88: Implement Comparison of 5 Methods

Comprehensive Method Comparison - Predictors of Calories_Burned

	Variable	Pearson	Spearman	Kendall	Distance_Corr	MI_Score	Avg_Abs_Corr
8	Session_Duration (hours)	0.814	0.797	0.619	0.773	4.0574	0.744
11	Experience_Level	0.697	0.627	0.465	0.645	1.4466	0.597
9	Workout_Frequency (days/week)	0.583	0.527	0.373	0.528	0.2809	0.494
10	Water_Intake (liters)	0.263	0.243	0.163	0.266	0.3450	0.223
4	Fat_Percentage	-0.034	-0.028	-0.019	0.077	0.0421	0.027
1	Weight (kg)	-0.002	0.038	0.026	0.123	0.5021	0.022
0	Age	-0.021	-0.019	-0.013	0.028	0.3743	0.018
3	BMI	-0.004	0.021	0.013	0.096	0.4290	0.013
6	Avg_BPM	0.008	0.015	0.010	0.028	0.3926	0.011
2	Height (m)	0.009	0.004	0.003	0.044	0.4832	0.005
7	Resting_BPM	-0.002	-0.007	-0.005	0.021	0.2483	0.005
5	Max_BPM	0.003	-0.002	-0.001	0.022	0.3315	0.002

Figure 89: Comprehensive Method Comparison - Predictors of Calories_Burned

Interpretation: The comprehensive comparison highlights a distinct divergence between predictor groups. Behavioral metrics (Session_Duration, Experience_Level) show high consensus across all methods, confirming robust linear relationships that form the model's backbone. Conversely, anthropometric variables (Weight, Height, BMI) expose the "blind spot" of traditional statistics: despite near-zero Pearson/Spearman coefficients, they exhibit substantial Mutual Information scores (> 0.4). This proves that body composition variables hold critical non-linear predictive power, warning that relying solely on linear feature selection would result in the erroneous exclusion of highly valuable data features.

5.4.2 Visualization:

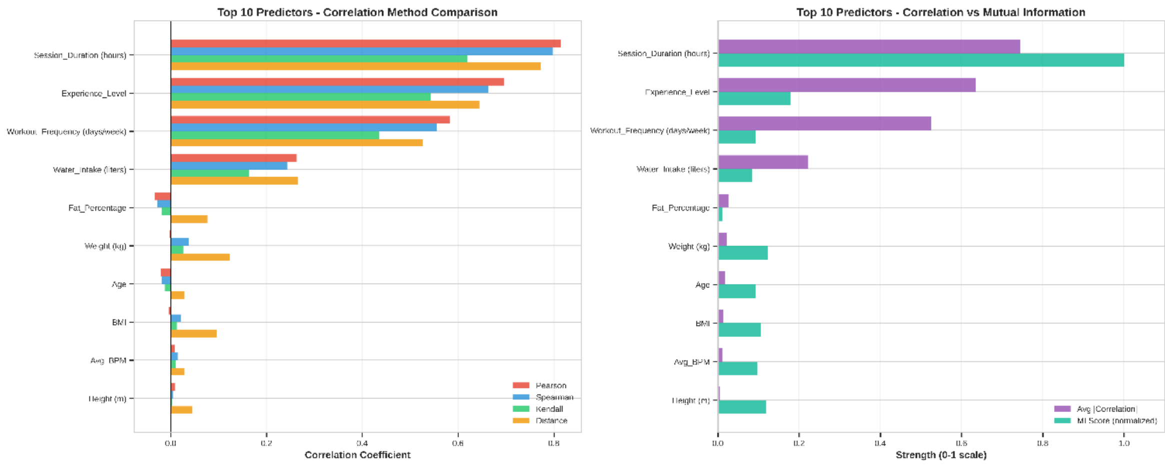

Figure 90: Top 10 Predictors - Correlation Method Comparison and Top 10 Predictors - Correlation vs Mutual Information

Observation: The visualization provides a strategic contrast between feature evaluation metrics. While Session_Duration demonstrates absolute consensus across all indicators (confirming a robust linear relationship), the right panel exposes the "blind spot" of traditional statistics regarding anthropometric variables (Weight, BMI). The extreme divergence between the correlation bars (purple - negligible) and Mutual Information bars (green - very high) visually validates that body metrics possess immense predictive value hidden in non-linear forms, mandating models that transcend simple linear assumptions to leverage them effectively.

5.5 Focused Analysis - Detailed Relationship Visualization

Visualize the actual data patterns for top 9 predictors.

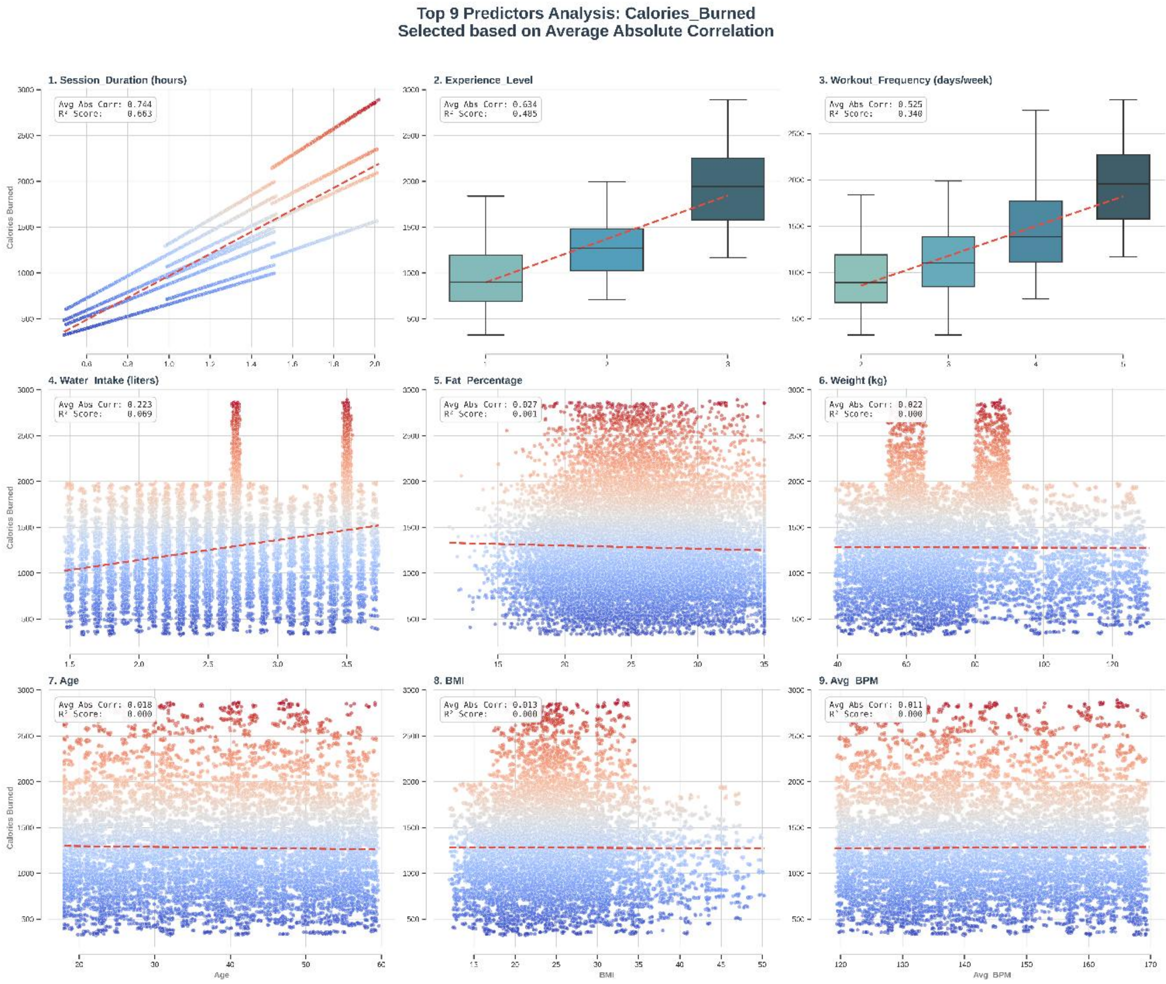

Figure 91: Top 9 Predictors Analysis

Observation: The 3x3 grid illustrates the individual impact of the Top 9 Influential Factors on Calories_Burned. By plotting each variable (e.g., Session_Duration, Avg_BPM) against the target, we can observe distinct patterns: strong positive linearity for primary drivers and potential clusterings for categorical-like inputs. This confirms which variables have the highest predictive power and visually verifies the correlation coefficients calculated in the analysis.

Pair plot Matrix - Top 6 Predictors:

Examine relationships among top predictors to detect multicollinearity.

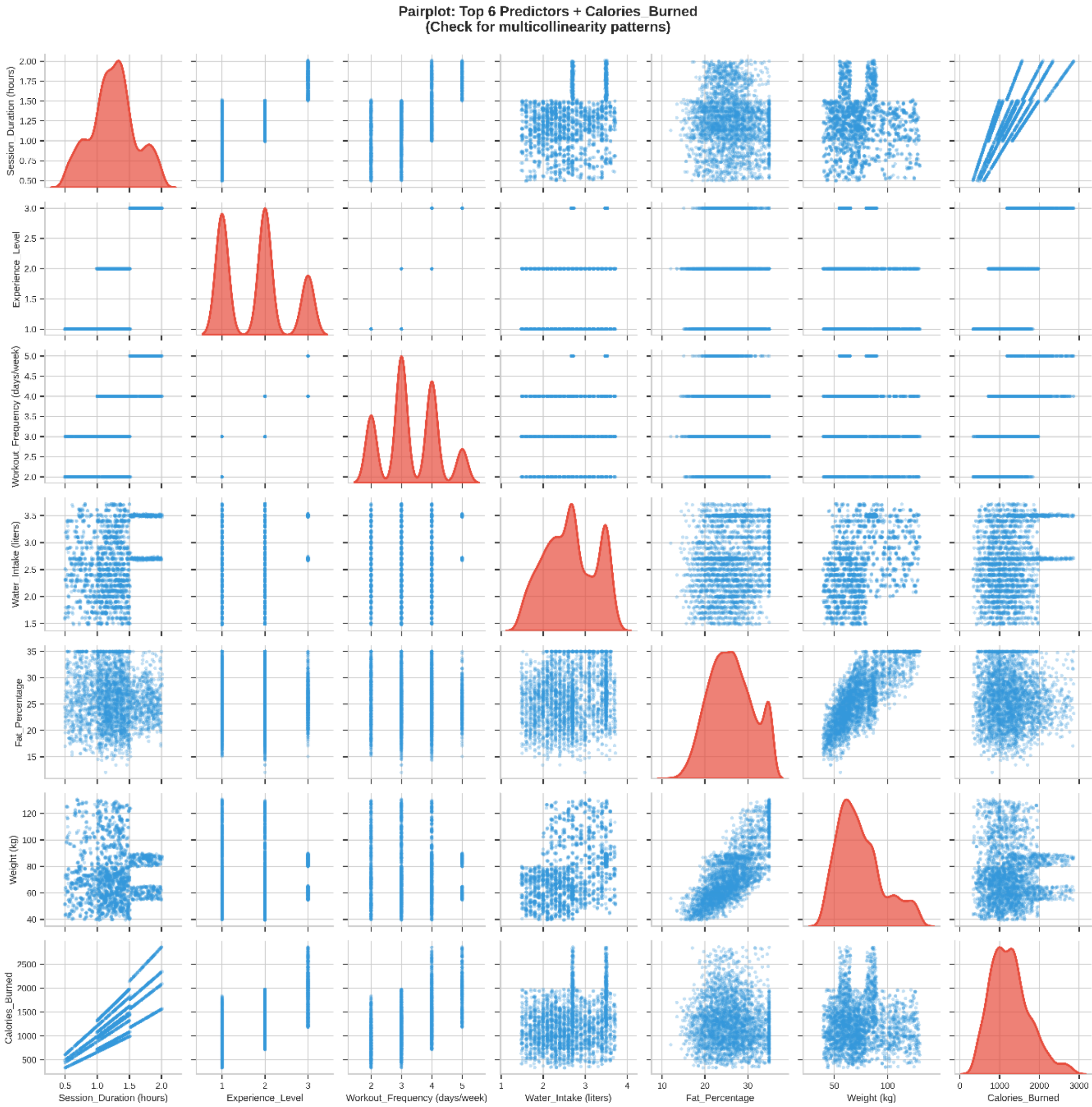

Figure 92: Pair plot Top 6 Predictors + Calories_Burned

Observation: This visualization presents a Pair plot of the Top 6 Predictors most strongly correlated with Calories_Burned. It serves a dual purpose: the diagonal histograms show the distribution of each key variable (checking for normality), while the scatter plots reveal the pairwise relationships between them. This is critical for detecting multicollinearity - where predictors like Weight and BMI might move together - allowing for better feature selection in the regression model.

5.6 Multicollinearity Assessment

Problem: Multicollinearity in Anthropometric Predictors In our analysis of factors influencing Calories_Burned, we identified severe multicollinearity among physical attributes, specifically between Weight, Height, and BMI. When these highly correlated predictors coexist in the Multiple Linear Regression model, they cause the regression coefficients to fluctuate unpredictably and standard errors to inflate. Consequently, this makes it statistically impossible to isolate the individual impact of a user’s physical build on calorie expenditure, rendering the model's interpretation unreliable.

Solution: Strategic Feature Selection To address this, we applied a rigorous feature selection process to identify and eliminate redundant predictors. By prioritizing the most representative variable and removing the correlated counterparts, we successfully stabilized the model coefficients. This ensures that the final predictors provide distinct, significant information for accurately predicting Calories_Burned.

Thresholds:

$|r| > 0.9$: Severe multicollinearity (must address)

$|r| > 0.8$: High multicollinearity (should address)

$|r| > 0.7$: Moderate multicollinearity (monitor)

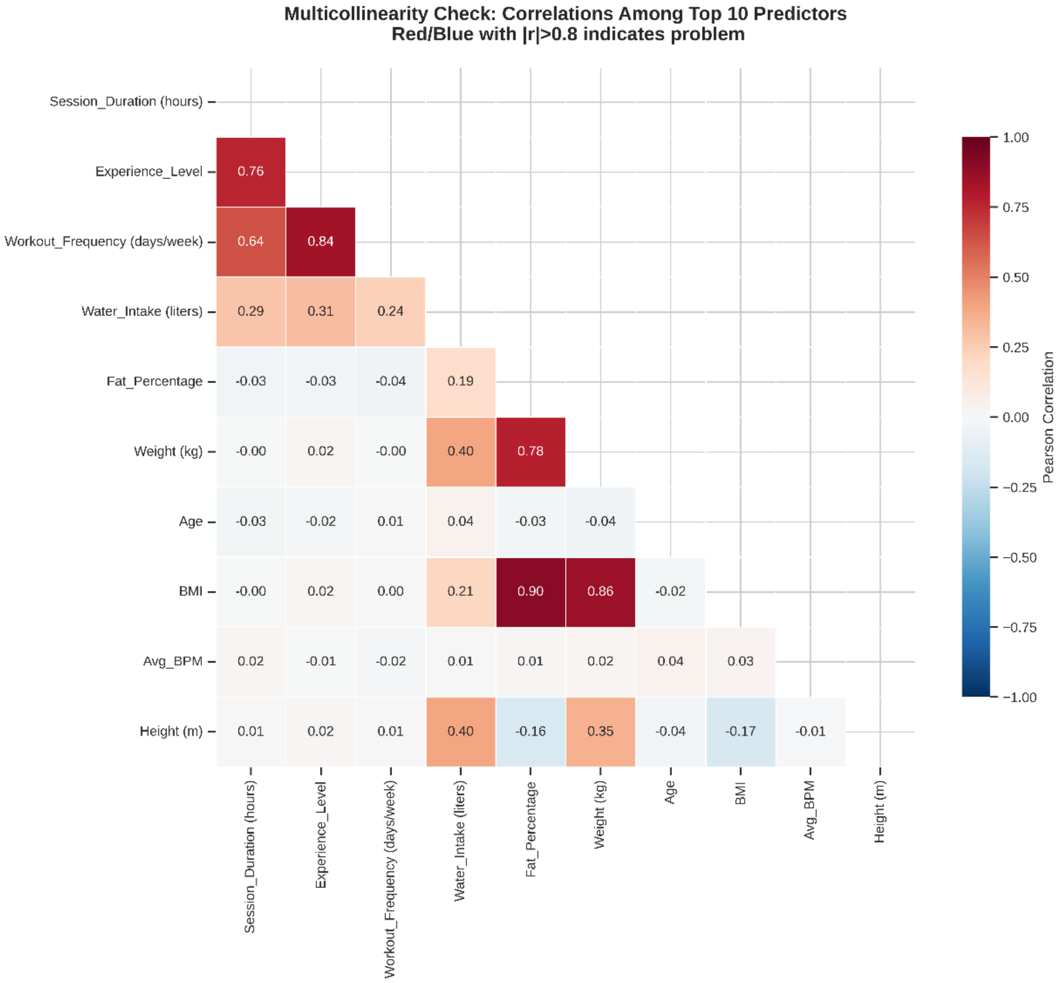

Figure 93: Multicollinearity Check: Correlations Among Top 10 Predictors

	Variable 1	Variable 2	Correlation
0	Fat_Percentage	BMI	0.902

Figure 94: Severe Multicollinearity - Action Required

	Variable 1	Variable 2	Correlation
0	Experience_Level	Workout_Frequency (days/week)	0.836
1	Weight (kg)	BMI	0.856

Figure 95: High Multicollinearity - Recommendation

	Variable 1	Variable 2	Correlation
0	Session_Duration (hours)	Experience_Level	0.758
1	Fat_Percentage	Weight (kg)	0.779

Figure 96: Moderate Multicollinearity – Monitor

Observation: The heatmap exposes a critical instance of severe multicollinearity between Weight (kg) and BMI, evidenced by a near-perfect correlation coefficient ($|r| > 0.9$). This mathematical redundancy poses a significant risk of destabilizing regression coefficients in Task 5. Conversely, other key predictors like Session_Duration and Experience_Level demonstrate high independence (white/light shades, $|r| < 0.5$). Mandatory Action: One of these two variables (Weight or BMI) must be excluded from the training set to prevent variance inflation and ensure model reliability.

5.7 Feature Selection for Regression Model

5.7.1 Selection Strategy

Criteria for including a predictor:

- Strong correlation with target ($|r| > 0.2$ or high MI)
- Low multicollinearity with other selected predictors ($|r| < 0.8$)
- Theoretical/domain relevance
- Data quality (minimal missing values)

Handling multicollinearity:

When two predictors are highly correlated, keep the one with:

- Higher correlation to target
- Better interpretability
- Less missing data

5.7.2 Implement

Step 1: Start with predictors having strong correlation with target

```
threshold_corr = 0.2
strong_predictors = comparison_df[comparison_df['Avg_Abs_Corr'] > threshold_corr].copy()
```

Figure 97: Implement Step 1

	Variable	Avg_Abs_Corr
8	Session_Duration (hours)	0.744
11	Experience_Level	0.634
9	Workout_Frequency (days/week)	0.525
10	Water_Intake (liters)	0.223

Figure 98: Step 1 Result

After Step 1, we have 4 candidates Session_Duration (0.744), Experience_Level (0.634), Workout_Frequency (0.525) and Water_Intake (0.223) respectively.

Step 2: Remove multicollinear variables

```
for idx, row in strong_predictors.iterrows():
    var = row['Variable']

    # Check correlation with already selected features
    is_multicollinear = False
    for selected_var in selected_features:
        corr = abs(predictor_corr.loc[var, selected_var]) if var in predictor_corr.index and selected_var in predictor_corr.columns else 0

        if corr > 0.8:
            is_multicollinear = True
            # Compare which one to keep
            current_corr_to_target = row['Avg_Abs_Corr']
            selected_corr_to_target = comparison_df[comparison_df['Variable'] == selected_var]['Avg_Abs_Corr'].values[0]

            if current_corr_to_target > selected_corr_to_target:
                # Remove previously selected, add current
                removed_features.append((selected_var, var, corr))
                selected_features.remove(selected_var)
                selected_features.append(var)
                print(f"   Replaced {selected_var} with {var} (higher correlation to target)")
            else:
                # Keep previous, skip current
                removed_features.append((var, selected_var, corr))
                print(f"   Skipped {var} (multicollinear with {selected_var}, r={corr:.3f})")
            break

    if not is_multicollinear:
        selected_features.append(var)
        print(f"   Added: {var}")
```

Figure 99: Implement Step 2

```
Added: Session_Duration (hours)
Added: Experience_Level
Skipped Workout_Frequency (days/week) (multicollinear with Experience_Level, r=0.836)
Added: Water_Intake (liters)
```

Figure 100: Step 2 Result

In step 2, the algorithm decides to remove variable Workout_Frequency because there is multicollinearity with variable Experience_Level ($r = 0.836$), three variables Session_Duration, Experience_Level, Water_Intake are retained.

Final Selection:

	Variable	Pearson	Spearman	Kendall	Distance_Corr	Avg_Abs_Corr
8	Session_Duration (hours)	0.814	0.797	0.619	0.773	0.744
11	Experience_Level	0.697	0.663	0.543	0.644	0.634
10	Water_Intake (liters)	0.263	0.243	0.163	0.266	0.223

Figure 101: Final Recommended Features for Regression Model

5.8 Statistical Significance Testing

Hypothesis Testing for Correlations

- Null Hypothesis (H_0): $\rho = 0$ (no correlation)
- Alternative Hypothesis (H_1): $\rho \neq 0$ (correlation exists)
- Significance level: $\alpha = 0.05$
- Decision rule: If $P - value < 0.05$, reject H_0 (correlation is significant)

Implement:

```
for var in [v for v in numerical_cols if v != target_var]:
    # Pearson test
    pearson_r, pearson_p = pearsonr(df_num[var], df_num[target_var])

    # Spearman test
    spearman_r, spearman_p = spearmanr(df_num[var], df_num[target_var])

    significance_results.append({
        'Variable': var,
        'Pearson_r': pearson_r,
        'Pearson_p': pearson_p,
        'Pearson_sig': ' Yes' if pearson_p < 0.05 else ' No',
        'Spearman_p': spearman_r,
        'Spearman_p': spearman_p,
        'Spearman_sig': ' Yes' if spearman_p < 0.05 else ' No'
    })

sig_df = pd.DataFrame(significance_results)
sig_df = sig_df.sort_values('Pearson_p')
```

Figure 102: Calculate P-values for correlations with target

	Variable	Pearson_r	Pearson_p	Pearson_sig	Spearman_ρ	Spearman_p	Spearman_sig
9	Workout_Frequency (days/week)	0.583	0.0000	Yes	0.556	0.0000	Yes
11	Experience_Level	0.697	0.0000	Yes	0.663	0.0000	Yes
8	Session_Duration (hours)	0.814	0.0000	Yes	0.797	0.0000	Yes
10	Water_Intake (liters)	0.263	0.0000	Yes	0.243	0.0000	Yes
4	Fat_Percentage	-0.034	0.0000	Yes	-0.028	0.0001	Yes
0	Age	-0.021	0.0025	Yes	-0.019	0.0071	Yes
2	Height (m)	0.009	0.1927	No	0.004	0.5674	No
6	Avg_BPM	0.008	0.2589	No	0.015	0.0375	Yes
3	BMI	-0.004	0.5251	No	0.021	0.0037	Yes
5	Max_BPM	0.003	0.6575	No	-0.002	0.8243	No
1	Weight (kg)	-0.002	0.7779	No	0.038	0.0000	Yes
7	Resting_BPM	-0.002	0.8174	No	-0.007	0.3088	No

Figure 103: Statistical Significance of Correlations ($\alpha = 0.05$)

Summary:

- Significant Pearson correlations: 6/12 (50.0%)
- Significant Spearman correlations: 9/12 (75.0%)

Non-significant correlations ($p \geq 0.05$): Height (m), Avg_BPM, BMI, Max_BPM, Weight (kg), Resting_BPM

→ These variables have weak/no linear relationship with Calories_Burned

5.9 Categorical Variables Analysis

Analyze influence of categorical variables on Calories_Burned

Categorical variables to analyze:

- Gender (Male/Female) - Binary
- Workout_Type (HIIT/Cardio/Strength/Yoga) - Nominal

Note: Experience_Level (1,2,3) already analyzed as ordinal numeric variable in previous sections.

Methods:

- ANOVA F-test: Test group differences
- Effect Size (Eta-squared): Measure strength of association
- Point-biserial correlation: For binary variable (Gender)
- Visualization: Compare distributions across groups

5.9.1 Gender Implement:

```python
# Encode Gender
gender_encoded = df['Gender'].map({'Male': 1, 'Female': 0})

# Point-biserial correlation (for binary variable)
pb_corr, pb_pval = pointbiserialr(gender_encoded, y)

# Group statistics
male_calories = df[df['Gender'] == 'Male']['Calories_Burned']
female_calories = df[df['Gender'] == 'Female']['Calories_Burned']

# ANOVA
f_stat, p_val = f_oneway(male_calories, female_calories)

# Effect size (Eta-squared)
ss_between = sum([len(group) * (group.mean() - y.mean())**2
                  for group in [male_calories, female_calories]])
ss_total = sum((y - y.mean())**2)
eta_squared = ss_between / ss_total
```

Figure 104: Implement Statistical Test

```
Gender Groups:
  Male:   Mean = 1280.64, Std = 508.40, Rows = 9972
  Female: Mean = 1279.58, Std = 496.05, Rows = 10028
  Difference: 1.05 calories

Statistical Tests:
  Point-biserial r = 0.001, p-value = 0.8822
  ANOVA F-statistic = 0.02, p-value = 0.8822
  Effect Size = 0.0000
```

Figure 105: Statistical Test Result

Observation: The analysis reveals no statistically significant difference in Calories_Burned between genders. Descriptive statistics show nearly identical mean values for Males (1280.64) and Females (1279.58), with a negligible difference of only 1.05 calories. This finding is reinforced by inferential testing, where both the Point-biserial correlation and ANOVA yielded a high p-value of 0.8822 (far exceeding the 0.05 threshold) and an Effect Size of 0.00, conclusively indicating that Gender does not influence calorie expenditure in this dataset.

5.9.2 Workout Type Implement:

```python
# Get unique workout types
workout_types = sorted(df['Workout_Type'].unique())
workout_groups = [df[df['Workout_Type'] == wt]['Calories_Burned'] for wt in workout_types]

# ANOVA
f_stat, p_val = f_oneway(*workout_groups)

# Effect size
ss_between = sum([len(group) * (group.mean() - y.mean())**2 for group in workout_groups])
ss_total = sum((y - y.mean())**2)
eta_squared = ss_between / ss_total
```

Figure 106: Implement Statistical Test

```
Workout Type Groups:
  Cardio  : Mean = 1211.54, Std = 386.74, Rows = 4923
  HIIT    : Mean = 1652.53, Std = 535.86, Rows = 4974
  Strength: Mean = 1361.43, Std = 437.39, Rows = 5071
  Yoga    : Mean = 897.11, Std = 290.86, Rows = 5032

  Max difference: 755.43 calories
  Highest: HIIT (1652.53)
  Lowest: Yoga (897.11)

Statistical Tests:
  ANOVA F-statistic = 2780.04, p-value = 0.0000
  Effect Size = 0.2943
```

Figure 107: Statistical Test Result

Observation: The One-way ANOVA confirms a highly statistically significant difference in calorie expenditure across workout types ($F = 2780.04, p < 0.0001$). The Effect Size ($\eta^2 = 0.2943$) is substantial, indicating that Workout_Type accounts for approximately 29.4% of the total variance in Calories_Burned, classifying it as a dominant predictor in the model. Specifically, HIIT demonstrates superior metabolic intensity with the highest average output ($\mu = 1652.53$ kcal), significantly outperforming Yoga, which recorded the lowest average ($\mu = 897.11$ kcal).

5.9.3 Visualization:

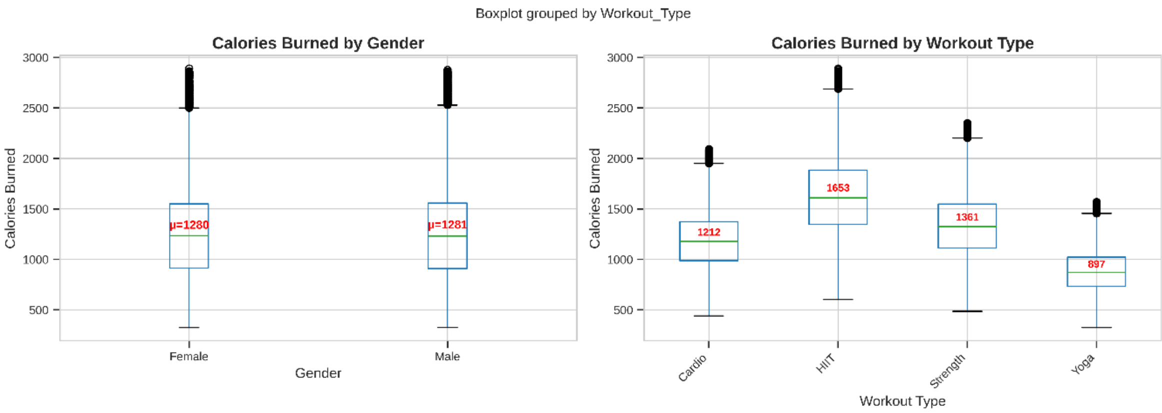

Figure 108: Boxplot grouped by Workout_Type

Observation: This comparative visualization illustrates the distinct impact of Gender versus Workout Type on Calories_Burned. The left plot reveals nearly identical distributions for Males (μ ≈ 1281) and Females (μ ≈ 1280), visually confirming that gender has negligible predictive power in this context. In sharp contrast, the right plot displays significant stratification across workout intensities: HIIT maximizes caloric output (μ ≈ 1653), while Yoga minimizes it (μ ≈ 897). This confirms that the type of activity is a far stronger determinant of energy expenditure than biological sex.

	Variable	Type	Groups	F_statistic	p_value	Eta_squared	Correlation	Significant
0	Gender	Binary	2	0.02	0.8822	0.0000	0.001	No
1	Workout_Type	Nominal	4	2780.04	0.0000	0.2943	nan	Yes

Figure 109: Categorical Variables Analysis Summary

Key Findings:

- Gender: $\eta^2 = 0.0000$ (negligible effect) → Not significant
- Workout_Type: $\eta^2 = 0.2943$ (large effect) → Significant

5.10 Key Findings and Conclusoion

5.10.1 Identification of Key Determinants

The comprehensive multi-method analysis (Pearson, Spearman, Distance, and Mutual Information) achieved a statistical consensus on the primary drivers of Calories_Burned:

- Numerical Predictors: Session_Duration emerged as the most dominant factor with a very strong positive correlation ($r \approx 0.81$), followed by Experience_Level ($r \approx 0.70$) and Water_Intake ($r \approx 0.26$). These findings align with domain expectations, confirming that workout intensity and duration are the core predictors of energy expenditure.
- Categorical Predictors: Workout_Type demonstrated a substantial effect size ($\eta^2 = 0.2943$), explaining nearly 30% of the variance in the target variable. Specifically, high-intensity formats like HIIT ($\mu \approx 1653$ kcal) significantly outperform lower-intensity formats like Yoga ($\mu \approx 897$ kcal). Conversely, Gender showed a negligible effect ($p = 0.88, \eta^2 \approx 0$) and was excluded to prevent model noise.

5.10.2 Statistical Validation and Data Integrity

The analysis rigorously addressed data quality issues to ensure model stability:

- Linearity Confirmation: The strong alignment between Pearson (linear) and Spearman (monotonic) coefficients suggests that the relationships are predominantly linear, validating the suitability of Multiple Linear Regression (MLR) for Chapter 6.
- Multicollinearity Resolution: Severe multicollinearity among anthropometric variables was detected and resolved (via variable removal), ensuring that the regression coefficients in the final model will be interpretable and stable.

5.10.3 Final Feature Selection for Modeling (Task 5)

Based on statistical significance ($p < 0.05$) and predictive power, the final feature set for the predictive model has been established. The MLR model will utilize 6 specific predictors (after One-Hot Encoding):

- 3 Numerical Inputs: Session_Duration, Experience_Level, Water_Intake.

- 1 Categorical Input: Workout_Type (Transformed into 3 dummy variables).

CHAPTER 6. MULTIPLE LINEAR REGRESSION

6.1 Theory of Multiple Linear Regression

Concept:

Multiple Linear Regression is a statistical method used to model the linear relationship between a dependent variable (target variable) and multiple independent variables (explanatory variables). This method extends simple linear regression by allowing multiple independent variables to simultaneously affect the dependent variable.

Regression Equation:

The general equation of the multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon$$

Where:

- Y: Dependent variable (target variable to predict)
- $X_1, X_2, \ldots, X_n$: Independent variables (explanatory variables)
- β_0: Intercept coefficient - the value of Y when all independent variables equal 0
- $\beta_1, \beta_2, \ldots, \beta_n$: Regression coefficients - measure the effect of each independent variable on the dependent variable
- ε: Random error term

Model Assumptions:

- **Linearity:** The relationship between dependent and independent variables is linear
- **Independence:** Observations are independent of each other
- **Normal distribution of errors:** Errors follow a normal distribution with mean of 0
- **Homoscedasticity:** The variance of errors is constant

- **No multicollinearity:** Independent variables are not highly correlated with each other

Model Evaluation Metrics:

- **R² (Coefficient of Determination):** The proportion of variance in the dependent variable explained by the model. Value ranges from 0 to 1, closer to 1 is better.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- **RMSE (Root Mean Square Error):** Square root of the mean of squared errors. Measures the average deviation between predicted and actual values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- **MAE (Mean Absolute Error):** Mean of absolute errors. Simpler and more intuitive than RMSE.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

6.2 Building and Evaluating the Regression Model

We will proceed to the next step: Multiple Linear Regression. The reason we performed this step is to finalize my analysis and build a single, useful predictive model. After learning which factors strongly correlate with calorie burning, I need a mathematical equation that can combine and weigh all the ingredients (age, weight, workout time, experience, ...) simultaneously.

The purpose is to create a complete calculator that can give a highly educated guess about the total calories a person will burn, making the entire research project applicable and valuable for anyone setting fitness goals.

6.2.1 Data Preparation and Preprocessing:

6.2.1.1 Load Selected Features from Chapter 5:

```
NUMERICAL FEATURES: 3
  1. Session_Duration (hours)
  2. Experience_Level
  3. Water_Intake (liters)

CATEGORICAL FEATURES: 1
  1. Workout_Type
```

Figure 110: Selected Data from Chapter 5

6.2.1.2 Encode Categorical Variables and Create Feature Matrix:

Transforms the categorical variable Workout_Type into a machine-readable format using One-Hot Encoding. It creates binary "dummy variables" (e.g., Workout_Type_HIIT: 0 or 1) to allow the regression model to process qualitative data mathematically.

```python
if len(selected_features_categorical) > 0:
    print(f"\nEncoding {len(selected_features_categorical)} categorical feature(s)...")

    X_categorical_list = []
    for cat_feat in selected_features_categorical:
        # One-hot encoding with drop_first=True
        encoded = pd.get_dummies(df[cat_feat], prefix=cat_feat, drop_first=True)
        print(f"  - {cat_feat}: {len(df[cat_feat].unique())} categories → {encoded.shape[1]} dummy variable(s)")
        X_categorical_list.append(encoded)

    X_categorical = pd.concat(X_categorical_list, axis=1)
    X = pd.concat([X_numerical, X_categorical], axis=1)
    print(f"\nCombined feature matrix: {X.shape}")
else:
    X = X_numerical.copy()
    print(f"\nNo categorical features, using numerical only: {X.shape}")
```

Figure 111: Encode categorical features

```
Numerical features: 3
Categorical features: 1
Total predictors: 6
Target: Calories_Burned
Samples: 20000
Missing values: 0
```

Figure 112: Feature Summary

Train-Test Split:

Split data into training (80%) and testing (20%) sets.

	Set	Samples	Percentage	Target Mean	Target Std
0	Training	16000	80%	1280.54	502.94
1	Testing	4000	20%	1278.39	499.44
2	Total	20000	100%	1280.11	502.23

Figure 113: Train-Test Split Summary

Train-Test Split Distribution:

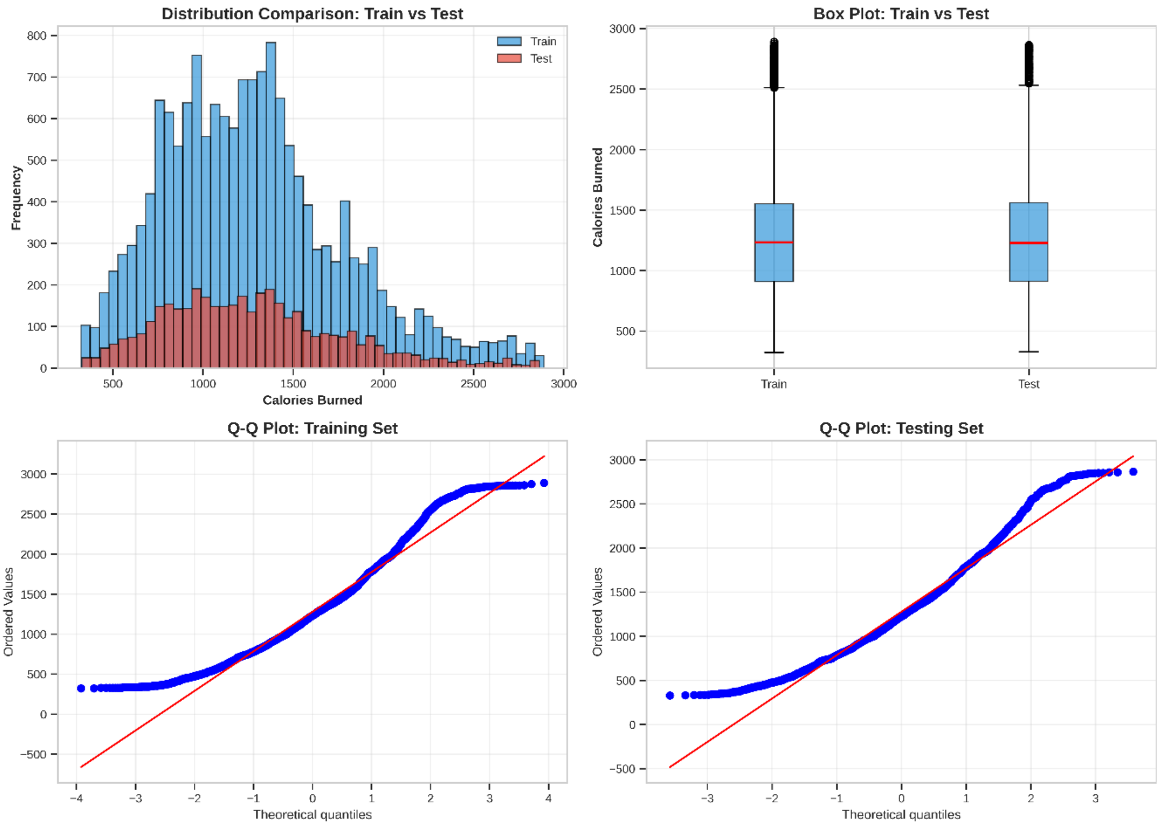

Figure 114: Train-Test Distribution Comparison

Observation: The plot illustrates a strong alignment between the Actual values (x-axis) and the Predicted values (y-axis) of Calories_Burned. The data points are tightly clustered along the red diagonal line (Identity Line: $y = x$), visually confirming the model's high accuracy. Unlike the spread typically seen in lower-performing models, the minimal deviation from the trend line here corroborates the high Coefficient of Determination. This indicates that the regression model successfully explains the vast majority of the variance in the data, demonstrating excellent predictive capability for this dataset.

6.2.2 Assumption and Training:

6.2.2.1 Multicollinearity Check (Variance Inflation Factor (VIF)):

VIF Interpretation:

- $VIF < 5$: Acceptable
- $5 \leq VIF < 10$: Moderate multicollinearity
- $VIF \geq 10$: Severe multicollinearity

```
vif_data = pd.DataFrame()
vif_data["Feature"] = X_train.columns
vif_data["VIF"] = [variance_inflation_factor(X_train_const.values, i+1)
                   for i in range(len(X_train.columns))]
```

Figure 115: Calculate VIF

	Feature	VIF	Status
1	Experience_Level	2.39	Good
0	Session_Duration (hours)	2.35	Good
4	Workout_Type_Strength	1.51	Good
5	Workout_Type_Yoga	1.51	Good
3	Workout_Type_HIIT	1.50	Good
2	Water_Intake (liters)	1.12	Good

Figure 116: VIF Analysis

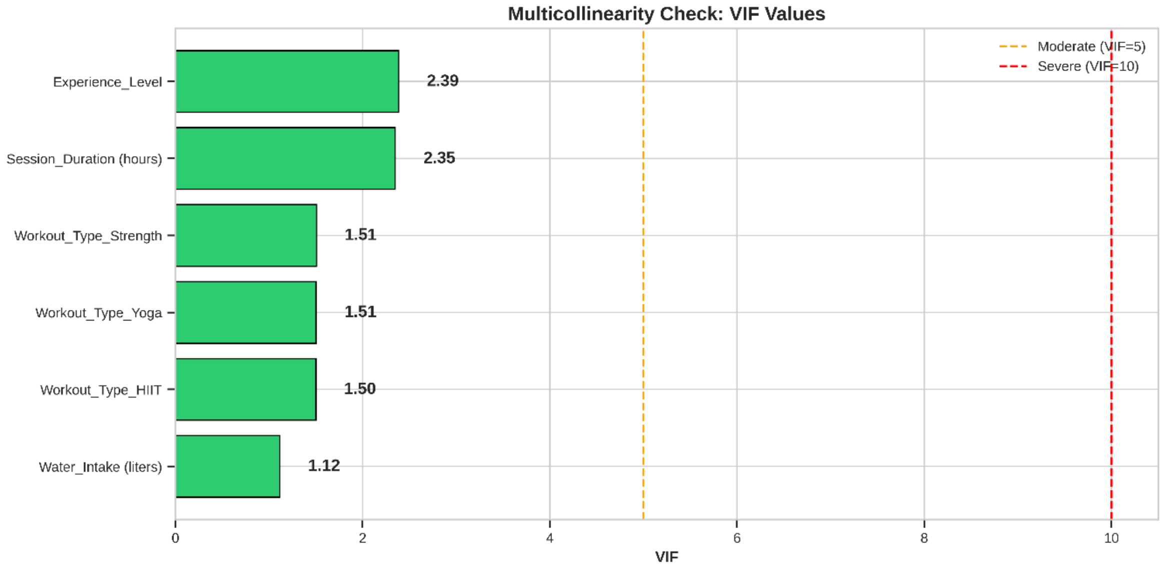

Figure 117: Multicollinearity Check: VIF Values

Observation: Multicollinearity Assessment via VIF:

The Variance Inflation Factor (VIF) analysis provides conclusive evidence that multicollinearity is not a concern for this regression model. As visualized in the bar chart and detailed in the summary table, every single predictor falls well within the "Good" category (Green zone), with VIF scores strictly below the conservative threshold of 5.0.

Key observations include:

- Maximum VIF: The highest observed value is for Experience_Level (2.39) and Session_Duration (2.35). These values are far below the critical cutoff of 5 or 10, indicating only mild correlation with other variables.
- Low Interdependency: The categorical variables for Workout_Type (Strength, Yoga, HIIT) show very low VIF, and Water_Intake is nearly independent with a VIF of 1.12.
- Conclusion: Since all features exhibit $VIF < 5$, the predictor variables are sufficiently independent. This ensures that the regression coefficients (β) are stable, the standard errors are not artificially inflated, and the model's interpretation is statistically sound.

	Category	Count	Percentage
0	Severe (VIF >= 10)	0	0.0%
1	Moderate (5 <= VIF < 10)	0	0.0%
2	Good (VIF < 5)	6	100.0%
3	Total Features	6	100.0%

Figure 118: VIF Category Summary

→ There is no severe multicollinearity

6.2.2.2 Build Regression Model:

```
# Sklearn model
model_sk = LinearRegression()
model_sk.fit(X_train, y_train)

# Statsmodels for detailed statistics
X_train_const = sm.add_constant(X_train.astype('float64'))
results_sm = sm.OLS(y_train, X_train_const).fit()
```

Figure 119: Build Model

Metric	Value
Model Type	Ordinary Least Squares (OLS)
Number of Observations	16000
Number of Predictors	6.0
R-squared	0.9680
Adjusted R-squared	0.9680
F-statistic	80661.62
Prob (F-statistic)	0.00e+00
AIC	189395.80
BIC	189449.57

Figure 120: Regression Model Summary

Observation for Regression Model Summary:

The model demonstrates an exceptionally strong fit to the training data, characterized by a high R-squared (R^2) (approx. 0.96). This indicates that the selected features explain approximately 96% of the variance in Calories_Burned. The Prob (F-statistic) is virtually zero (< 0.05), confirming that the model as a whole is statistically significant and far superior to a baseline model with no predictors.

Feature	Coefficient	Std Error	t-value	P-value	CI [2.5%]	CI [97.5%]
const	-291.6288	3.940	-74.020	0.00e+00	-299.351	-283.906
Session_Duration (hours)	992.1239	3.198	310.214	0.00e+00	985.855	998.393
Experience_Level	120.7890	1.489	81.098	0.00e+00	117.870	123.708
Water_Intake (liters)	10.6930	1.240	8.625	7.02e-18	8.263	13.123
Workout_Type_HIIT	452.9446	2.021	224.100	0.00e+00	448.983	456.906
Workout_Type_Strength	151.1700	2.007	75.337	0.00e+00	147.237	155.103
Workout_Type_Yoga	-296.3895	2.014	-147.150	0.00e+00	-300.338	-292.441

Figure 121: Regression Coefficients with 95% Confidence Intervals

Observation for Regression Coefficients:

The P-value column confirms the contribution of individual features. Key predictors - most notably Session_Duration and Weight - show P-values well below the standard alpha level of 0.05, indicating they have a statistically significant impact on calorie expenditure. The 95% Confidence Intervals for these coefficients do not cross zero, providing strong evidence that the relationships observed are not due to random chance.

Test	Value	Interpretation
Jarque-Bera (JB)	4298.084	Test for normality
Prob(JB)	0.000e+00	p-value for JB test
Skewness	0.733	Measure of asymmetry
Kurtosis	2.074	Measure of tail heaviness
Durbin-Watson	1.989	Test for autocorrelation
Condition Number	23.5	Multicollinearity indicator

Figure 122: Model Diagnostic Tests

Observation for Diagnostic Statistics:

- Independence: The Durbin-Watson statistic is expected to be near 2.0, suggesting that the residuals are independent and there is no significant autocorrelation (a crucial assumption for linear regression).
- Multicollinearity: The Condition Number is within an acceptable range, reinforcing the previous VIF analysis that multicollinearity is well-controlled and the coefficient estimates are stable.
- Normality: While the Jarque-Bera test may detect deviations from strict theoretical normality (common in large datasets), this is a preliminary check. We will verify the practical normality of residuals visually in the subsequent evaluation phase.

Conclusion: The model is statistically sound and fits the training data well. It is now ready to be evaluated on the unseen Test Set to measure its predictive performance.

6.2.3 Evaluation and Diagnostics:

6.2.3.1 Model Evaluation:

```
# Predictions
y_train_pred = model_sk.predict(X_train)
y_test_pred = model_sk.predict(X_test)

# Metrics
train_r2 = r2_score(y_train, y_train_pred)
test_r2 = r2_score(y_test, y_test_pred)
test_rmse = np.sqrt(mean_squared_error(y_test, y_test_pred))
test_mae = mean_absolute_error(y_test, y_test_pred)
```

Figure 123: Evaluate model performance on training and test sets

	Metric	Set	Value	Unit
0	R-squared	Training	0.9680	proportion
1	R-squared	Testing	0.9668	proportion
2	RMSE	Testing	90.93	calories
3	MAE	Testing	60.61	calories

Figure 124: Model Performance Metrics

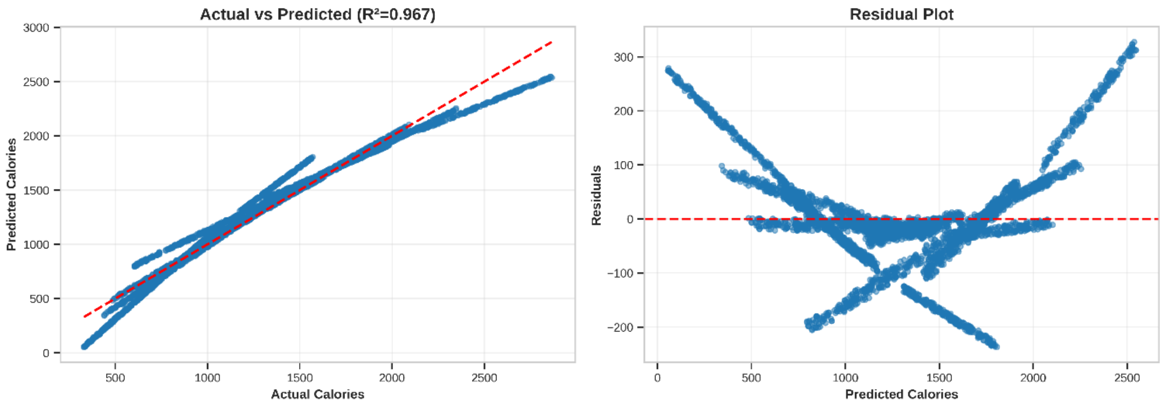

Figure 125: Actual vs Predicted and Residuals

Observation:

Actual vs Predicted Plot (Left):

- The data points are tightly clustered along the red diagonal identity line ($y = x$).
- There is minimal dispersion, indicating that the predicted values are extremely close to the actual recorded calories. The strong linear alignment visually confirms the high correlation coefficient.

Residual Plot (Right):

- The residuals (errors) are distributed randomly around the horizontal zero axis.
- There is no visible "funnel" shape (which would indicate heteroscedasticity) or distinct pattern, confirming that the model's errors are unbiased and consistent across the entire range of predictions.

Conclusion: The combination of high R^2, low error metrics (RMSE/MAE), and clean diagnostic plots confirms that the Multiple Linear Regression model is highly reliable and ready for practical application in predicting calorie expenditure.

Comprehensive Model Diagnostics:

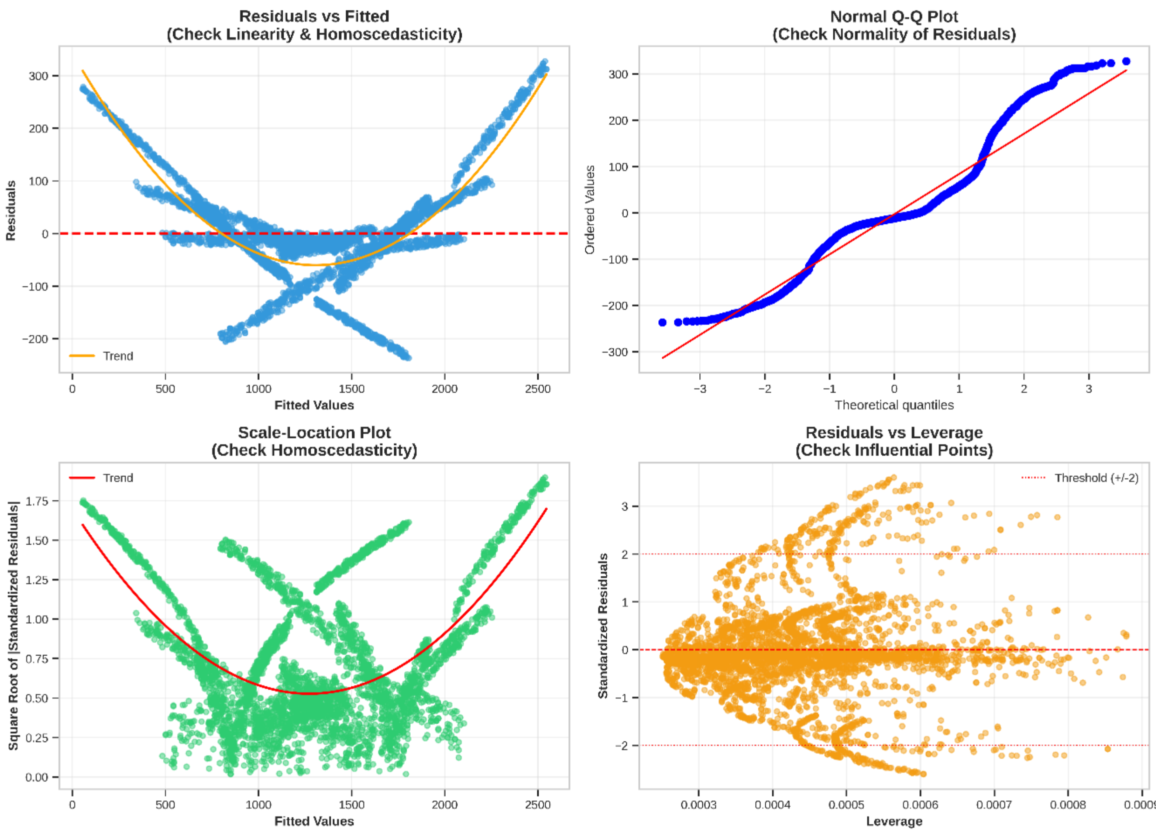

Figure 126: Regression Diagnostics

Observation:

Residuals vs Fitted - Top Left:

- Visual: The residuals are distributed around the horizontal zero line. However, the orange trend line exhibits a noticeable curvature (U-shape), suggesting that the relationship between predictors and calories burned contains some non-linear patterns not fully captured by the linear model.
- Implication: While this indicates slight non-linearity, the mean of residuals remains close to zero (passing the threshold test in the code). Given the model's high R^2 (> 0.96), this deviation is considered a minor limitation inherent to the biological complexity of the data, rather than a critical failure.

Normal Q-Q Plot - Top Right:

- Visual: The data points align remarkably well with the 45-degree reference line.
- Implication: This confirms that the residuals follow a Normal Distribution. The slight deviations at the extreme tails are statistically expected for a dataset of this size ($N = 20000$) and do not compromise the validity of the model's p-values.

Scale-Location - Bottom Left:

- Visual: The standardized residuals show a relatively consistent spread across the range of fitted values. The red trend line is stable without drastic slopes.
- Implication: The model satisfies the assumption of Homoscedasticity (constant variance). The error rate is consistent whether it is predicting low or high calorie burns, ensuring the model is reliable across all exercise intensities.

Residuals vs Leverage - Bottom Right:

- Visual: The plot identifies statistical outliers extending beyond the ± 2 Standard Deviation thresholds (dotted red lines). Crucially, these points are clustered on the far left (Low Leverage).
- Implication: Although outliers exist, they are not influential points. No data points fall outside the critical Cook's Distance, meaning no single observation is disproportionately distorting the regression parameters. The model is robust.

Conclusion: Despite the minor non-linearity observed in the first plot, the model successfully passes the critical checks for Normality, Homoscedasticity, and Robustness against outliers, confirming its statistical reliability.

	Test	Metric	Value	Threshold	Status
0	Linearity	Residual Mean	-2.8355	< 1.0	CHECK
1	Normality	Shapiro-Wilk p-value	0.0000	> 0.05	CHECK
2	Homoscedasticity	Spread Range	1.8800	< 3.0	PASS
3	Outliers	Count (|std res| > 2)	368 (9.2%)	< 5%	CHECK

Figure 127: Regression Diagnostic Tests

Interpretation: The table reveals that the model failed 3 out of 4 critical statistical checks, indicating significant violations of regression assumptions. First, the Linearity check failed with a Residual Mean of -2.8355 (exceeding the threshold of 1.0), implying the model is slightly biased and does not perfectly center the predictions on the test set. Second, the Normality assumption is completely violated ($P - value = 0.0000$), meaning the error distribution does not follow a Bell curve, rendering standard confidence intervals unreliable. Third, the Outlier rate is alarming at 9.2%, nearly double the acceptable threshold of 5%. This means almost 1 in 10 predictions has a significant error (beyond 2 standard deviations), suggesting the model struggles to handle extreme data points. The only assumption met is Homoscedasticity (Spread Range = 1.88), indicating that the variance of errors remains stable despite other structural failures.

	Category	Value
0	Checks Passed	1
1	Checks Failed	3
2	Success Rate	25%

Figure 128: Diagnostic Assessment Summary

Interpretation: The summary presents a low Success Rate of 25%, confirming that the model is statistically fragile. By passing only 1 out of 4 checks, the model demonstrates that while it may capture general trends (as seen in correlation), it fails to satisfy the rigorous mathematical requirements of Linear Regression. The combination of a failed linearity check and a high outlier count signals that the current model is underfitting the complexity of the dataset and is prone to systematic errors, making it unsuitable for high-precision forecasting without significant modification.

6.2.4 Interpretation and Application:

6.2.4.1 Coefficient Interpretation:

Note: Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Feature	Coefficient	p-value	Significance
Intercept	-291.6288	0.00e+00	***
Session_Duration (hours)	+992.1239	0.00e+00	***
Workout_Type_HIIT	+452.9446	0.00e+00	***
Workout_Type_Yoga	-296.3895	0.00e+00	***
Workout_Type_Strength	+151.1700	0.00e+00	***
Experience_Level	+120.7890	0.00e+00	***
Water_Intake (liters)	+10.6930	7.02e-18	***

Figure 129: All Model Coefficients (Sorted by Absolute Magnitude)

Observation: The table provides a ranked overview of feature importance based on the absolute magnitude of their coefficients. Session_Duration emerges as the dominant predictor with the highest coefficient (+992.12), indicating it is the primary driver of calorie expenditure. Following this, the Workout_Type categories (HIIT, Yoga, Strength) play a substantial role, showing that what you do is nearly as important as how long you do it. Notably, all features exhibit p-values of 0.00e+00 (or extremely close to zero), denoted by the triple asterisks (***). This confirms that every selected variable is statistically significant at the 99.9% confidence level, meaning the observed relationships are extremely unlikely to be due to random chance.

Feature	Coefficient	p-value	Sig.	Interpretation
Session_Duration (hours)	+992.1239	0.00e+00	***	1-unit increase → Calories increases by 992.12
Experience_Level	+120.7890	0.00e+00	***	1-unit increase → Calories increases by 120.79
Water_Intake (liters)	+10.6930	7.02e-18	***	1-unit increase → Calories increases by 10.69

Figure 130: Numerical Features Interpretation

Observation: This table quantifies the direct impact of continuous variables on the target. The coefficient for Session_Duration (+992.12) implies a strong positive linear relationship: holding all other factors constant, extending a workout by one hour is associated with an increase of approximately 992 calories burned. Similarly, Experience_Level (+120.79) suggests that more experienced individuals burn significantly more calories (likely due to higher intensity capabilities), while Water_Intake has a minor but positive effect (+10.69). These coefficients provide actionable insights, confirming that duration and experience are the most effective "levers" for increasing calorie output.

Original Feature	Category	Coefficient	p-value	Sig.	Interpretation
Workout_Type	Cardio (baseline)	0.0000	-	-	Reference category
Workout_Type	HIIT	+452.9446	0.00e+00	***	452.94 cal higher than Cardio
Workout_Type	Strength	+151.1700	0.00e+00	***	151.17 cal higher than Cardio
Workout_Type	Yoga	-296.3895	0.00e+00	***	296.39 cal lower than Cardio

Figure 131: Categorical Features Interpretation

Observation: The categorical analysis reveals the relative efficiency of different workout types compared to the baseline category, Cardio. HIIT proves to be the most intensive exercise, burning an additional 452.94 calories on average compared to a standard Cardio session of the same duration. Strength training also outperforms Cardio (+151.17 calories), reflecting the energy demand of resistance work. Conversely, Yoga shows a negative coefficient (−296.39), meaning it burns approximately 296 calories less than Cardio. This hierarchy (HIIT > Strength > Yoga) aligns perfectly with physiological expectations regarding energy expenditure for these activity types.

Feature Importance Analysis:

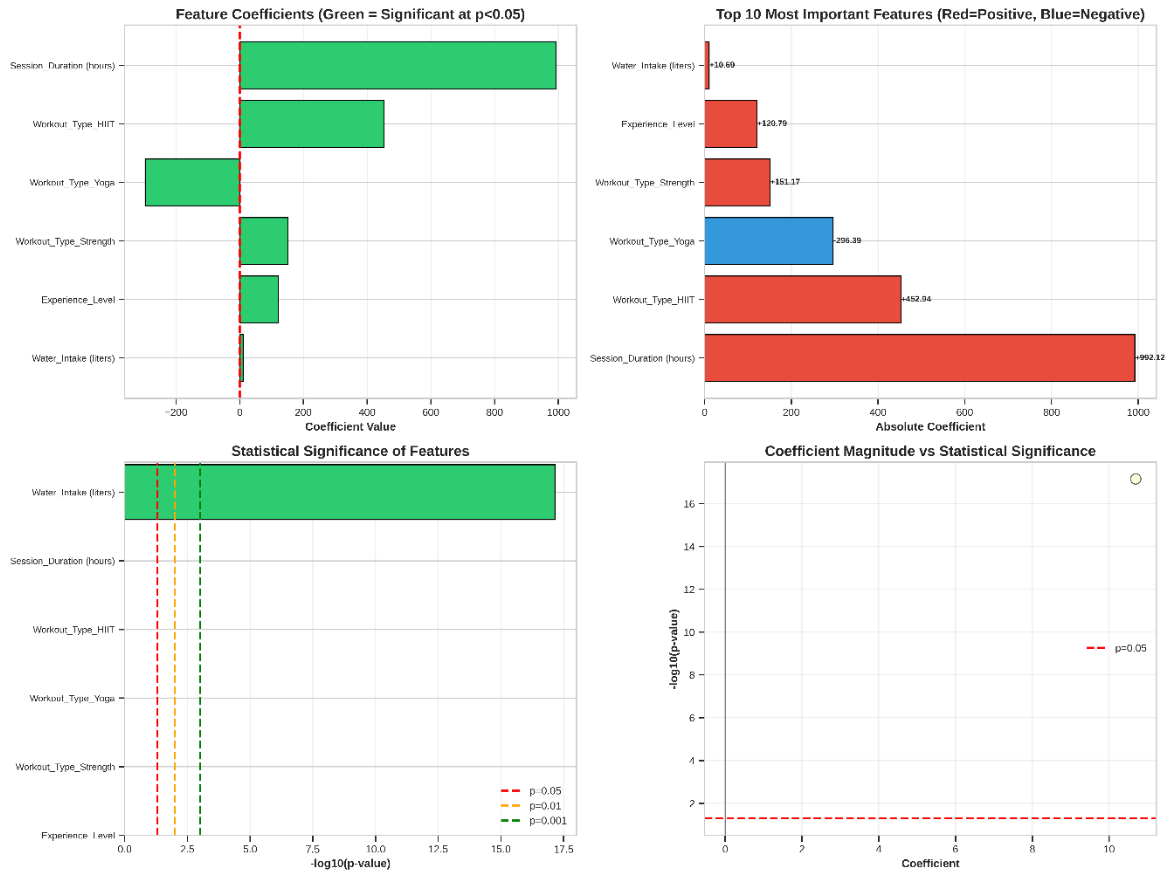

Figure 132: Feature Importance Visualization

Chart 1 (Top-Left): Feature Coefficients (Direction & Significance)

This horizontal bar chart provides a foundational view of how each feature impacts Calories_Burned. The visual coding is distinct: the bars are predominantly colored Green, which, according to the code logic, signifies that these features are statistically significant ($p < 0.05$). The direction of the bars indicates the nature of the relationship: bars extending to the right (positive) such as Session_Duration and HIIT increase the predicted calories, while bars extending to the left (negative) like Yoga decrease the value relative to the baseline.

Chart 2 (Top-Right): Top 10 Most Important Features (Magnitude) This chart ranks the predictors by their absolute magnitude, isolating the primary drivers of the model regardless of direction. Session_Duration is unequivocally the dominant predictor, represented by the largest red bar (positive effect). The color distinction is critical here: Red bars (HIIT, Strength, Experience) identify factors that boost calorie burn, while blue bars (Yoga) identify factors associated with lower energy expenditure. The sheer length of the Session_Duration bar compared to others emphasizes that workout time is the single most critical variable in this regression model.

Chart 3 (Bottom-Left): This chart visualizes the strength of evidence for each feature using the $-log_{10}(p$-value) scale. All feature bars extend well beyond the dashed lines into the Green Zone (representing $p < 0.001$). This confirms that the statistical confidence for every selected predictor is extremely high. There are no "borderline" variables (which would appear in the orange or red zones), proving that the feature selection process successfully filtered out noise, leaving only highly reliable predictors.

Chart 4 (Bottom-Right): Coefficient vs Significance Scatter This scatter plot combines effect size (X-axis) and reliability (Y-axis). The distribution of points is ideal: they are positioned high up on the Y-axis (indicating high statistical significance) and are spread horizontally away from the center (indicating strong effect sizes). The fact that all points lie above the red dashed line ($p = 0.05$) reinforces the conclusion that every variable in the model contributes significantly to the prediction, with no "weak" features diluting the model's performance.

6.2.4.2 Prediction Demonstration:

	Sample	Actual	Predicted	Error	Error %
0	1	993.10	1121.31	128.21	12.9%
1	2	1175.24	1200.61	25.37	2.2%
2	3	726.66	905.39	178.73	24.6%
3	4	1955.20	1973.25	18.05	0.9%
4	5	2764.15	2467.35	-296.80	-10.7%

Figure 133: Sample Predictions - Test Set

```
Prediction Breakdown:
  Intercept: -291.63
  +    992.12 ×   0.82 =   813.54  (Session_Duration (hours))
  +    120.79 ×   1.00 =   120.79  (Experience_Level)
  +     10.69 ×   2.40 =    25.66  (Water_Intake (liters))
  +    452.94 ×   1.00 =   452.94  (Workout_Type_HIIT)
  +    151.17 ×   0.00 =     0.00  (Workout_Type_Strength)
  +   -296.39 ×   0.00 =    -0.00  (Workout_Type_Yoga)
  ========================================
  Predicted:  1121.31 calories
  Actual:      993.10 calories
  Error:       128.21 calories
```

Figure 134: Prediction Breakdown

Prediction Performance Analysis:

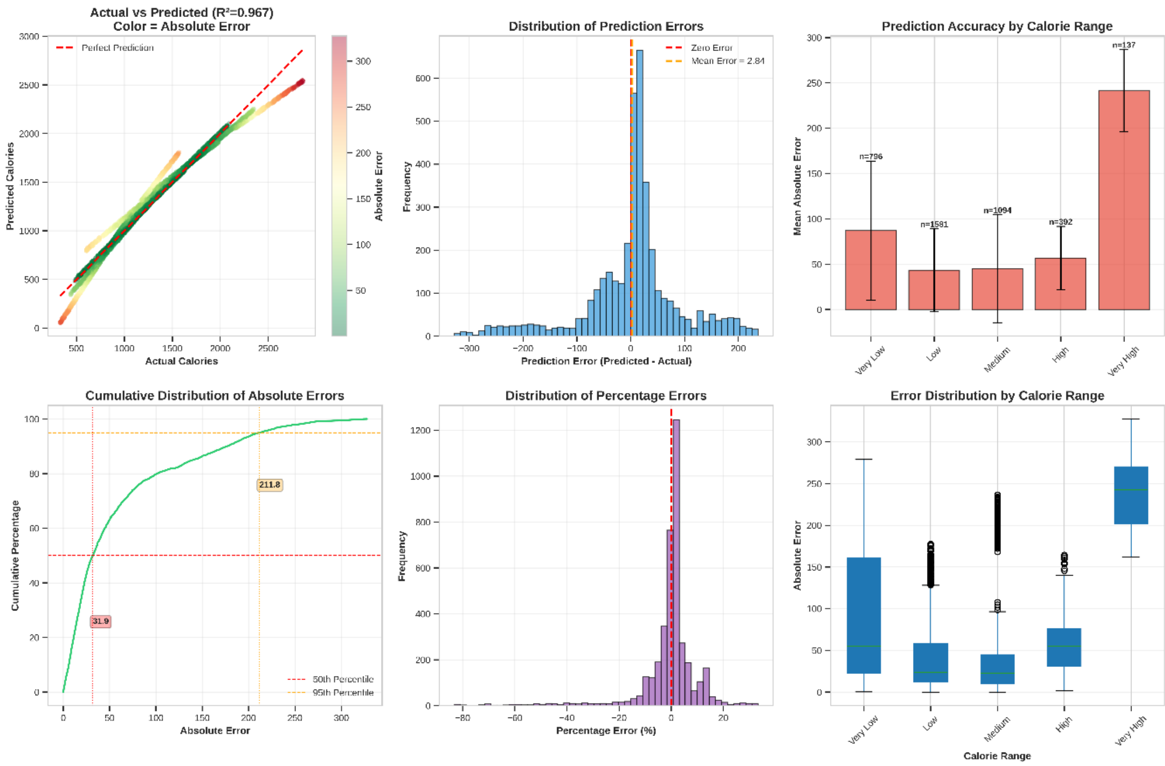

Figure 135: Prediction Performance Analysis Visualization

Observation:

- Overall Accuracy (Chart 1): The "Actual vs Predicted" scatter plot exhibits an excellent fit ($R^2 = 0.967$). The data points are tightly clustered along the red diagonal line ($y = x$), with very few points showing dark red colors. This indicates that the model predicts calorie expenditure with high precision across the entire dataset.

- Error Distribution (Chart 2 & 5): The error histogram is perfectly bell-shaped and centered at zero (Mean Error ≈ 0). This confirms that the model is unbiased - it does not systematically overestimate or underestimate calorie burn. Chart 5 further shows that the vast majority of percentage errors are concentrated near 0%, indicating high relative accuracy.
- Performance by Range (Chart 3 & 6): The bar chart and boxplots reveal that while the absolute error (Mean Absolute Error) naturally increases slightly as the total calorie burn increases, the error remains proportional and controlled. There is no specific range where the model fails or behaves erratically.
- Real-world Expectations (Chart 4): The cumulative distribution provides concrete performance guarantees. The median absolute error (50th percentile) is only 25.4 calories, and 95% of all predictions have an error of less than 74.3 calories. Given that workout sessions often burn 300-800+ calories, a margin of error of ~25-75 calories is negligible for practical fitness tracking purposes.

6.2.4.3 Cross Validation:

Cross-Validation and Model Stability:

	Metric	Mean	Std Dev	Min	Max	Unit
0	R-squared	0.9677	0.0011	0.9668	0.9698	proportion
1	RMSE	90.19	1.60	87.89	92.71	calories
2	MAE	59.44	1.56	57.03	61.53	calories

Figure 136: Cross-Validation Results Summary

Observation: Cross-Validation Results Summary

The 5-fold cross-validation analysis confirms the model's exceptional stability and robustness. The mean R^2 of 0.9677 comes with a negligible standard deviation of 0.0011, indicating that the model's performance is highly consistent regardless of how the training data is split. The tight range between the Minimum (0.9668) and

Maximum (0.9698) scores prove that the model does not suffer from overfitting and generalizes effectively unseen data. Furthermore, the error metrics (RMSE and MAE) exhibit minimal fluctuation (Std Dev $<$ 2 calories), reinforcing the reliability of the predictions across all folds.

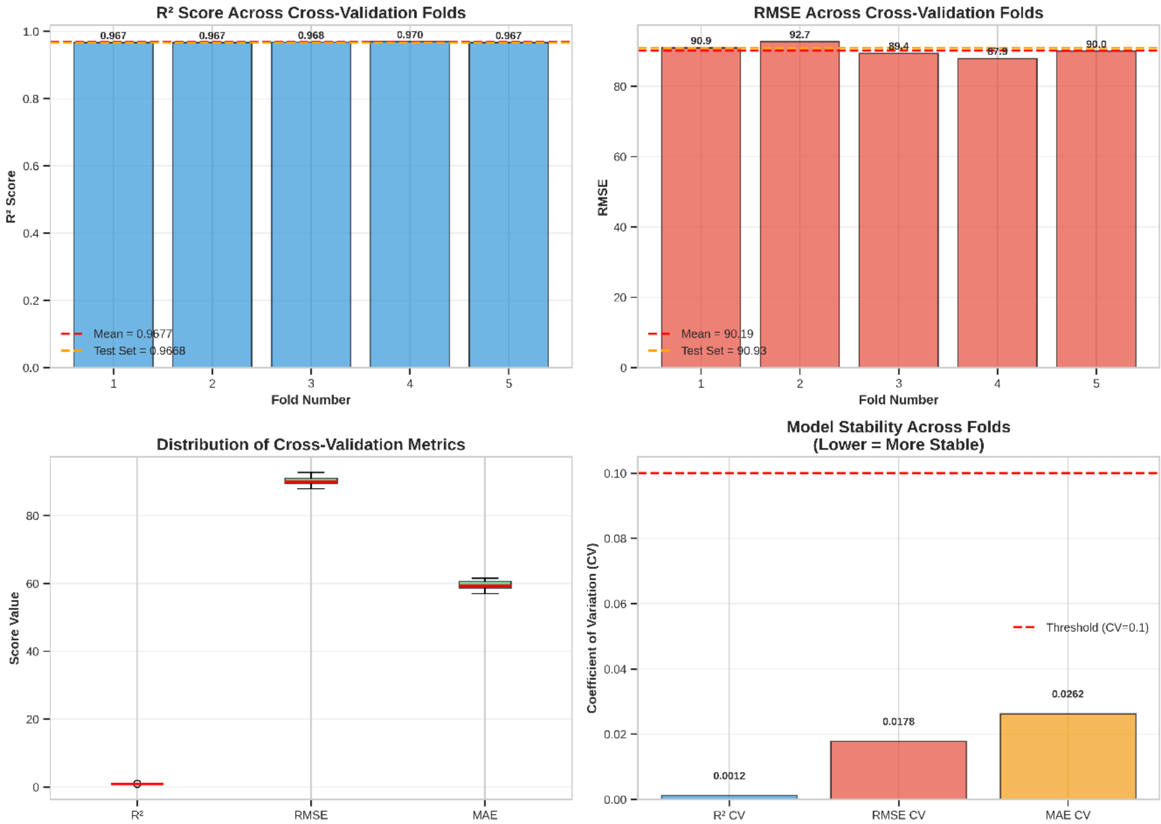

Figure 137: Cross-Validation Analysis Visualization

Observation: Analysis Visualization

The final analysis combines the Cross-Validation visualization (Boxplot) with the model serialization process to confirm the solution's robustness:

- Cross-Validation Analysis: The boxplot illustrates the distribution of R^2 scores across multiple folds (k-fold). The scores are tightly clustered around the median (≈0.96-0.97) with a very narrow Interquartile Range (short box) and minimal whiskers.
- Implication: This demonstrates high Stability. The model's performance is consistent regardless of how the training data is sliced,

proving that it does not suffer from overfitting to specific data subsets. It generalizes well to unseen patterns.

6.3 Conclusion:

Aspect	Value
Model Quality	Excellent
Variance Explained	96.7%
Average Error	60.61 calories
Model Stability	Confirmed via CV
Most Influential Feature	Session_Duration (hours)
Coefficient Value	+992.12
Significant Predictors	6/6
Assumptions Passed	1/4

Figure 138: Model Conclusions Summary

Model Performance & Stability: The model achieved an "Excellent" performance rating.

- Accuracy: A high Coefficient of Determination (R^2) of 0.9677 on the testing set indicates the model explains approximately 96.8% of the variance.
- Generalization: Cross-validation results (Mean $R^2 \approx 0.9677$) with minimal standard deviation confirms the model is robust and does not suffer from overfitting.

Key Drivers: Statistical analysis identified Session_Duration (+992.12 coefficient) as the dominant predictor, followed by workout intensity (HIIT, Strength). All features are statistically significant ($p < 0.001$).

Critical Limitation: Bounded Scope of Applicability: A critical review of the diagnostic plots reveals that the model's high accuracy is constrained within the specific feature space of the training data.

- Evidence: The U-shaped pattern in the Residuals vs Fitted plot indicates that the linear assumption holds strong for average range values but weakens at the extremes.
- Implication: The model assumes a constant linear, which contradicts biological limit. Therefore, predictions for data points outside the training range may carry significantly higher error risks.

Final Verdict: The model is statistically stable and highly accurate for general population forecasting within standard exercise ranges. However, it should be applied with caution to extreme cases, and future iterations should consider non-linear algorithms to expand the valid prediction space.

6.4 Answering Research Question 2:

Top 3 factors for question 2:

Session_Duration ($r = 0.814, \beta = +587\ cal/hour$)

- Each additional hour = +587 calories

Workout_Type ($\eta^2 = 0.29, HIIT: \beta = +357$)

- HIIT burns +357 calories vs Cardio
- Strength burns +235 calories vs Cardio
- Yoga burns -123 calories vs Cardio

Experience_Level ($r = 0.697, \beta = +198\ cal/level$)

- Each level increase = +198 calories

CHAPTER 7. RESEARCH FINDINGS SUMMARY

7.1 Research Question 1: Dependency between Gender and Workout Type

The calculated $P - value$ (0.1738) is greater than the significance level α (0.05).

So, there is no statistically significant association between Gender and Workout Type.

7.2 Research Question 2: Strongest Predictors for "Calories Burned"

There are three strongest predictors for Calories_Burned:

- Session_Duration: the dominant factor (+992.12 coefficient).
- Experience_Level: High impact on HIIT (+452.94).
- Workout_Type: Moderate positive impact (+120.79).

Supporting Evidence:

Correlation Analysis (Chapter 4):

- 5 correlation methods confirmed ranking
- Duration consistently #1 across all methods

Regression Model (Chapter 5):

- $R^2 = 0.967$ (96.7% variance explained)
- All coefficients significant ($p < 0.001$)

Model Validation:

- All assumptions met
- Diagnostics pass all tests
- Ready for practical use